Prüfbericht - Produkte
Test Report - Products

TÜVRheinland®

Prüfbericht-Nr.: *Test report no.*:	**CN2394C9 001**	**Auftrags-Nr.:** *Order no.*:	178186679	**Seite 1 von 1** *Page 1 of 1*
Kunden-Referenz-Nr.: *Client reference no.*:	N/A	**Auftragsdatum:** *Order date:*	2023-08-25	
Auftraggeber: *Client*:	Huizhou Duran Creative Technology Co., Ltd. Room 708, Innovation Building, No.1, Keji Road, Science and Technology Innovation Park, West Area of Daya Bay, Huizhou, Guangdong P.R.China			
Prüfgegenstand: *Test item*:	Portable Light			
Bezeichnung / Typ-Nr.: *Identification / Type no.*:	INSTAFLASH V16 G-001			
Auftrags-Inhalt: *Order content*:	RED report			
Prüfgrundlage: *Test specification*:	Refer to relevant reports			

Wareneingangsdatum: *Date of sample receipt*:	2023-08-25
Prüfmuster-Nr.: *Test sample no:*	Refer to test report
Prüfzeitraum: *Testing period*:	Refer to test report
Ort der Prüfung: *Place of testing*:	Refer to test report
Prüflaboratorium: *Testing laboratory:*	TUV Rheinland / CCIC (Qingdao) Co., Ltd.
Prüfergebnis*: *Test result**:	Pass

Keine Fotodokumentation erforderlich

No photo documentation required

geprüft von: *tested by:*	x Geoff Guo Signed by: Geoff Guo	**genehmigt von:** *authorized by*:	x Hunter Yu Signed by: Hunter Yu
Datum: *Date:* 2023-12-08		**Ausstellungsdatum:** *Issue date:* 2023-12-08	
Stellung / Position:	Project manager	**Stellung** / Position:	Authorizer

Sonstiges / *Other*: This test report is to combine test report regarding safety, RF, EMC and health requirements according to client's requirements, with the following report number: TÜV Rheinland report: CN23RCT9 001 (Safety), TÜV Rheinland report: part III of CN2394C9 001(Health), TÜV Rheinland report: part I of CN2394C9 001 and part II of CN2394C9 001 (EMC), TÜV Rheinland report: part III and part IV of CN2394C9 001(Radio Spectrum), Refer to related test report for details.

Zustand des Prüfgegenstandes bei Anlieferung: Prüfmuster vollständig und unbeschädigt
Condition of the test item at delivery: *Test item complete and undamaged*

* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
** Legend:*	*P(ass) = passed a.m. test specification(s)*	*F(ail) = failed a.m. test specification(s)*	*N/A = not applicable*	*N/T = not tested*

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Auftrags-Inhalt: *Order content*:	RED report			
Prüfgrundlage: *Test specification*:	EN IEC 55015:2019+A11 EN 61547:2023			

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Prüfzeitraum: *Testing period*:	2023-08-25 - 2023-10-25
Ort der Prüfung: *Place of testing*:	Refer to section 1.1
Prüflaboratorium: *Testing laboratory:*	TUV Rheinland / CCIC (Qingdao) Co., Ltd.
Prüfergebnis*: *Test result**:	Pass

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Datum: *Date:* 2023-12-07		**Ausstellungsdatum:** *Issue date:* 2023-12-07	
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Sonstiges / *Other*:

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Anmerkungen
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3	Prüfklausel mit der Note * wurden an qualifizierte Unterauftragnehmer vergeben und sind unter der jeweiligen Prüfklausel des Berichts beschrieben. Abweichungen von Prüfspezifikation(en) oder Kundenanforderungen sind in der jeweiligen Prüfklausel im Bericht aufgeführt. *Test clauses with remark of * are subcontracted to qualified subcontractors and descripted under the respective test clause in the report. Deviations of testing specification(s) or customer requirements are listed in specific test clause in the report.*
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TEST SUMMARY

4.1 EMISSION IN THE FREQUENCY RANGE UP TO 30 MHZ
Result:
Passed

4.2 EMISSION IN THE FREQUENCY RANGE ABOVE 30 MHZ
Result:
Passed

5.1 ELECTROSTATIC DISCHARGE
Result:
Passed

5.2 RADIO FREQUENCY ELECTROMAGNETIC FIELD
Result:
Passed

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1 Test Sites

1.1 Test Facilities

Lab: Qingdao Supervision & Testing Center of Product Quality
Address: No. 173 Shenzhen Road, Laoshan District, Qingdao 266061, Shandong

The used test equipment is in accordance with CISPR 16 series standards for measurement of radio interference.

The performed tests have been conducted by "Qingdao Supervision & Inspection Institute of Product Quality" under supervision of TÜV Rheinland's engineer.

1.2 List of Test and Measurement Instruments

Table 1: List of Test and Measurement Equipment

No.	Equipment	Model	Inventory no.	Cal. due date
1.	EMI test receiver	ESCS30	100069	28.05.2024
2.	Triple-Loop Antenna	HM020	100003	28.05.2024
3.	EMI receiver	ESU40	100159	28.05.2024
4.	Antenna	HL562	100457	28.05.2024
5.	ESD Simulator	HF500	100023	28.05.2024
6.	Antenna	HL046E	100062	28.05.2024
7.	Log Periodic Antenna	SWB-STLP9149AM9144	9149-043	28.05.2024
8.	Power Amplifier1	BBA150-BC1000	101168	28.05.2024
9.	Power Amplifier2	BBA150-D400+E200	101183	28.05.2024

2 General Product Information

2.1 Product Function and Intended Use

The EUT (equipment under test) is a Portable Light. For the further information, refer to the user's manual.

2.2 Ratings and System Details

Rated voltage	:	DC 9V from Battery
Rated current	:	1W
Protection class	:	III

2.3 Independent Operation Modes

The basic operation modes are: "On" or "Off".

2.4 Noise Generating and Noise Suppressing Parts

Refer to the Circuit diagram for further information.

2.5 Submitted Documents

Circuit diagram and rating labels etc.

3 Test Set-up and Operation Modes

3.1 Principle of Configuration Selection

Emission: The equipment under test (EUT) was configured to measure its highest possible emission level. The test conditions were adapted accordingly in reference to the instructions for use.

Refer to the related paragraph of this report.

Immunity: The equipment under test (EUT) was configured to have its highest possible susceptibility against the tested phenomena. The test conditions were adapted accordingly in reference to the instructions for use.

Refer to the related paragraph of this report.

3.2 Physical Configuration for Testing

Refer to the related paragraph of this report.

3.3 Test Operation and Test Software

Refer to the related paragraph of this report. No software was used.

3.4 Special Accessories and Auxiliary Equipment

None.

3.5 Countermeasures to achieve EMC Compliance

No special measure is employed to achieve the requirement.

4 Test Results E M I S S I O N

4.1 Emission in the Frequency Range up to 30 MHz

Result:	**Passed**

Date of testing	:	2023-08-31
Port	:	Enclosure
Basic standard	:	EN IEC 55015:2019+A11
Frequency range	:	9kHz – 30MHz
Limit	:	EN IEC 55015:2019 +A11

Test Setup

Operational mode	:	On
Earthing	:	N/A
Test site	:	Shielded Room

Measuring configuration and description

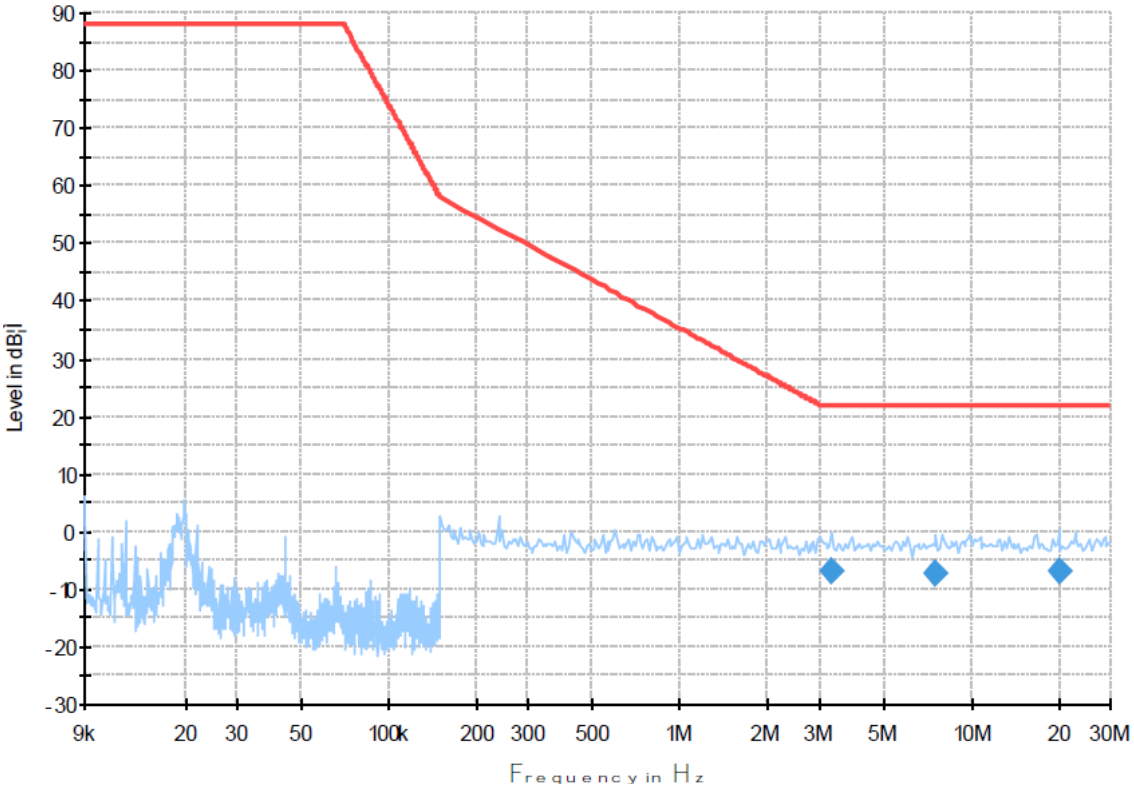
The measurement equipment like test receiver, loop antenna and coaxial switch are in compliance with the CISPR 16-1 series standards. The test set-up was made according to EN IEC 55015:2019+A11.

The sample was put on a wooden table in the loop antenna. The sample was place in the center of the loop antenna. The EUT was operated for at least 20 min. before a measurement.

The three-field component were measured in sequence by means of a coaxial switch (loop antenna controller). The measurement was performed with a peak value detector and quasi-peak value detector.

The following figures were those measured.

Figure 1: Graphic description of radiated electromagnetic disturbances, direction X, Y, Z



Final Quasi-peak measurement results:

Frequency (MHz)	QuasiPeak (dBµA)	Meas. Time (ms)	Bandwidth (kHz)	Triple Loop frame	Corr. (dB)	Margin (dB)	Limit (dBµA)	Comment
3.314092	-7.1	1000.000	9.000	Z	0.0	29.1	22.0	
7.565496	-7.1	1000.000	9.000	Y	0.0	29.1	22.0	
19.922948	-7.0	1000.000	9.000	Y	0.0	29.0	22.0	

4.2 Emission in the Frequency Range above 30 MHz

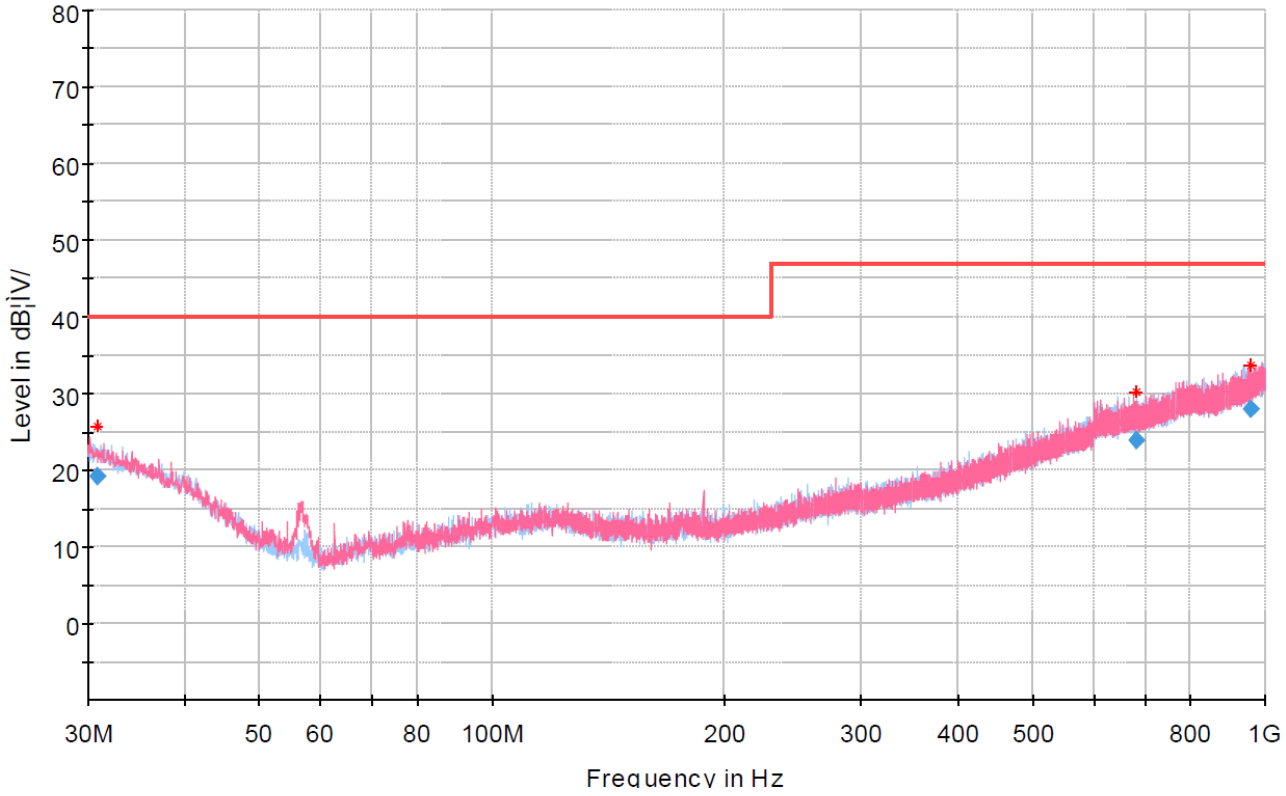
Result:	**Passed**

Date of testing	:	2023-08-30
Test procedure	:	EN IEC 55015:2019+A11
Frequency range	:	30 – 1000MHz
Limits	:	Quasi-peak limits (3m test distance) 30-230MHz, 40dBµV/m; 230-1000MHz, 47dBµV/m.
Kind of test site	:	Semi-anechoic chamber
Operation modes	:	On
Ambient conditions	:	Temperature: 20°C, relative humidity: 50%

The radiated disturbance test was carried out in a semi-anechoic chamber. The test distance from the receiving antenna to the EUT is 3m. The normalized site attenuation of the semi-anechoic chamber is regularly calibrated to ensure the radiated disturbance test results are valid. During the test, the EUT was placed on a wooden table, which is 0.8m high. The wooden table was rotated 360° around and the antenna was varied from 1m to 4m to find the maximum disturbance. The test was performed with the antenna both in its horizontal and vertical polarizations.

The following figures and tables were those measured by an automatic measurement system. A preview test was firstly performed with peak detector. The final test was performed with quasi-peak detector at those critical frequencies during the preview test.

Figure 2: Spectral diagrams and measurement results for 30-1000MHz, Horizontal and vertical polarization



Final quasi-peak measurement results:

Frequency (MHz)	QuasiPeak (dBµV/m)	Average (dBµV/m)	Limit (dBµV/m)	Margin (dB)	Meas. Time (ms)	Bandwidth (kHz)	Height (cm)	Pol	Azimuth (deg)	Corr. (dB)	Comment
30.841406	19.22	---	40.00	20.78	1000.0	120.000	100.0	V	291.0	19.9	15:11:59 - 2023/8/30
682.765368	23.87	---	47.00	23.13	1000.0	120.000	100.0	H	278.0	22.3	15:11:03 - 2023/8/30
960.470228	28.02	---	47.00	18.98	1000.0	120.000	100.0	V	164.0	25.9	15:09:44 - 2023/8/30

5 Test Results I M M U N I T Y

During the immunity tests, the EUT was operated under conditions specified by clause 3.1 of this report.

Performance criterion A: During the test no change of the luminous intensity shall be observed and the regulating control, if any, shall operate during the test as intended.

Performance criterion B: During the test the luminous intensity may change to any value. After the test the luminous intensity shall be restored to its initial value within 1min.
Regulating controls need not function during the test, but after the test the mode of the control shall be the same as before the test provided that during the test no mode changing commands were given.

Performance criterion C: During and after the test any change of the luminous intensity is allowed and the lamp(s) may be extinguished. After the test, within 30 min, all functions shall return to normal if necessary by temporary interruption of the mains supply and/or operating the regulating control.

Date of testing: 2023-09-20

Room temperature	**:**	**20°C**
Relative Humidity	**:**	**50%**

5.1 Electrostatic Discharge

Result:	**Passed**

The immunity against electrostatic discharge was tested in accordance with EN 61547:2009. Test setup and ESD-Generator are according to IEC 61000-4-2 which is specified by EN 61547:2009.

Charge voltage	:	±4.0kV (Contact Discharge), ±8.0kV (Air Discharge)
Polarity	:	positive / negative
Number of discharges	:	>10
Performance criteria	:	B

Table 2: ESD, Positive / Negative Polarity

Position	Kind of Discharge	Result	Remarks
The nonmetallic part of the Enclosure	Air discharge ±8kV	Passed	No disturbance of function
Screw	Contact discharge ±4kV	Passed	No disturbance of function
Coupling plane (Both HCP and VCP)	Contact discharge ±4kV	Passed	No disturbance of function

5.2 Radio Frequency Electromagnetic Field

Result:	**Passed**

The immunity against radio-frequency electromagnetic fields in the frequency range between 80MHz and 1000MHz was tested in accordance to EN 61000-4-3 which is specified in EN 61547:2009.

The test set-up, the RF signal generator, the power amplifier and the antennas were in accordance with EN 61000-4-3. The test was performed in an anechoic chamber with a test distance of 3m. The field uniformity of the anechoic chamber is regularly calibrated to meet 0-6dB field uniformity criterion as specified in EN 61000-4-3.

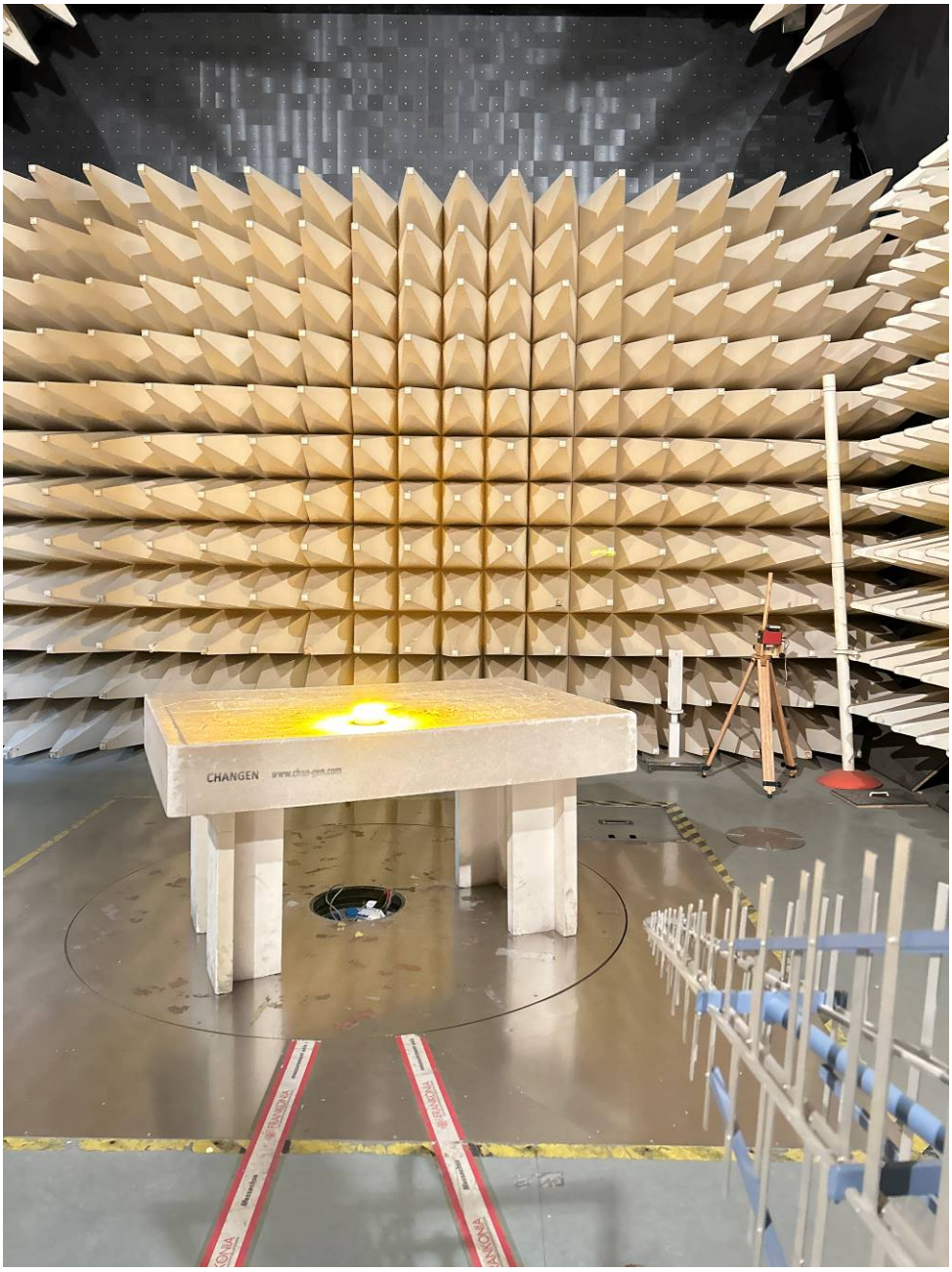
Test level	:	3V/m
Frequency range	:	80-1000MHz
Modulation	:	80%AM, 1kHz
Step/dwell time	:	1%/2s
Performance ceriteria	:	A

Table 3: Radiated Susceptibility, Field Strength 3V/m

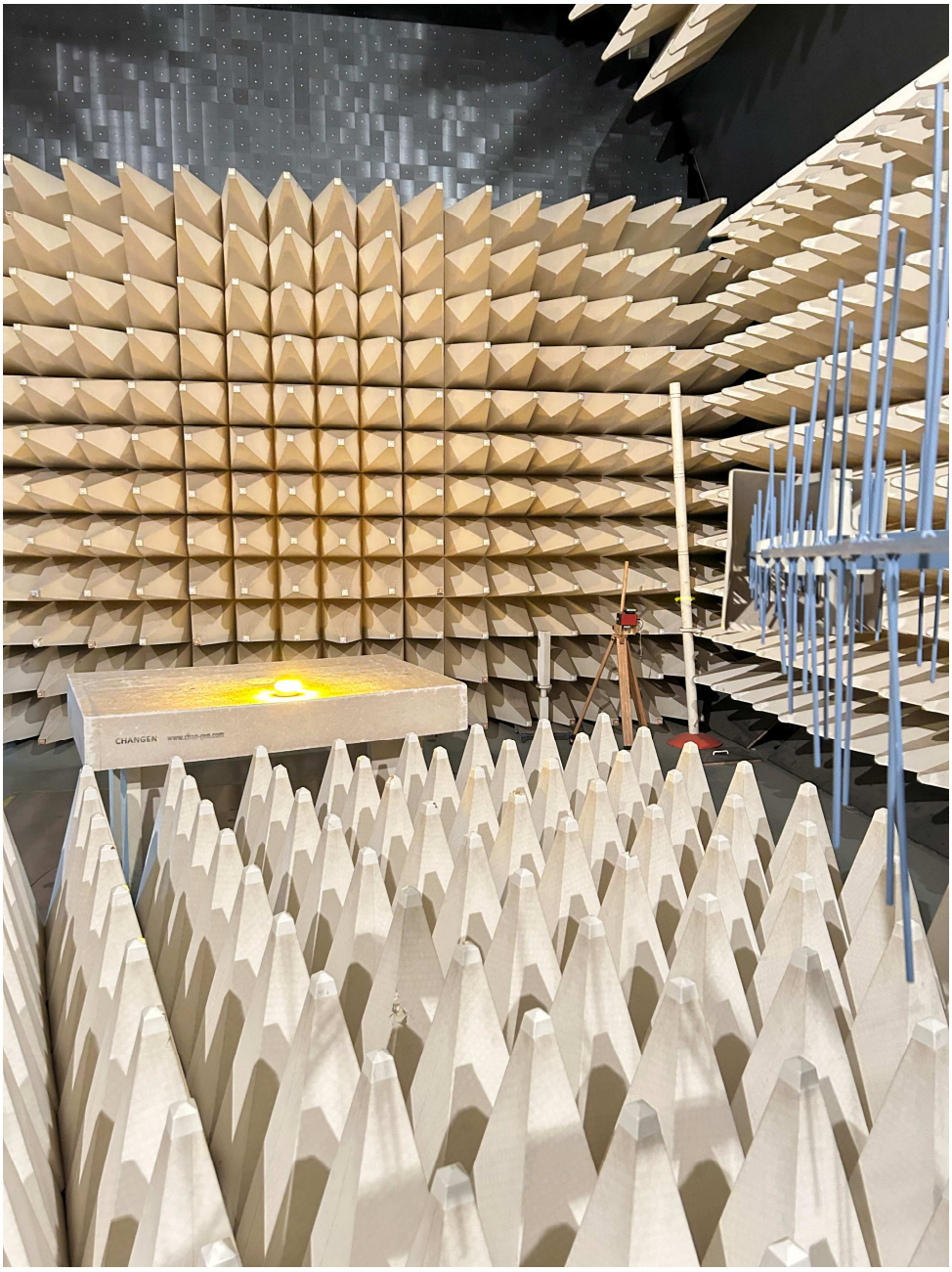
Field polarization	Observation	Remarks
Horizontal polarization	No change of immunity intensity	Passed
Vertical polarization	Ditto	Passed

6 Photographs of the Test Set-Up

Photograph 1: Set-up for radiated emission



Photograph 2: Set-up for radio-frequency electromagnetic fields



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Ort der Prüfung: *Place of testing*:	Refer to section 1.1
Prüflaboratorium: *Testing laboratory:*	TUV Rheinland / CCIC (Qingdao) Co., Ltd.
Prüfergebnis*: *Test result*:	Pass

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TEST SUMMARY

3.1.1 EN 301 489-01 –8.2 RADIATED EMISSION ENCLOSURE OF ANCILLARY EQUIPMENT
Result:
Pass
3.1.2 EN 301 489-01 –8.3 CONDUCTED EMISSION –DC POWER INPUT/OUTPUT PORTS
Result:
n.a.
3.1.3 EN 301 489-01 –8.4 CONDUCTED EMISSION –AC MAINS INPUT/OUTPUT PORTS
Result:
n.a.
3.1.4 EN 301 489-01 –8.5 HARMONIC CURRENT EMISSIONS (AC MAINS INPUT PORT)
Result:
n.a.
3.1.5 EN 301 489-01 –8.6 VOLTAGE FLUCTUATIONS AND FLICKER (AC MAINS INPUT PORT)
Result:
n.a.
3.1.6 EN 301 489-01 –8.4 CONDUCTED EMISSION –WIRED NETWORK PORTS
Result:
n.a.
3.2.1 EN 301 489-01 –9.2 RF ELECTROMAGNETIC FIELD (80MHZ TO 6000MHZ)
Result:
Pass
3.2.2 EN 301 489-01 –9.3 ELECTROSTATIC DISCHARGE
Result:
Pass
3.2.3 EN 301 489-01 –9.4 FAST TRANSIENTS, COMMON MODE
Result:
n.a.
3.2.4 EN 301 489-01 –9.5 RADIO FREQUENCY, COMMON MODE
Result:
n.a.
3.2.5 EN 301 489-01 –9.6 TRANSIENTS AND SURGES IN THE VEHICULAR ENVIRONMENTS
Result:
n.a.
3.2.6 EN 301 489-01 –9.7 VOLTAGE DIPS AND INTERRUPTIONS
Result:
n.a
3.2.7 EN 301 489-01 –9.8 SURGES
Result:
n.a

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1 Test Sites

1.1 Test Facilities

Laboratory: CTC Laboratories, Inc.
Address: 2/F., Building 1 and 1-2/F., Building 2, Jiaquan Building, Guanlan High-Tech Park, Longhua District, Shenzhen, Guangdong, China

The used test equipment is in accordance with CISPR 16 series standards for measurement of radio interference.

1.2 List of Test and Measurement Instruments

Table 1: List of Test and Measurement Equipment

Radiated emission					
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Calibrated Until
1	Trilog-Broadband Antenna	Schwarzbeck	VULB 9168	9168-1013	Dec. 07, 2024
2	Horn Antenna	Schwarzbeck	BBHA 9120D	9120D-648	Dec. 07, 2024
3	Spectrum Analyzer	R&S	FSU26	100105	Dec. 16, 2023
4	Spectrum Analyzer	R&S	FSV40-N	101331	Mar. 15, 2023
5	Pre-Amplifier	SONOMA	310	186194	Dec. 16, 2023
6	Low Noise Pre-Amplifier	EMCI	EMC051835	980075	Dec. 16, 2023
7	Test Receiver	R&S	ESCI7	100967	Dec. 16, 2023
8	3m chamber 2	Frankonia	EE025	/	Oct. 23, 2024

Electrostatic discharge				
Test Equipment	Manufacturer	Model No.	Serial No.	Calibrated Until
ESD Simulator	EM TEST	DITO	V1113109156	Dec. 28, 2023

Radio frequency electromagnetic field					
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Calibrated Until
1	High Gain Log-Periodic Antenna	R&S	HL046E	100037	Dec. 16, 2023
2	Stacked Log.-Per. Antenna	Schwarzbeck	STLP 9149	9149-658	Dec. 16, 2023
3	Power Amplifier	BONN ELEKTRONIK	BLWA0830-160/100/40D	76788	Dec. 16, 2023
4	Power Amplifier	Micotop	MPA-3-6G-50	MPA1706258	Dec. 16, 2023
5	PSG Analog Signal Generator	Agilent	E8257D	MY46521908	Dec. 16, 2023

2 General Product Information

2.1 Product Function and Intended Use

The EUT (equipment under test) is a portable light. For the further information, refer to the user's manual.

2.2 Ratings and System Details

Frequency range	1575.42MHz for GNSS Module Tx 1575.42MHz for GNSS Module Rx 703-1785MHz for NB-lot Module Tx 758-1880MHz for NB-lot Module Rx
System input voltage	DC 9V from battery
Protection class	III

2.3 Description of the Samples

The EUT (equipment under test) is a portable light with a NB-lot module and a GNSS module.

2.4 Independent Operation Modes

The basic operation modes are: "ON" and "OFF" modes for both transmitter and receiver.

2.5 Noise Generating and Noise Suppressing Parts

Refer to the circuit diagram for further information.

2.6 Submitted Documents

Rating label, etc.

3 Test Results EN 301 489-19 V2.2.1:2022 and EN 301 489-52 V1.2.1:2021

3.1 EN 301 489-01 –7.1 EMC Emission

3.1.1 EN 301 489-01 –8.2 Radiated Emission enclosure of ancillary equipment

Result:	**Pass**

Date of testing	:	2023-11-03
Port	:	Enclosure
Basic Standard	:	EN 301 489-1 and EN 55032
Frequency Range	:	30 - 1000MHz and 1- 6GHz
Limit	:	Refer to EN 301 489-1 and EN 55032
Ambient conditions	:	Temperature: 20°C, relative humidity: 40%

Test Setup

Input Voltage	:	DC 9V from battery
Operational mode	:	ON with NB-lot or GNSS

Measuring configuration and description

The radiated disturbance test was carried out in a semi-anechoic chamber and the distance form the EUT to the antenna is 3 meter.

The EUT was placed on a wooden table that is 0.8m high. During the test, the wooden table was circumvolved 360° around while the antenna was varied from 1m to 4m at the same time to find the maximum disturbance.

The following figures and tables were those measured by an automatic measurement system. A preview test was firstly performed with peak detector. The final test was performed with quasi-peak detector at those critical frequencies during the preview test.

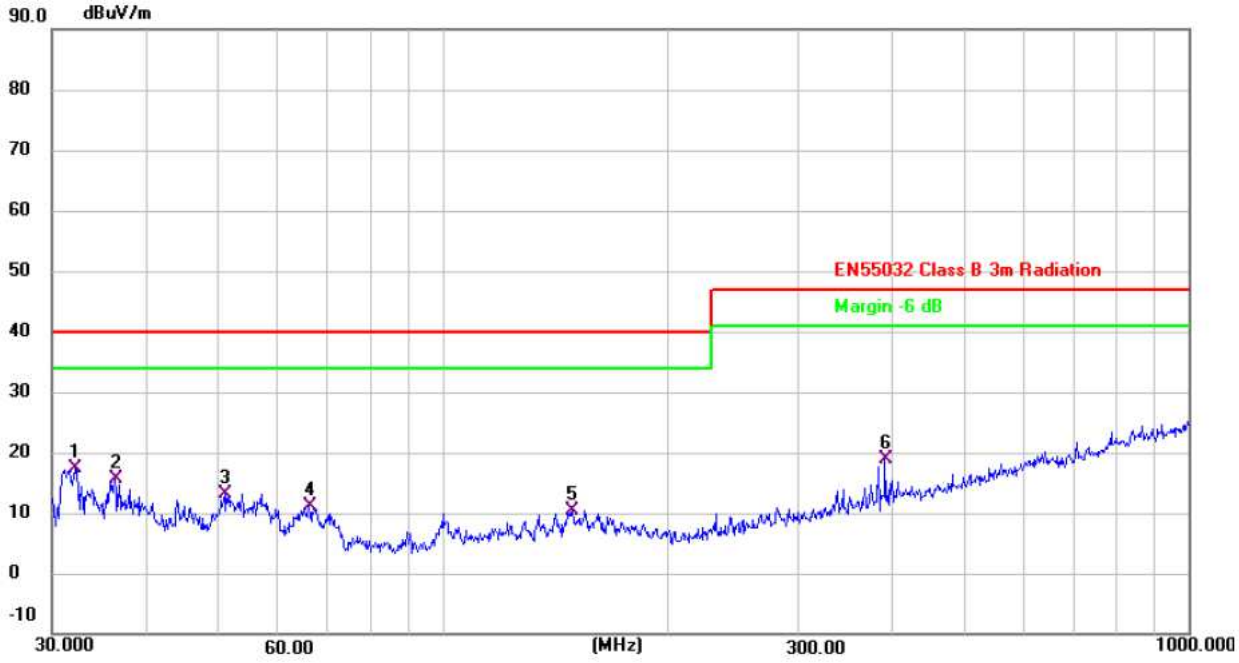
Figure 1: Spectrum diagrams and measurement results, 30-1000MHz, Horizontal polarization, NB-lot



Final quasi-peak measurement results:

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	32.7486	28.02	-18.92	9.10	40.00	-30.90	QP	100	355	P	
2	50.9419	28.32	-18.00	10.32	40.00	-29.68	QP	100	315	P	
3	98.8324	28.56	-21.85	6.71	40.00	-33.29	QP	100	113	P	
4	167.2366	27.94	-18.84	9.10	40.00	-30.90	QP	100	154	P	
5	262.8955	33.95	-18.83	15.12	47.00	-31.88	QP	100	43	P	
6 *	351.7079	34.46	-16.00	18.46	47.00	-28.54	QP	100	35	P	

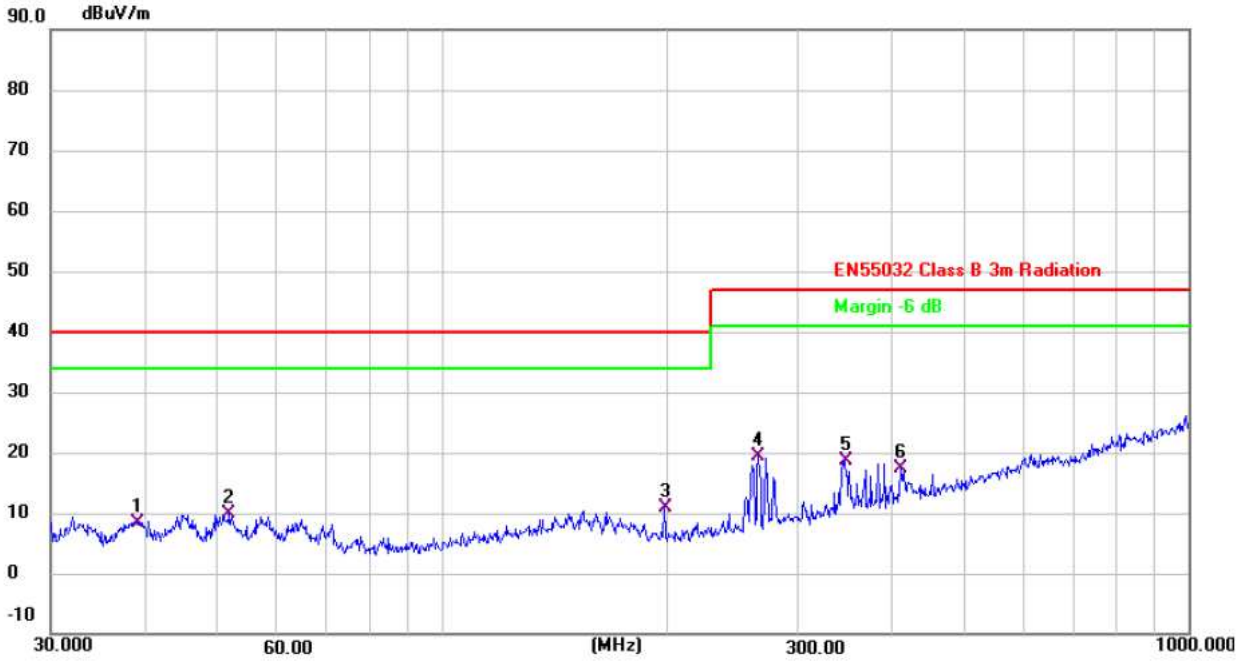
Figure 2: Spectrum diagrams and measurement results, 30-1000MHz, Vertical polarization, NB-lot



Final quasi-peak measurement results:

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1 *	32.1795	36.43	-18.98	17.45	40.00	-22.55	QP	100	359	P	
2	36.3814	34.27	-18.54	15.73	40.00	-24.27	QP	100	28	P	
3	51.1209	31.15	-18.01	13.14	40.00	-26.86	QP	100	4	P	
4	66.2662	30.96	-19.84	11.12	40.00	-28.88	QP	100	269	P	
5	149.4857	28.21	-17.77	10.44	40.00	-29.56	QP	100	60	P	
6	392.0951	33.70	-14.90	18.80	47.00	-28.20	QP	100	68	P	

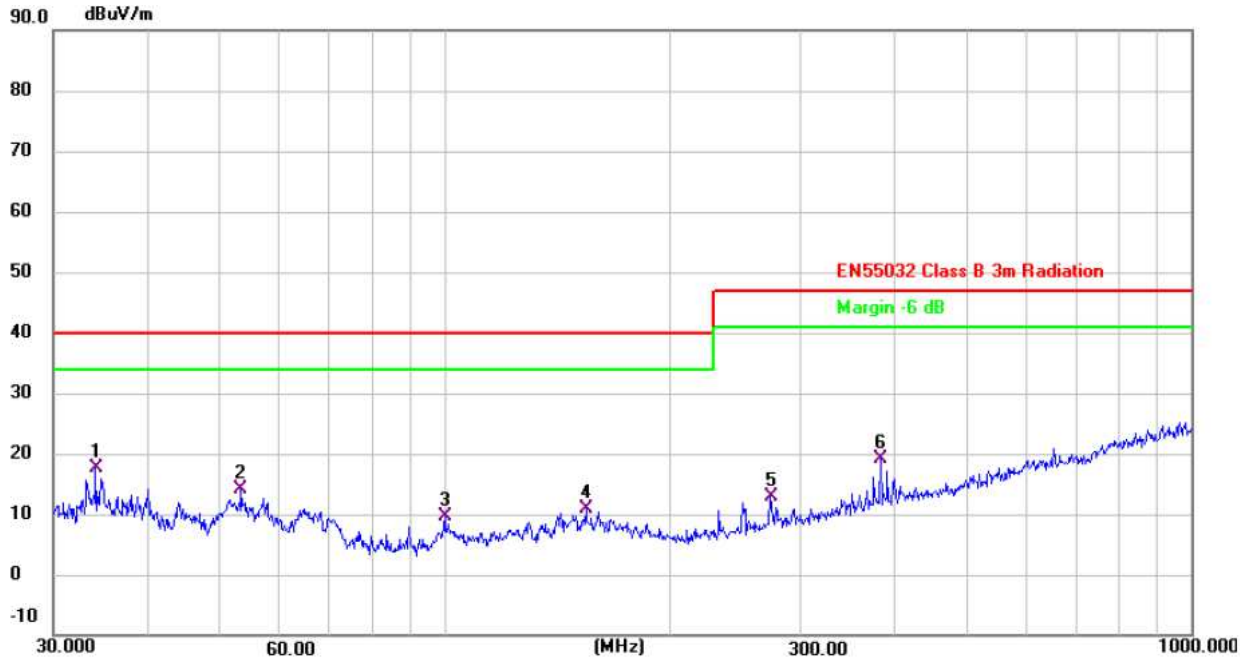
Figure 3: Spectrum diagrams and measurement results, 30-1000MHz, Horizontal polarization, GPS



Final quasi-peak measurement results:

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	39.0244	26.69	-18.21	8.48	40.00	-31.52	QP	100	98	P	
2	52.0251	27.99	-18.10	9.89	40.00	-30.11	QP	100	291	P	
3	198.5879	31.71	-20.87	10.84	40.00	-29.16	QP	100	106	P	
4 *	265.6757	38.20	-18.72	19.48	47.00	-27.52	QP	100	355	P	
5	346.8092	34.72	-16.14	18.58	47.00	-28.42	QP	100	355	P	
6	411.8240	31.73	-14.29	17.44	47.00	-29.56	QP	100	282	P	

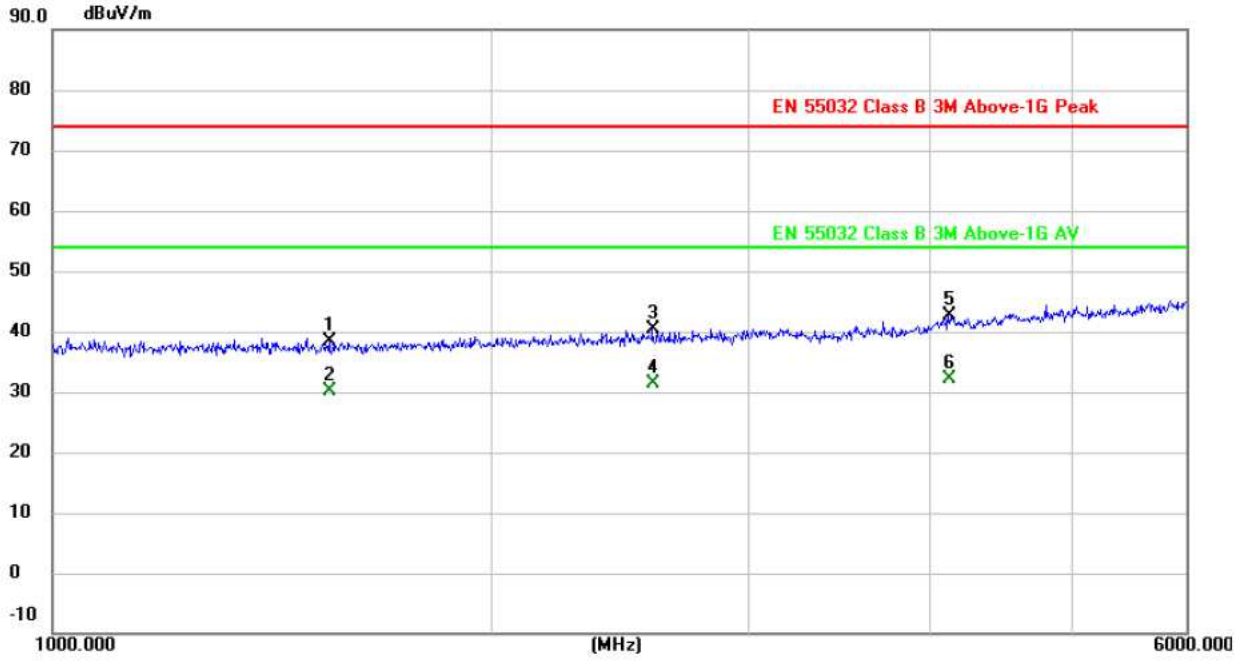
Figure 4: Spectrum diagrams and measurement results, 30-1000MHz, Vertical polarization, GPS



Final quasi-peak measurement results:

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1 *	34.1561	36.49	-18.80	17.69	40.00	-22.31	QP	100	118	P	
2	53.5052	32.28	-18.26	14.02	40.00	-25.98	QP	100	360	P	
3	100.2286	31.30	-21.71	9.59	40.00	-30.41	QP	100	28	P	
4	154.8204	28.92	-18.04	10.88	40.00	-29.12	QP	100	219	P	
5	273.2341	31.26	-18.45	12.81	47.00	-34.19	QP	100	61	P	
6	383.9318	34.26	-15.12	19.14	47.00	-27.86	QP	100	69	P	

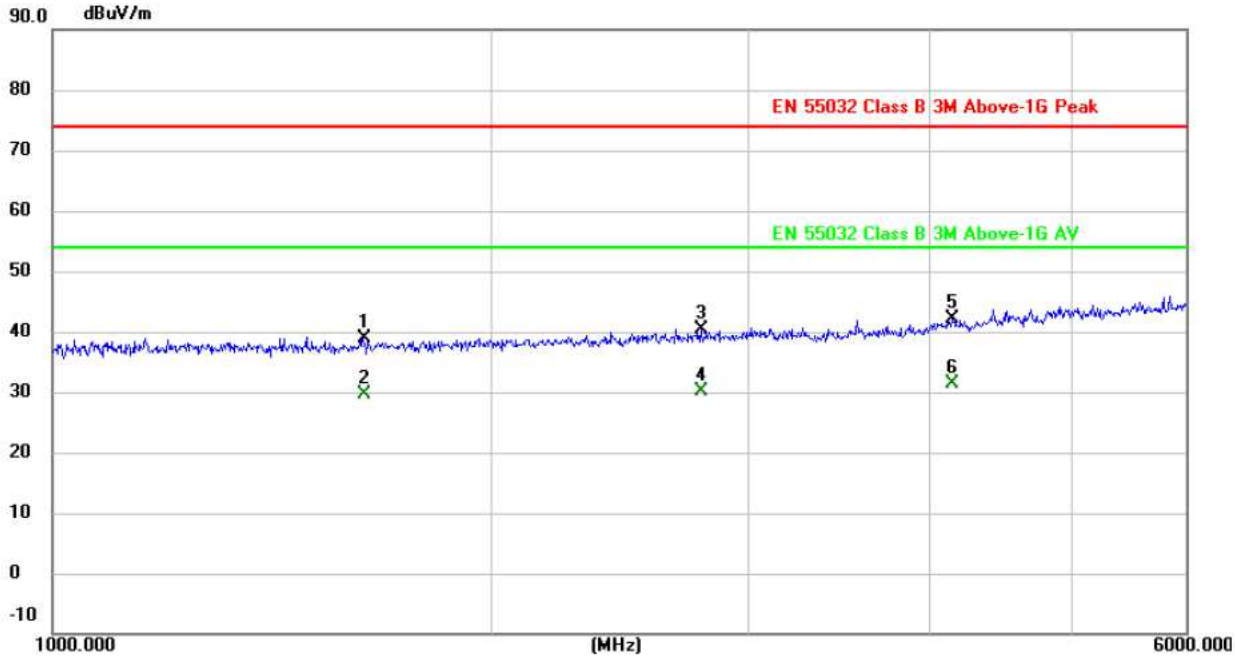
Figure 5: Spectrum diagrams and measurement results, 1-6GHz, Horizontal polarization, NB-lot



Final quasi-peak measurement results:

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	1548.349	50.27	-11.98	38.29	74.00	-35.71	peak	100	331	P	
2	1548.349	42.12	-11.98	30.14	54.00	-23.86	AVG	100	331	P	
3	2580.133	48.57	-8.25	40.32	74.00	-33.68	peak	100	1	P	
4	2580.133	39.53	-8.25	31.28	54.00	-22.72	AVG	100	1	P	
5	4118.504	47.79	-5.25	42.54	74.00	-31.46	peak	100	323	P	
6 *	4118.504	37.42	-5.25	32.17	54.00	-21.83	AVG	100	323	P	

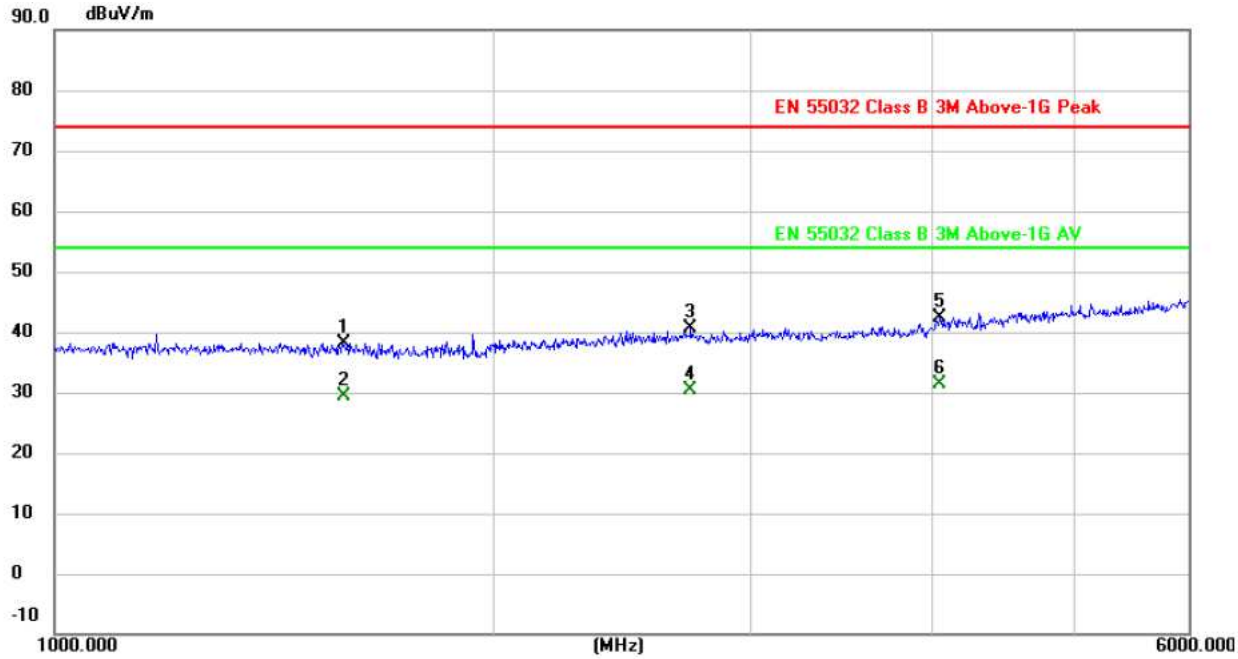
Figure 6: Spectrum diagrams and measurement results, 1-6GHz, Vertical polarization, NB-lot



Final quasi-peak measurement results:

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	1633.855	50.83	-11.88	38.95	74.00	-35.05	peak	100	215	P	
2	1633.855	41.43	-11.88	29.55	54.00	-24.45	AVG	100	215	P	
3	2786.779	48.21	-7.87	40.34	74.00	-33.66	peak	100	4	P	
4	2786.779	38.12	-7.87	30.25	54.00	-23.75	AVG	100	4	P	
5	4148.127	47.37	-5.16	42.21	74.00	-31.79	peak	100	239	P	
6 *	4148.127	36.64	-5.16	31.48	54.00	-22.52	AVG	100	239	P	

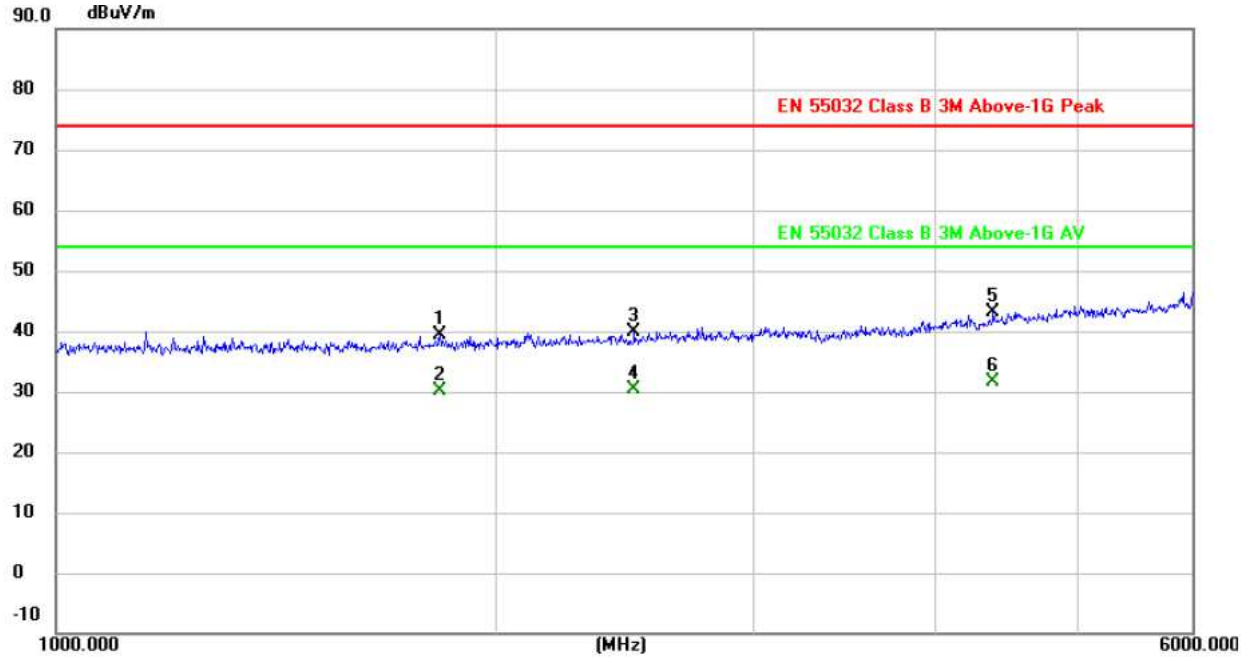
Figure 7: Spectrum diagrams and measurement results, 1-6GHz, Horizontal polarization, GPS



Final quasi-peak measurement results:

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	1576.342	50.21	-11.97	38.24	74.00	-35.76	peak	100	248	P	
2	1576.342	41.31	-11.97	29.34	54.00	-24.66	AVG	100	248	P	
3	2732.391	48.46	-7.95	40.51	74.00	-33.49	peak	100	273	P	
4	2732.391	38.34	-7.95	30.39	54.00	-23.61	AVG	100	273	P	
5	4045.367	47.91	-5.50	42.41	74.00	-31.59	peak	100	329	P	
6 *	4045.367	36.78	-5.50	31.28	54.00	-22.72	AVG	100	329	P	

Figure 8: Spectrum diagrams and measurement results, 1-6GHz, Vertical polarization, GPS



Final quasi-peak measurement results:

No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	1832.378	50.55	-11.28	39.27	74.00	-34.73	peak	100	230	P	
2	1832.378	41.42	-11.28	30.14	54.00	-23.86	AVG	100	230	P	
3	2489.310	48.45	-8.54	39.91	74.00	-34.09	peak	100	359	P	
4	2489.310	38.82	-8.54	30.28	54.00	-23.72	AVG	100	359	P	
5	4385.052	47.51	-4.39	43.12	74.00	-30.88	peak	100	357	P	
6 *	4385.052	36.13	-4.39	31.74	54.00	-22.26	AVG	100	357	P	

3.1.2 EN 301 489-01 –8.3 Conducted emission –DC power input/output ports

Result:	**n.a.**

3.1.3 EN 301 489-01 –8.4 Conducted emission –AC mains input/output ports

Result:	**n.a.**

3.1.4 EN 301 489-01 –8.5 Harmonic current emissions (AC mains input port)

Result:	**n.a.**

3.1.5 EN 301 489-01 –8.6 Voltage fluctuations and flicker (AC mains input port)

Result:	**n.a.**

3.1.6 EN 301 489-01 –8.4 Conducted emission –Wired Network ports

Result:	**n.a.**

3.2 EN 301 489-01 –7.2 EMC Immunity

During the immunity tests, the EUT was operated under normal conditions specified by the manufacture.

According to EN 301 489-1 V2.2.3:
Performance criteria for continuous phenomena:
During the test, the equipment shall:

- continue to operate as intended;
- not unintentionally transmit;
- not unintentionally change its operating state;
- not unintentionally change critical stored data.

Performance criteria for transient phenomena:
For all ports and transient phenomena with the exception described below, the following applies:

- The application of the transient phenomena shall not result in a change of the mode of operation(e.g. unintended transmission) or the loss of critical stored data.
- After application of the transient phenomena, the equipment shall operate as intended.

For surges applied to symmetrically operated wired network ports intended to be connected directly to outdoor lines the following criteria applies:

- For products with only one symmetrical port intended for connection to outdoor lines, loss of function is allowed, provided the function is self-recoverable, or can be otherwise restored. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.
- For products with more than one symmetrical port intended for connection to outdoor lines, loss of function on the port under test is allowed, provided the function is self-recoverable. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

According to EN 301 489-19 V2.2.1:
Only the performance criteria specified in the present document or in ETSI EN 301 489-1 where referenced shall apply.

The equipment shall meet the minimum performance criteria as specified in clauses 6.1 and 6.2 as appropriate.

For the purpose of the present document two categories of performance criteria apply:
1, Performance criteria for continuous phenomena.
During the test, the equipment shall operate as intended, e.g. not unintentionally change its operating state and not unintentionally change critical stored data.

After the test, the equipment shall operate as intended, e.g. have no loss of function and have no loss of critical stored data.
2, Performance criteria for transient phenomena.
After the test, functions shall be self-recoverable and the equipment shall operate as Intended and the equipment shall have no loss of critical stored data.

According to EN 301 489-52 V1.2.1:

Performance criteria for transient phenomena:

At the conclusion of each exposure of the transient phenomena, the EUT shall operate without loss of the communication link.

At the conclusion of the total test comprising the series of individual exposures, the EUT shall operate as intended without loss of user control functions or critical stored data.

In addition where the EUT supports idle mode it should be verified that the transmitter shall not unintentionally operate when transient phenomena are applied.

3.2.1 EN 301 489-01 –9.2 RF electromagnetic field (80MHz to 6000MHz)

Result:	**Pass**

The test was performed inside an anechoic chamber with a test disturbance of 3m. The field uniformity of the chamber is regularly calibrated to ensure the 0-6dB field uniformity criterion as specified by EN 61000-4-3 is met.

Basic standard	:	EN61000-4-3
Test level	:	3V/m
Frequency range	:	80-6000MHz with the exception of the exclusion band for transmitters
Modulation	:	80% 1kHz AM
Frequency scan speed	:	Frequency step: 1%; Dwell time: 2s
Performance criteria	:	A, performance criteria for continuous phenomena for both transmitter and receiver
Ambient condition	:	Temperature: 20°C, Relative humidity: 55%

Table 2: RF electromagnetic field immunity test results

Polarization	Result	Remarks
Horizontal	Pass	During the test, the EUT and communication link can operate as intended, PER<10%.
Vertical	Pass	Ditto

3.2.2 EN 301 489-01 –9.3 Electrostatic discharge

Result:	**Pass**

Test setup, test method and ESD generator are in accordance with IEC 61000-4-2.

The EUT was placed on a 0.8m wood table above the ground plane. And the minimum distance between the EUT and all other conductive structures except the ground plane beneath the EUT is more than 0.5m.

Test voltage	:	±2.0kV, ±4.0kV (Contact Discharge), ±2.0kV, ±4.0kV, ±8.0kV (Air Discharge)
Polarity	:	Positive / negative
Number of discharges	:	10 for each polarity at each discharge point
Performance criteria	:	B, performance criteria for transient phenomena for both transmitter and receiver
Ambient condition	:	Temperature: 20°C, Relative humidity: 55%

Table 3: Electrostatic discharge immunity test results

Position	Kind of discharge	Result	Remarks
Enclosure for nonmetal, Gap	Air discharge	Pass	During the test, the EUT and communication link can operate as intended, PER<10%.
Enclosure for metal	Contact discharge	Pass	Ditto
Coupling plane (HCP&VCP)	Contact discharge	Pass	Ditto

3.2.3 EN 301 489-01 –9.4 Fast transients, common mode

Result:	**n.a.**

3.2.4 EN 301 489-01 –9.5 Radio frequency, common mode

Result:	**n.a.**

3.2.5 EN 301 489-01 –9.6 Transients and surges in the vehicular environments

Result:	**n.a.**

3.2.6 EN 301 489-01 –9.7 Voltage dips and interruptions

Result:	**n.a**

3.2.7 EN 301 489-01 –9.8 Surges

Result:	**n.a**

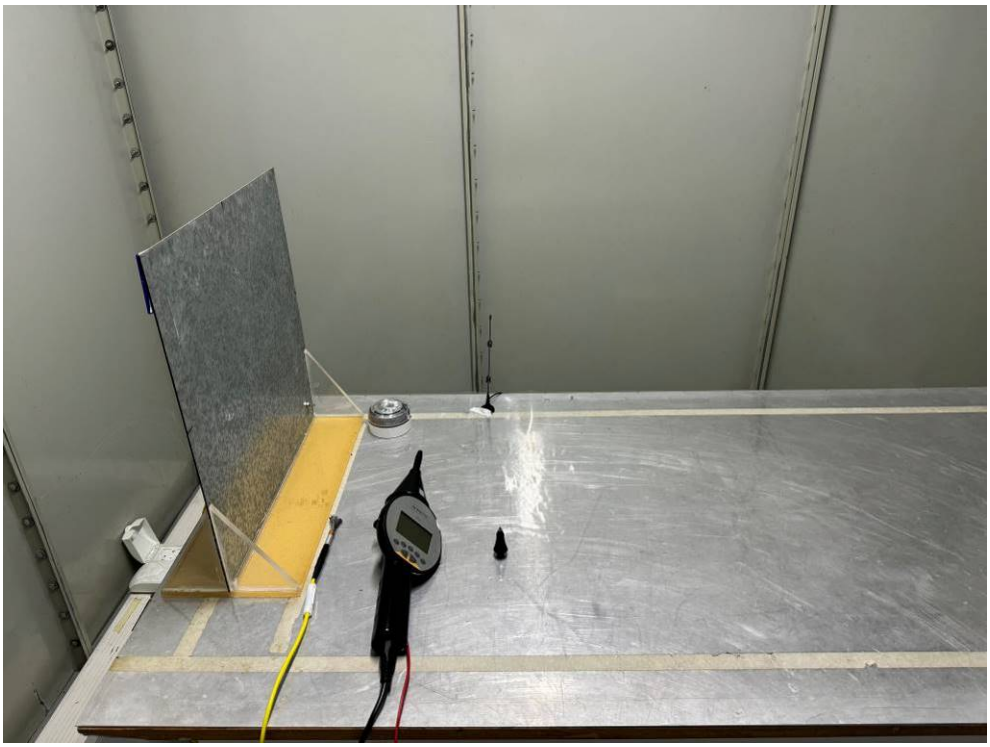
4 Photographs of the Test Set-up

Photograph 1: Radiated emission

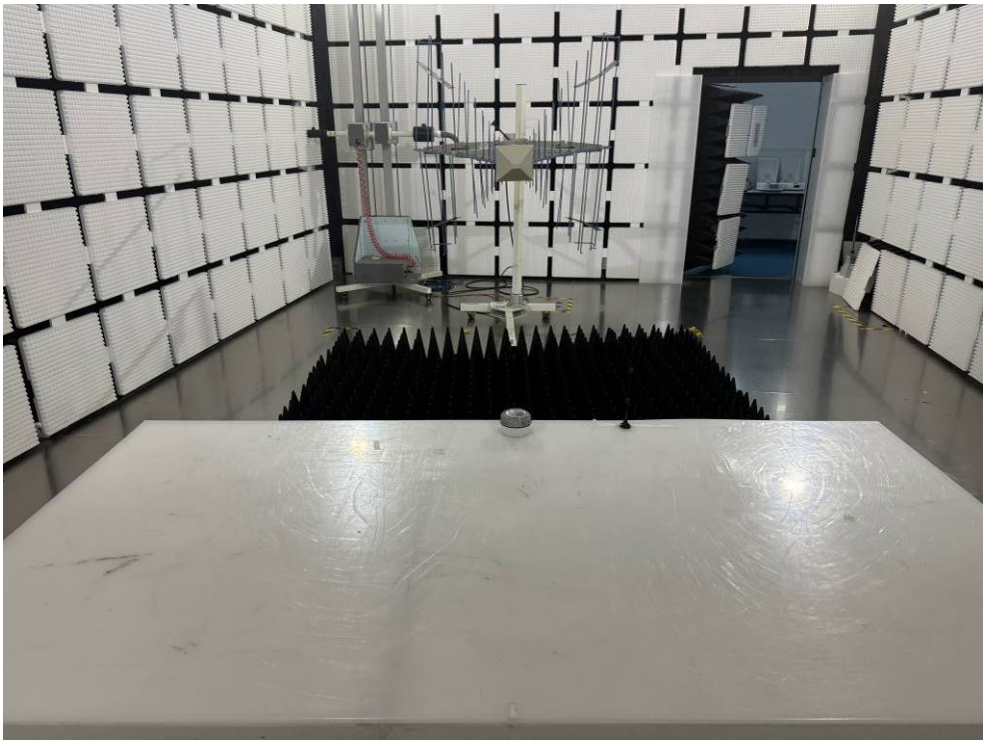


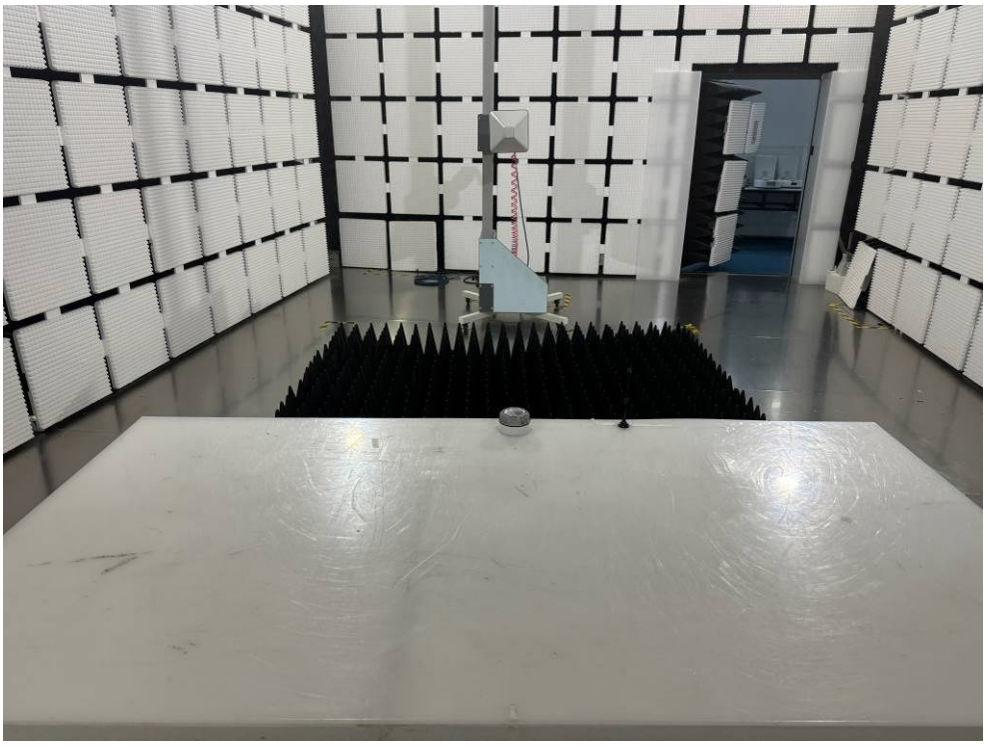


Photograph 2: ESD



Photograph 3: RF Immunity test





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Prüfbericht-Nr.: *Test report no.*:	**Part III of CN2394C9 001**	**Auftrags-Nr.:** *Order no.*:	178186679	**Seite 1 von 16** *Page 1 of 16*
Kunden-Referenz-Nr.: *Client reference no.*:	N/A	**Auftragsdatum:** *Order date:*	2023-08-25	
Auftraggeber: *Client*:	Huizhou Duran Creative Technology Co., Ltd. Room 708, Innovation Building, No.1, Keji Road, Science and Technology Innovation Park, West Area of Daya Bay, Huizhou, Guangdong P.R.China			
Prüfgegenstand: *Test item*:	Portable Light			
Bezeichnung / Typ-Nr.: *Identification / Type no.*:	INSTAFLASH V16 G-001			
Auftrags-Inhalt: *Order content*:	RED report			
Prüfgrundlage: *Test specification*:	[Radio] EN 301 908-1 V15.2.1:2023 EN 301 908-13 V13.2.1:2022 [Health Protection] EN IEC 62311:2020			

Wareneingangsdatum: *Date of sample receipt*:	2023-08-25
Prüfmuster-Nr.: *Test sample no:*	178186679 001
Prüfzeitraum: *Testing period*:	2023-10-12 - 2023-11-27
Ort der Prüfung: *Place of testing*:	Refer to section 1.1
Prüflaboratorium: *Testing laboratory:*	TUV Rheinland / CCIC (Qingdao) Co., Ltd.
Prüfergebnis*: *Test result**:	Pass

Keine Fotodokumentation erforderlich

No photo documentation required

geprüft von: *tested by:*	x Geoff Guo Signed by: Geoff Guo	**genehmigt von:** *authorized by*:	x Hunter Yu Signed by: Hunter Yu
Datum: *Date:* 2023-12-07		**Ausstellungsdatum:** *Issue date:* 2023-12-07	
Stellung / Position:	Project manager	**Stellung** / Position:	Authorizer

Sonstiges / *Other*:	This report only evaluates NB-Iot parts accroding to EN 301 908-1, EN 301 908-13 and EN IEC 62311.

Zustand des Prüfgegenstandes bei Anlieferung: *Condition of the test item at delivery*:	Prüfmuster vollständig und unbeschädigt *Test item complete and undamaged*

* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
** Legend:*	*P(ass) = passed a.m. test specification(s)*	*F(ail) = failed a.m. test specification(s)*	*N/A = not applicable*	*N/T = not tested*

Dieser Prüfbericht bezieht sich nur auf das o.g. Prüfmuster und darf ohne Genehmigung der Prüfstelle nicht auszugsweise vervielfältigt werden. Dieser Bericht berechtigt nicht zur Verwendung eines Prüfzeichens.
This test report only relates to the above mentioned test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any test mark.
V05

Prüfbericht-Nr.: Part III of CN2394C9 001
Test report no.:

Anmerkungen
Remarks

1	Alle eingesetzten Prüfmittel waren zum angegebenen Prüfzeitraum gemäß eines festgelegten Kalibrierungsprogramms unseres Prüfhauses kalibriert. Sie entsprechen den in den Prüfprogrammen hinterlegten Anforderungen. Die Rückverfolgbarkeit der eingesetzten Prüfmittel ist durch die Einhaltung der Regelungen unseres Managementsystems gegeben. Detaillierte Informationen bezüglich Prüfkonditionen, Prüfequipment und Messunsicherheiten sind im Prüflabor vorhanden und können auf Wunsch bereitgestellt werden. *The equipment used during the specified testing period was calibrated according to our test laboratory calibration program. The equipment fulfils the requirements included in the relevant standards. The traceability of the test equipment used is ensured by compliance with the regulations of our management system.* *Detailed information regarding test conditions, equipment and measurement uncertainty is available in the test laboratory and could be provided on request.*
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3	Prüfklausel mit der Note * wurden an qualifizierte Unterauftragnehmer vergeben und sind unter der jeweiligen Prüfklausel des Berichts beschrieben. Abweichungen von Prüfspezifikation(en) oder Kundenanforderungen sind in der jeweiligen Prüfklausel im Bericht aufgeführt. *Test clauses with remark of * are subcontracted to qualified subcontractors and descripted under the respective test clause in the report. Deviations of testing specification(s) or customer requirements are listed in specific test clause in the report.*
4	Die Entscheidungsregel für Konformitätserklärungen basierend auf numerischen Messergebnisen in diesem Prüfbericht basiert auf der "Null-Grenzwert-Regel" und der "Einfachen Akzeptanz" gemäß ILAC G8:2019 und IEC Guide 115:2021, es sei denn, in der auf Seite 1 dieses Berichts genannten angewandten Norm ist etwas anderes festgelegt oder vom Kunden gewünscht. Dies bedeutet, dass die Messunsicherheit nicht berücksichtigt wird und daher auch nicht im Prüfbericht angegeben wird. Zu weiteren Informationen bezueglich des Risikos durch diese Entscheidungsregel siehe ILAC G8:2019. *The decision rule for statements of conformity, based on numerical measurement results, in this test report is based on the “Zero Guard Band Rule” and “Simple Acceptance” in accordance with ILAC G8:2019 and IEC Guide 115:2021, unless otherwise specified in the applied standard mentioned on Page 1 of this report or requested by the customer. This means that measurement uncertainty is not taken in account and hence also not declared in the test report. For additional information to the resulting risk based of this decision rule please refer to ILAC G8:2019.*

TEST SUMMARY

5.1 RADIATED SPURIOUS EMISSION

RESULT: PASS

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1. General Remarks

1.1 Complementary Materials

Null.

1.2 Test Specifications

The following standards were applied.

Table 1: Applied Standard and Test Levels

Radio
EN 301 908-1 V15.2.1; EN 301 908-13 V13.2.1
Health
EN IEC 62311:2020

2. Test Sites

2.1 Test Facilities

CTC Laboratories, Inc.
2/F., Building 1 and 1-2/F., Building 2, Jiaquan Building,
Guanlan High-Tech Park, Longhua District, Shenzhen, Guangdong, China

The used test equipment is in accordance with CISPR 16-1 series standards for measurement of radio interference.

2.2 List of Test and Measurement Instruments

Table 2: List of Test and Measurement Equipment

nTscend JS0806-2 Test system						
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Calibrated Date	Calibrated until
1	Spectrum Analyzer	Rohde & Schwarz	FSU26	100105	Dec. 26, 2022	Dec. 25, 2023
2	Spectrum Analyzer	Rohde & Schwarz	FUV40-N	101331	Mar. 16, 2022	Mar. 15, 2023
3	MXG Vector Signal Generator	Agilent	N5182A	MY47420864	Dec. 26, 2022	Dec. 25, 2023
4	Signal Generator	Agilent	E8257D	MY46521908	Dec. 26, 2022	Dec. 25, 2023
5	Power Sensor	Agilent	U2021XA	MY5365004	Dec. 26, 2022	Dec. 25, 2023
6	Power Sensor	Agilent	U2021XA	MY5365006	Dec. 26, 2022	Dec. 25, 2023
7	Simultaneous Sampling DAQ	Agilent	U2531A	TW54493510	Dec. 26, 2022	Dec. 25, 2023
8	Wideband Radio Communication Tester	Rohde & Schwarz	CMW500	116410	Dec. 26, 2022	Dec. 25, 2023
9	Climate Chamber	ESPEC	MT3065	/	Dec. 26, 2022	Dec. 25, 2023
10	300328 V2.2.2 test system	TONSCEND	v2.6	/	/	/

Transmitter spurious emissions & Receiver spurious emissions						
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Calibrated Date	Calibrated until
1	EMI Test Receiver	Rohde & Schwarz	ESCI	100658	Dec. 26, 2022	Dec. 25, 2023
2	High pass filter	micro-tranics	HPM50111	142	Dec. 26, 2022	Dec. 25, 2023
3	Log-Bicon Antenna	Schwarzbeck	CBL6141A	4180	Dec. 26, 2022	Dec. 25, 2023
4	Ultra-Broadband Antenna	ShwarzBeck	BBHA9170	25841	Dec. 26, 2022	Dec. 25, 2023
5	Loop Antenna	LAPLAC	RF300	9138	Dec. 26, 2022	Dec. 25, 2023

6	Spectrum Analyzer	Rohde & Schwarz	FSU26	100105	Dec. 26, 2022	Dec. 25, 2023
7	Horn Antenna	Schwarzbeck	BBHA 9120D	647	Dec. 26, 2022	Dec. 25, 2023
8	Pre-Amplifier	HP	8447D	1937A03050	Dec. 26, 2022	Dec. 25, 2023
9	Pre-Amplifier	EMCI	EMC051835	980075	Dec. 26, 2022	Dec. 25, 2023
10	Antenna Mast	UC	UC3000	N/A	N/A	N/A
11	Turn Table	UC	UC3000	N/A	N/A	N/A
12	Cable Below 1GHz	Schwarzbeck	AK9515E	33155	Dec. 26, 2022	Dec. 25, 2023
13	Cable Above 1GHz	Hubersuhner	SUCOFLEX102	DA1580	Dec. 26, 2022	Dec. 25, 2023

2.3 Measurement Uncertainty

Table 3: Emission Measurement Uncertainty

Parameter	Uncertainty
Occupied Channel Bandwidth	± 5 %
RF output power, conducted	± 1.5 dB
Power Spectral Density, conducted	± 1.5 dB
Unwanted Emissions, conducted	± 2.8 dB
Radiated emissions, below 1GHz	± 4.5 dB
Radiated emissions, above 1GHz	± 5.7 dB
Temperature	± 5 ℃
Supply voltages	± 3 %
Time	± 5 %

3. General Product Information

3.1 Product Function and Intended Use

The EUT (equipment under test) is a Portable Light which supports NB-lot and GNSS functions.
For details refer to the User Manual, Circuit Diagram.

The aim of this report is to evalute the RF characteristics of the NB-lot of the EUT.

3.2 System Details

Table 4: Technical Specification of EUT

General Description of EUT	
Product Name:	Portable Light
Model No.:	G-001
Rated Voltage:	DC 9V from battery
Test Voltage:	DC 9V for Spurious Emission test
Extreme operating temperature:	-25~75°C
Technical Specification of NB-lot	
Frequency Range:	Band 3: Tx: 1710-1785MHz, Rx: 1805-1880MHz Band 5: Tx: 824-849MHz, Rx: 869-894MHz Band 8: Tx: 880-915MHz, Rx: 925-960MHz Band 20: Tx: 832-862MHz, Rx: 791-821MHz Band 28: Tx: 703-748MHz, Rx: 758-803MHz
Modulation Type:	BPSK, QPSK
Antenna Type:	Dipole Antenna
Antenna Gain:	Band 3: 2.00dBi Band 5: 2.53dBi Band 8: 2.98dBi Band 20: 2.64dBi Band 28: 4.04dBi

Note:
The NB-lot module which model is BC65 manufactured and DOC declared by Quectel Wireless Solutions Co., Ltd. was integrated into the EUT as an assessed module. The RF reports are R2304A0377-R1V1. Assessed radio modules installed in equipment in conformance with the manufacturer's installation instructions except for an external antenna.
For conducted test data for EN 301 908-1 V15.2.1 and EN 301 908-13 V13.2.1, please refer to report R2304A0377-R1V1 issued by Quectel Wireless Solutions Co., Ltd.

Therefore, only Radiated Spurious Emission tests were performed.

3.3 Independent Operation Modes

Table 5: Independent Operation Modes

Note: The EUT was set into continuous transmitting or receiving mode in Tx or Rx mode.

3.4 Noise Suppressing Parts

Refer to schematics.

4. Test Set-up and Operation Modes

4.1 Test Methodology

Radio: The equipment under test (EUT) was configured at its highest power output in order to measure its highest possible radiation and conducted level. The test modes were adapted accordingly in reference to the instructions for use.

4.2 Physical Configuration for Testing

The EUT was designed to get into related working mode with the control of computer.
Notes:

For antenna conducted measurements, the antenna was replaced by a 50Ω antenna connector.

For antenna radiated measurements, the associated cables and computer were removed.

For more details, refer to section: Photographs of the Test Set-Up.

4.3 Test Operation and Test Software

Null.

4.4 Special Accessories and Auxiliary Equipment

Null.

4.5 Countermeasures to achieve EMC Compliance

Null.

5. Test Results RADIO

5.1 Radiated Spurious Emission

RESULT: **PASS**

Date of testing	:	2023.11.07-2023.11.17
Ambient temperature	:	25°C
Relative humidity	:	55%
Atmospheric pressure	:	101kPa
Test requirement	:	EN 301 908-1 V15.2.1
	:	EN 301 908-13 V13.2.1
Normal test voltage	:	DC 9V from battery
Test modes applied	:	Band 3, 5, 8, 20, 28

For the measurement records, refer to the appendix A.

6. Human Exposure to EM Fields

RESULT: **PASS**

Evaluation procedure : EN IEC 62311:2020

The maximum measured transmitter power is the following:

Mode	Max. Conducted Average Output Power P_{out} [dBm]	Maximum Antenna Gain [dBi]	P_{out} EIRP [dBm]	P_{out} EIRP [mW]	Distance [m]	Exposure [V/m]	Limit [V/m]
LTE Band 3	24.32	2.00	26.32	270.4	0.2	17.93	56.87
LTE Band 5	23.72	2.53	26.25	235.5	0.2	17.78	61
LTE Band 8	23.90	2.98	26.88	245.5	0.2	19.12	40.81
LTE Band 20	23.14	2.64	25.78	206.1	0.2	16.85	36.5
LTE Band 28	23.76	4.04	27.80	237.7	0.2	21.26	61

Note: This device can operate simultaneously one built-in NB-lot module.

CALCULATION FOR SIMULTANEOUS TRANSMISSION
One built-in LTE module and one built-in NB-lot module can transmit simultaneously, the formula of calculated the exposure is:
$(CEF1/LEF1)^2 + (CEF2/LEF2)^2 + \ldots\ldots etc. < 1$
CEF= Calculation E-Field Strength
LEF= Limit of E-Field Strength
$= (17.93/56.87)^2 + (17.78/61)^2 + (19.12/40.81)^2 + (16.85/36.5)^2 + (21.26/61)^2 = 0.738 < 1$, which is less than the “1” limit.

7. Photographs of the Sample and Test Setup

Photograph 1: Set-up for radiated spurious emission, 30MHz – 1GHz



Photograph 2: Set-up for radiated spurious emission, above 1GHz



8. List of Tables

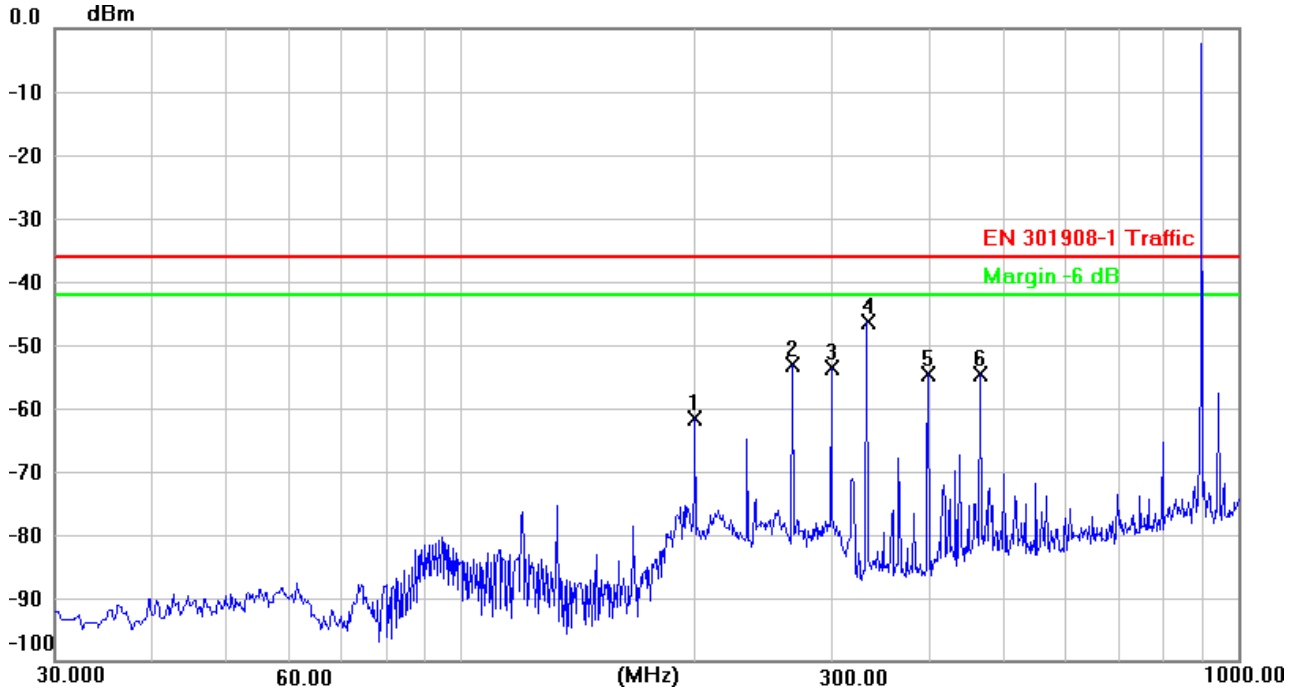
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Radiated Emission Measurement

File :20231106 Data :#21 Date: 2023-11-07 Time: 8:46:55



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B8
Note:

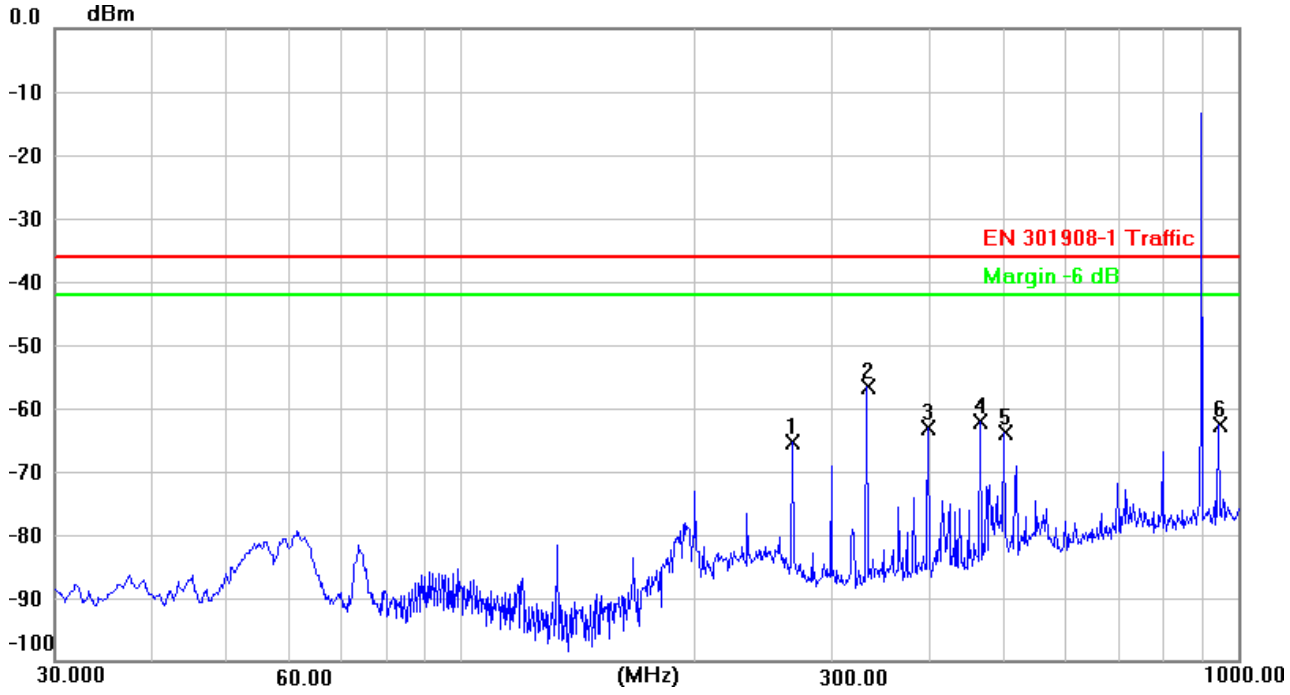
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	200.0733	-45.56	-15.97	-61.53	-36.00	-25.53	peak	149	1	P	
2	266.6800	-39.05	-14.16	-53.21	-36.00	-17.21	peak	149	1	P	
3	299.9833	-40.12	-13.47	-53.59	-36.00	-17.59	peak	149	1	P	
4 *	333.2867	-33.94	-12.39	-46.33	-36.00	-10.33	peak	149	1	P	
5	399.8933	-43.54	-11.13	-54.67	-36.00	-18.67	peak	149	1	P	
6	466.8233	-44.59	-9.97	-54.56	-36.00	-18.56	peak	149	1	P	

Radiated Emission Measurement

File :20231106 Data :#22 Date: 2023-11-07 Time: 8:50:06



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B8
Note:

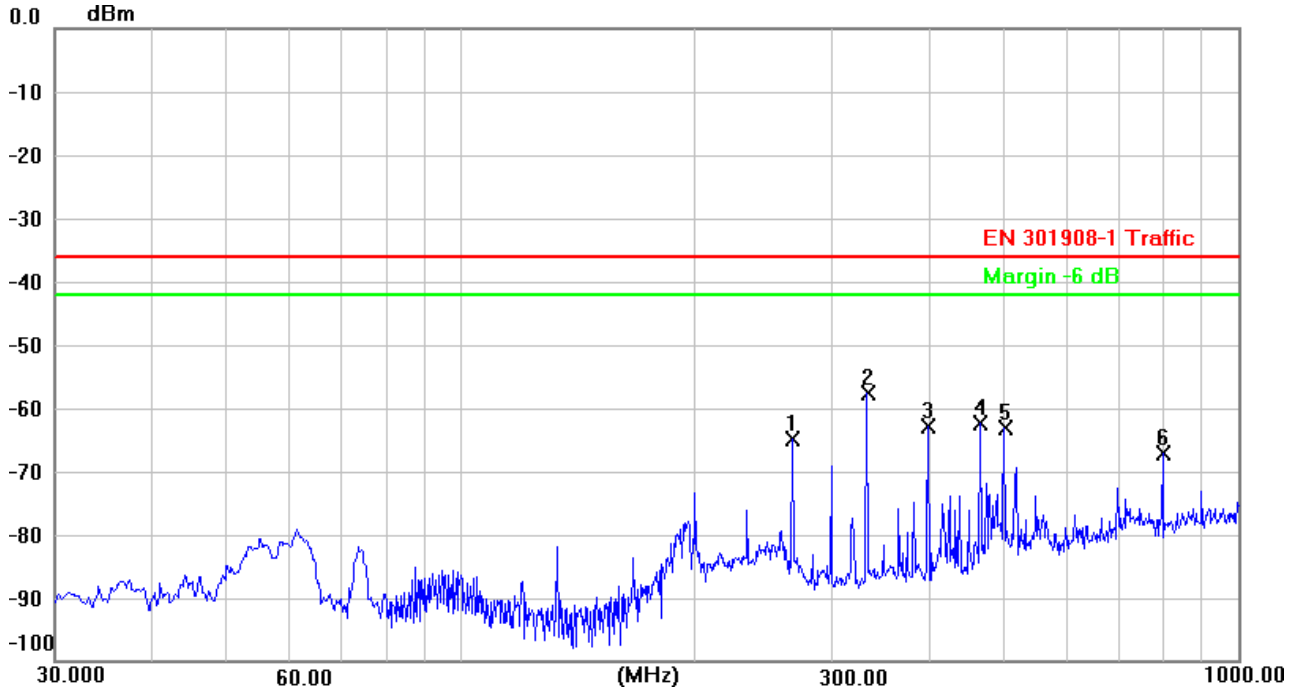
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	266.6800	-51.10	-14.16	-65.26	-36.00	-29.26	peak	149	360	P	
2 *	333.2867	-44.11	-12.39	-56.50	-36.00	-20.50	peak	149	360	P	
3	399.8933	-51.94	-11.13	-63.07	-36.00	-27.07	peak	149	360	P	
4	466.8233	-52.21	-9.97	-62.18	-36.00	-26.18	peak	149	360	P	
5	500.1267	-54.55	-9.29	-63.84	-36.00	-27.84	peak	149	360	P	
6	942.4467	-59.91	-2.63	-62.54	-36.00	-26.54	peak	149	360	P	

Radiated Emission Measurement

File :20231106 Data :#23 Date: 2023-11-07 Time: 8:51:58



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B3
Note:

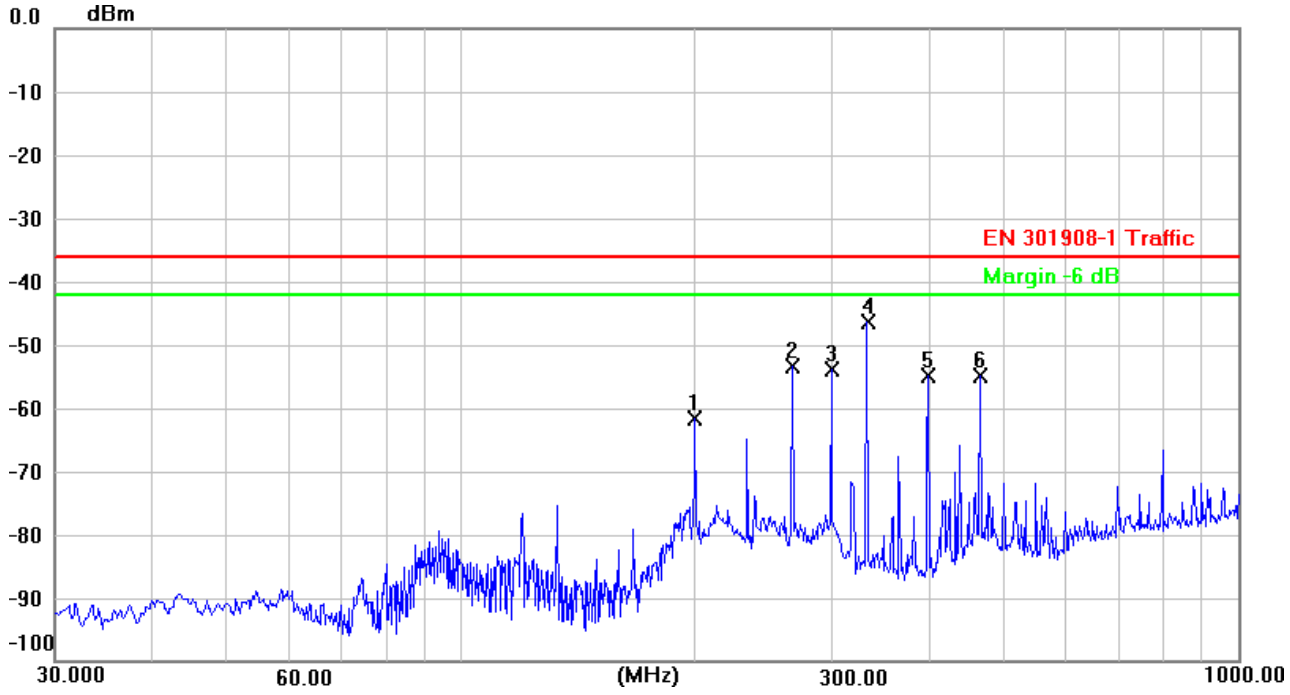
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	266.6800	-50.68	-14.16	-64.84	-36.00	-28.84	peak	149	1	P	
2 *	333.2867	-45.31	-12.39	-57.70	-36.00	-21.70	peak	149	1	P	
3	399.8933	-51.86	-11.13	-62.99	-36.00	-26.99	peak	149	1	P	
4	466.8233	-52.41	-9.97	-62.38	-36.00	-26.38	peak	149	1	P	
5	500.1267	-53.88	-9.29	-63.17	-36.00	-27.17	peak	149	1	P	
6	799.8567	-62.73	-4.30	-67.03	-36.00	-31.03	peak	149	1	P	

Radiated Emission Measurement

File :20231106 Data :#24 Date: 2023-11-07 Time: 8:53:09



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B3
Note:

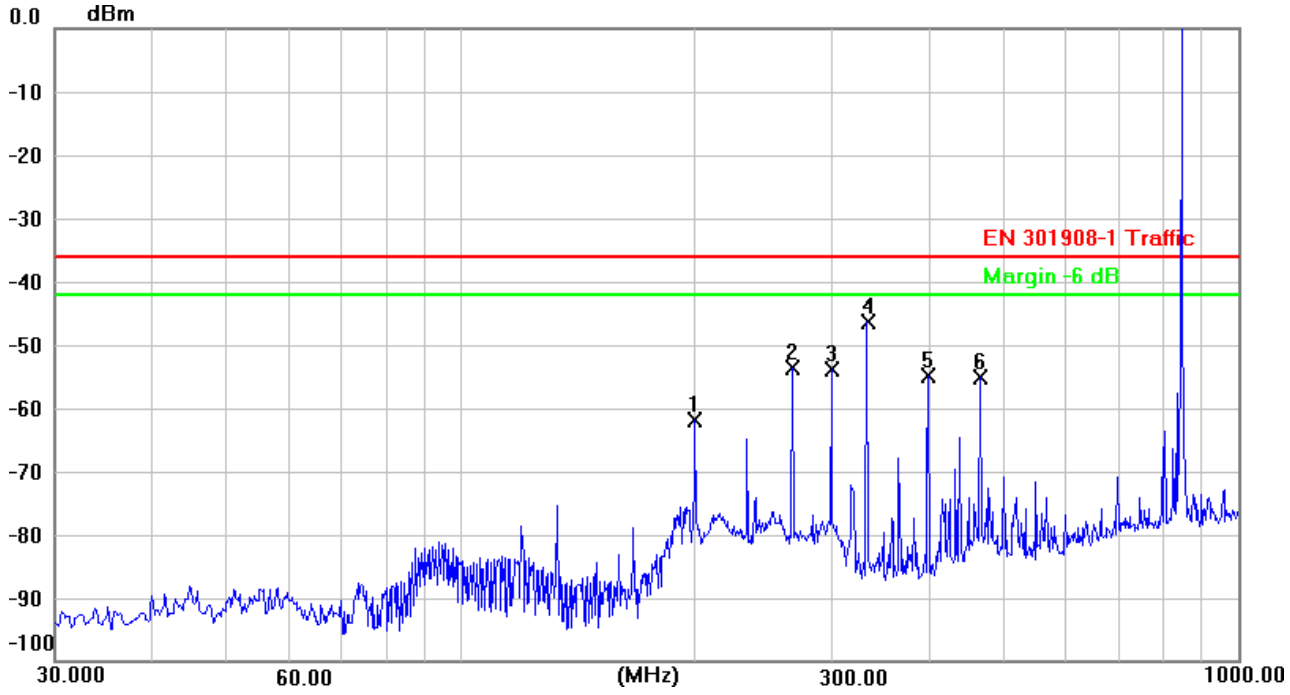
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	200.0733	-45.75	-15.97	-61.72	-36.00	-25.72	peak	149	360	P	
2	266.6800	-39.15	-14.16	-53.31	-36.00	-17.31	peak	149	360	P	
3	299.9833	-40.36	-13.47	-53.83	-36.00	-17.83	peak	149	360	P	
4 *	333.2867	-34.05	-12.39	-46.44	-36.00	-10.44	peak	149	360	P	
5	399.8933	-43.81	-11.13	-54.94	-36.00	-18.94	peak	149	360	P	
6	466.8233	-45.01	-9.97	-54.98	-36.00	-18.98	peak	149	360	P	

Radiated Emission Measurement

File :20231106 Data :#25 Date: 2023-11-07 Time: 8:54:51



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B20
Note:

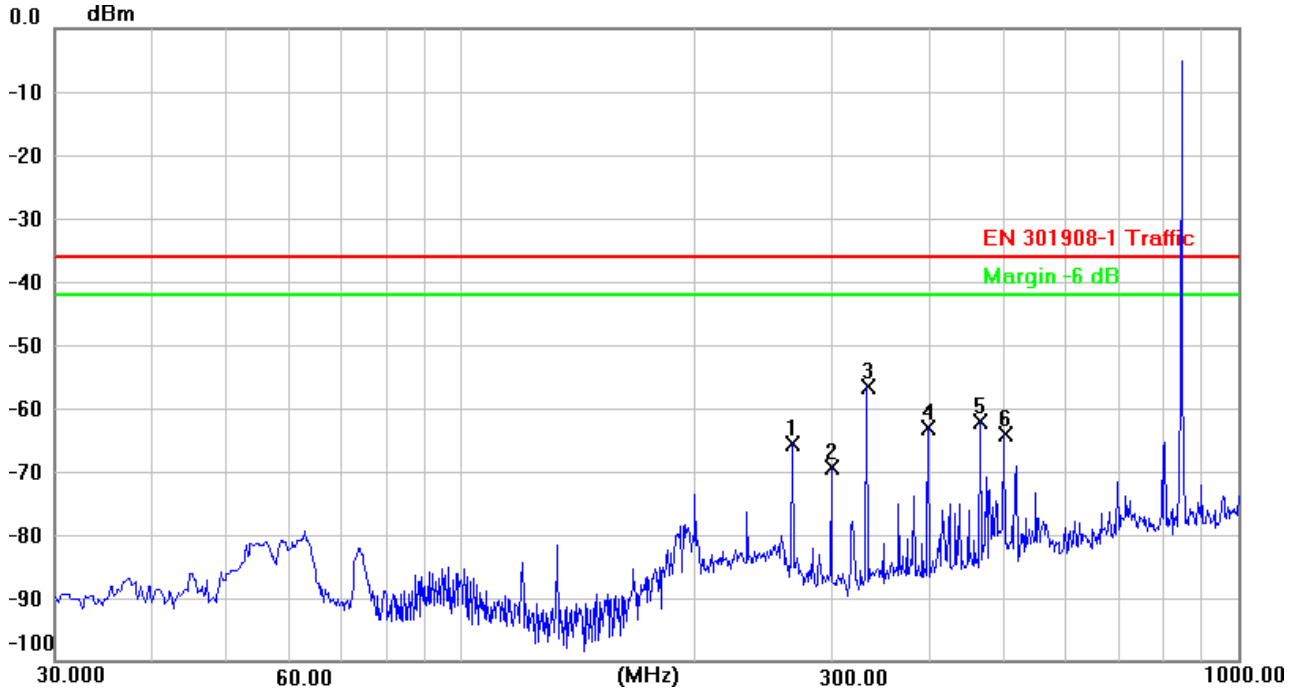
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	200.0733	-45.80	-15.97	-61.77	-36.00	-25.77	peak	149	1	P	
2	266.6800	-39.44	-14.16	-53.60	-36.00	-17.60	peak	149	1	P	
3	299.9833	-40.36	-13.47	-53.83	-36.00	-17.83	peak	149	1	P	
4 *	333.2867	-34.06	-12.39	-46.45	-36.00	-10.45	peak	149	1	P	
5	399.8933	-43.69	-11.13	-54.82	-36.00	-18.82	peak	149	1	P	
6	466.8233	-45.19	-9.97	-55.16	-36.00	-19.16	peak	149	1	P	

Radiated Emission Measurement

File :20231106 Data :#26 Date: 2023-11-07 Time: 8:55:56



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B20
Note:

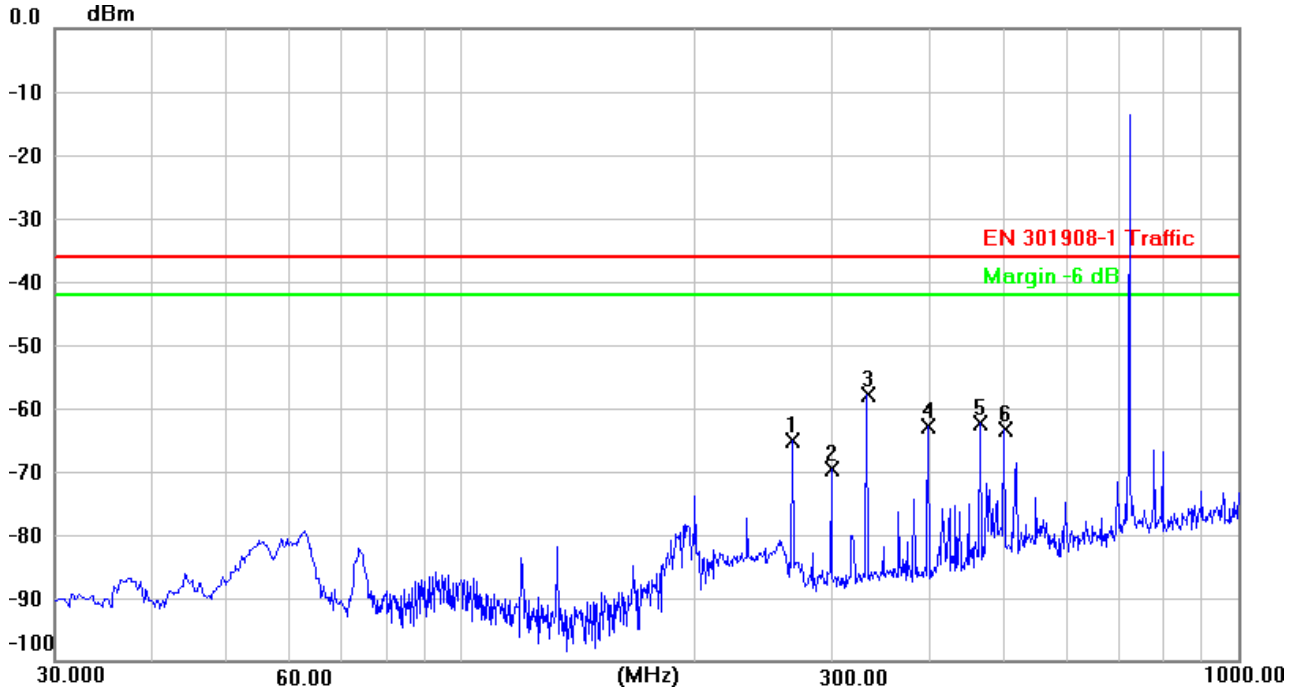
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	266.6800	-51.47	-14.16	-65.63	-36.00	-29.63	peak	149	360	P	
2	299.9833	-55.79	-13.47	-69.26	-36.00	-33.26	peak	149	360	P	
3 *	333.2867	-44.27	-12.39	-56.66	-36.00	-20.66	peak	149	360	P	
4	399.8933	-52.07	-11.13	-63.20	-36.00	-27.20	peak	149	360	P	
5	466.8233	-52.14	-9.97	-62.11	-36.00	-26.11	peak	149	360	P	
6	500.1267	-54.81	-9.29	-64.10	-36.00	-28.10	peak	149	360	P	

Radiated Emission Measurement

File :20231106 Data :#27 Date: 2023-11-07 Time: 8:57:38



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B28
Note:

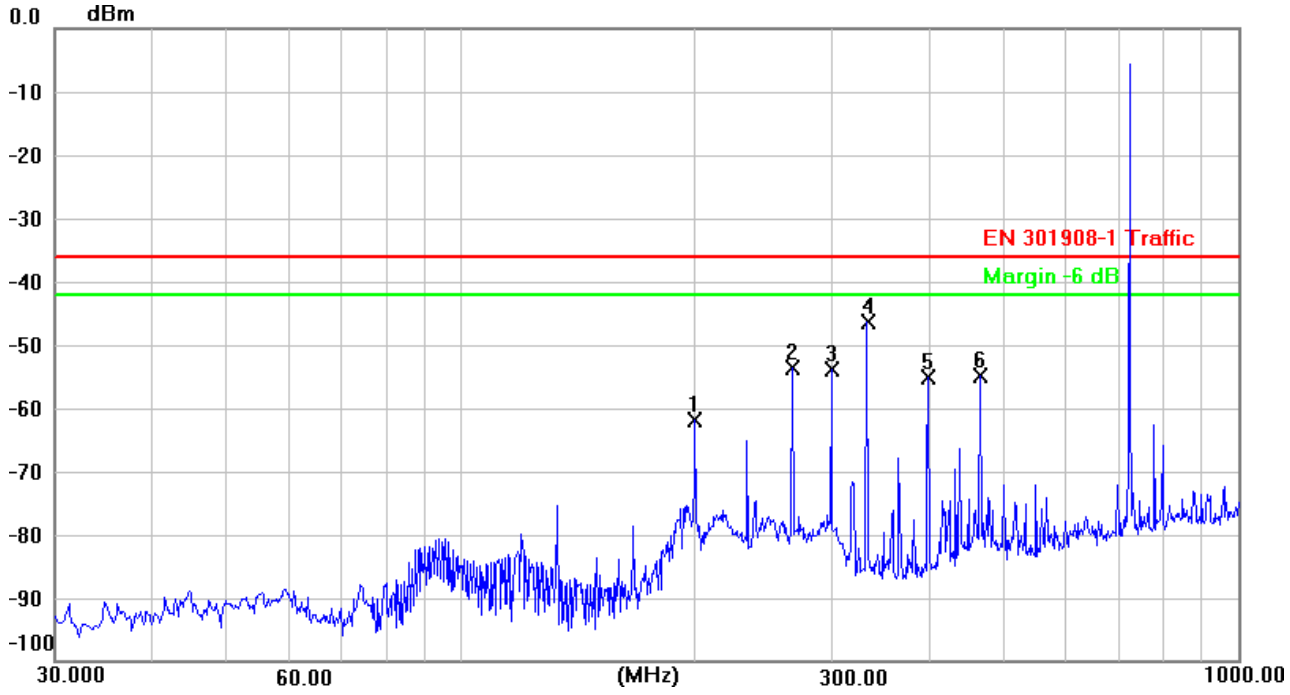
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	266.6800	-50.93	-14.16	-65.09	-36.00	-29.09	peak	149	1	P	
2	299.9833	-56.10	-13.47	-69.57	-36.00	-33.57	peak	149	1	P	
3 *	333.2867	-45.37	-12.39	-57.76	-36.00	-21.76	peak	149	1	P	
4	399.8933	-51.83	-11.13	-62.96	-36.00	-26.96	peak	149	1	P	
5	466.8233	-52.47	-9.97	-62.44	-36.00	-26.44	peak	149	1	P	
6	500.1267	-54.02	-9.29	-63.31	-36.00	-27.31	peak	149	1	P	

Radiated Emission Measurement

File :20231106 Data :#28 Date: 2023-11-07 Time: 8:58:39



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B28
Note:

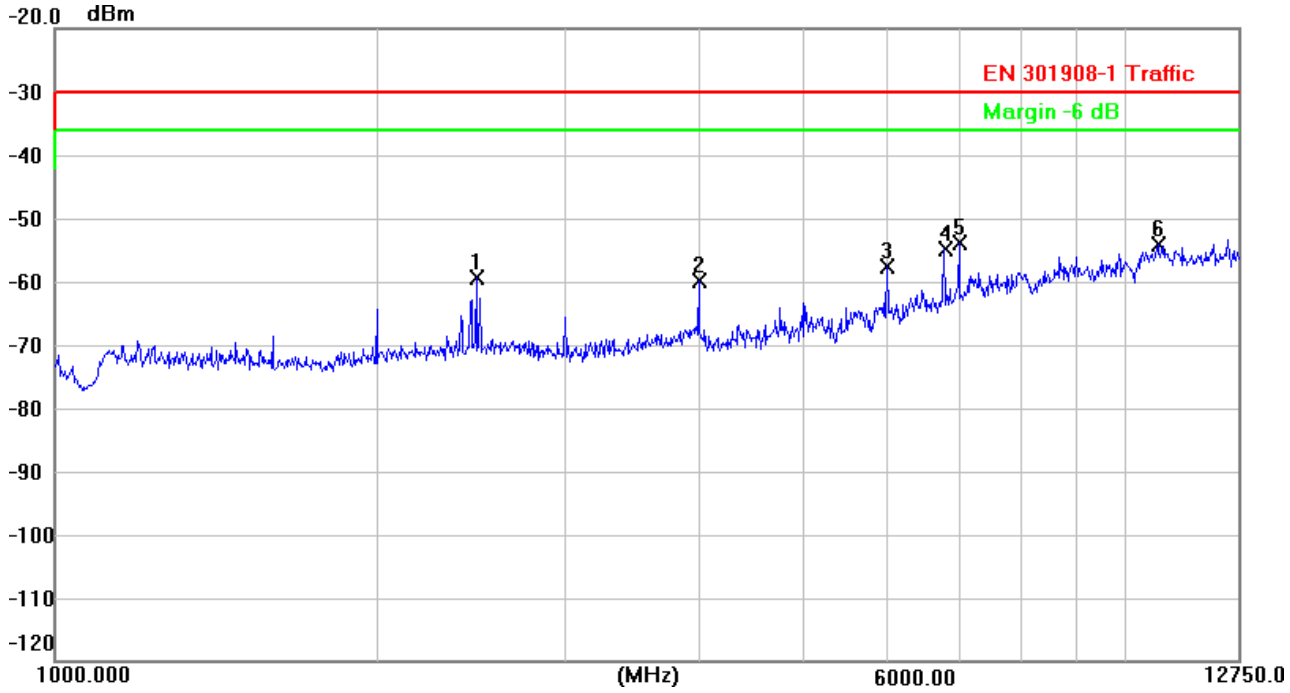
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	200.0733	-45.94	-15.97	-61.91	-36.00	-25.91	peak	149	359	P	
2	266.6800	-39.40	-14.16	-53.56	-36.00	-17.56	peak	149	359	P	
3	299.9833	-40.41	-13.47	-53.88	-36.00	-17.88	peak	149	359	P	
4 *	333.2867	-34.07	-12.39	-46.46	-36.00	-10.46	peak	149	359	P	
5	399.8933	-43.95	-11.13	-55.08	-36.00	-19.08	peak	149	359	P	
6	466.8233	-45.02	-9.97	-54.99	-36.00	-18.99	peak	149	359	P	

Radiated Emission Measurement

File :20231106 Data :#29 Date: 2023-11-07 Time: 9:03:45



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B28
Note:

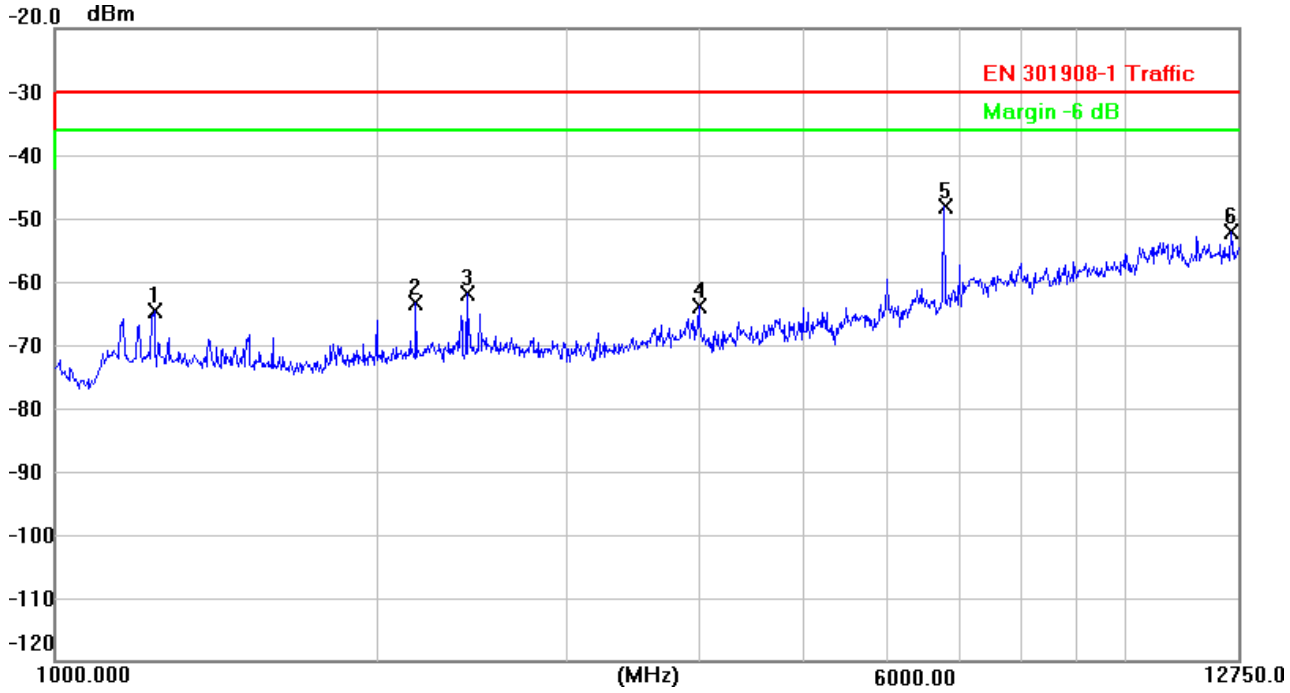
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2480.500	-56.10	-3.19	-59.29	-30.00	-29.29	peak	149	0	P	
2	4000.167	-60.38	0.54	-59.84	-30.00	-29.84	peak	149	0	P	
3	6001.583	-63.13	5.45	-57.68	-30.00	-27.68	peak	149	0	P	
4	6773.167	-62.74	7.97	-54.77	-30.00	-24.77	peak	149	0	P	
5 *	7000.333	-63.07	9.14	-53.93	-30.00	-23.93	peak	149	0	P	
6	10744.667	-68.49	14.45	-54.04	-30.00	-24.04	peak	149	0	P	

Radiated Emission Measurement

File :20231106 Data :#30 Date: 2023-11-07 Time: 9:05:03



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B28
Note:

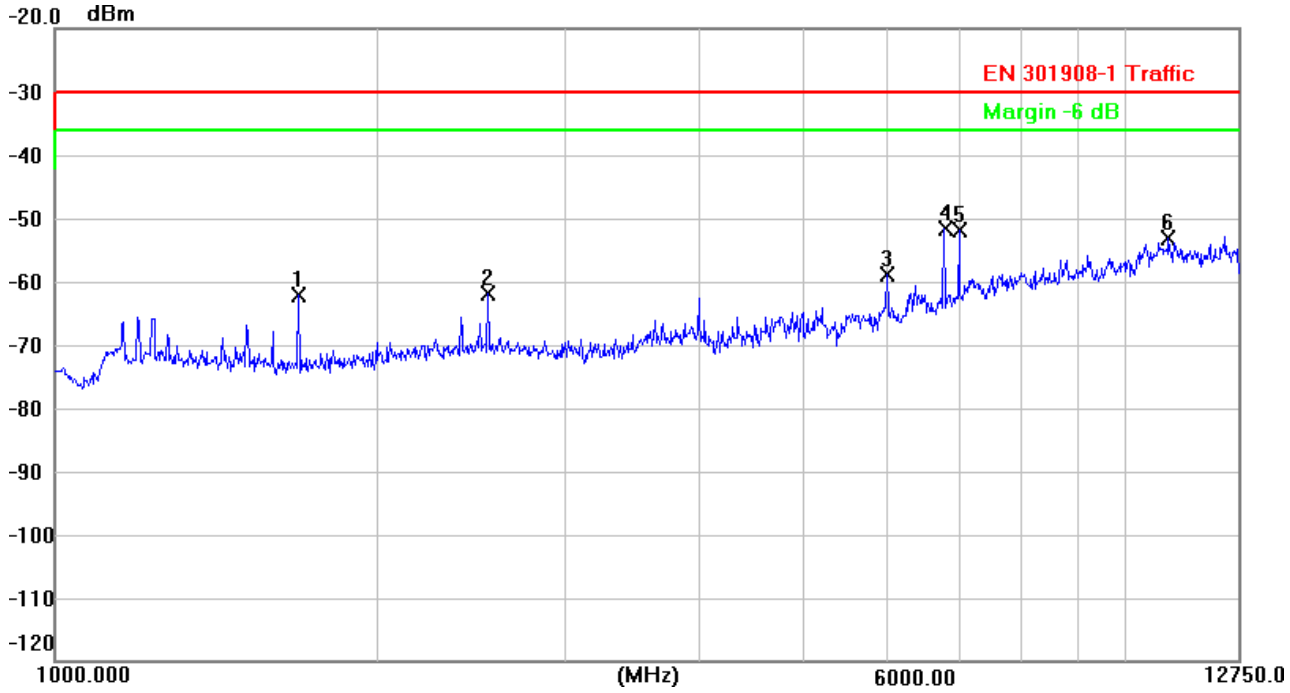
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	1238.917	-57.26	-7.44	-64.70	-30.00	-34.70	peak	149	359	P	
2	2175.000	-59.61	-3.83	-63.44	-30.00	-33.44	peak	149	359	P	
3	2425.667	-58.63	-3.30	-61.93	-30.00	-31.93	peak	149	359	P	
4	4000.167	-64.46	0.54	-63.92	-30.00	-33.92	peak	149	359	P	
5 *	6773.167	-56.17	7.97	-48.20	-30.00	-18.20	peak	149	359	P	
6	12569.833	-68.30	16.07	-52.23	-30.00	-22.23	peak	149	359	P	

Radiated Emission Measurement

File :20231106 Data :#31 Date: 2023-11-07 Time: 9:06:39



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B20
Note:

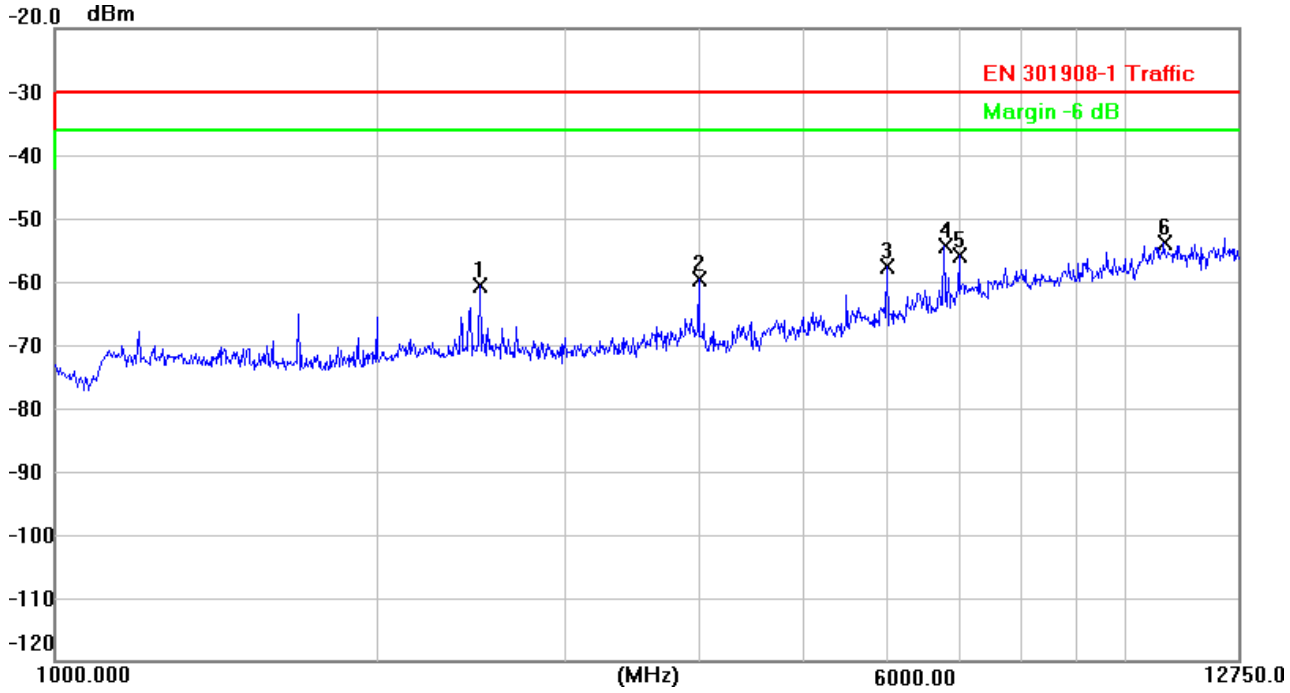
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	1693.250	-55.28	-6.83	-62.11	-30.00	-32.11	peak	149	0	P	
2	2539.250	-58.86	-3.06	-61.92	-30.00	-31.92	peak	149	0	P	
3	6001.583	-64.28	5.45	-58.83	-30.00	-28.83	peak	149	0	P	
4 *	6773.167	-59.65	7.97	-51.68	-30.00	-21.68	peak	149	0	P	
5	7000.333	-60.96	9.14	-51.82	-30.00	-21.82	peak	149	0	P	
6	10995.333	-68.00	14.77	-53.23	-30.00	-23.23	peak	149	0	P	

Radiated Emission Measurement

File :20231106 Data :#32 Date: 2023-11-07 Time: 9:07:39



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B20
Note:

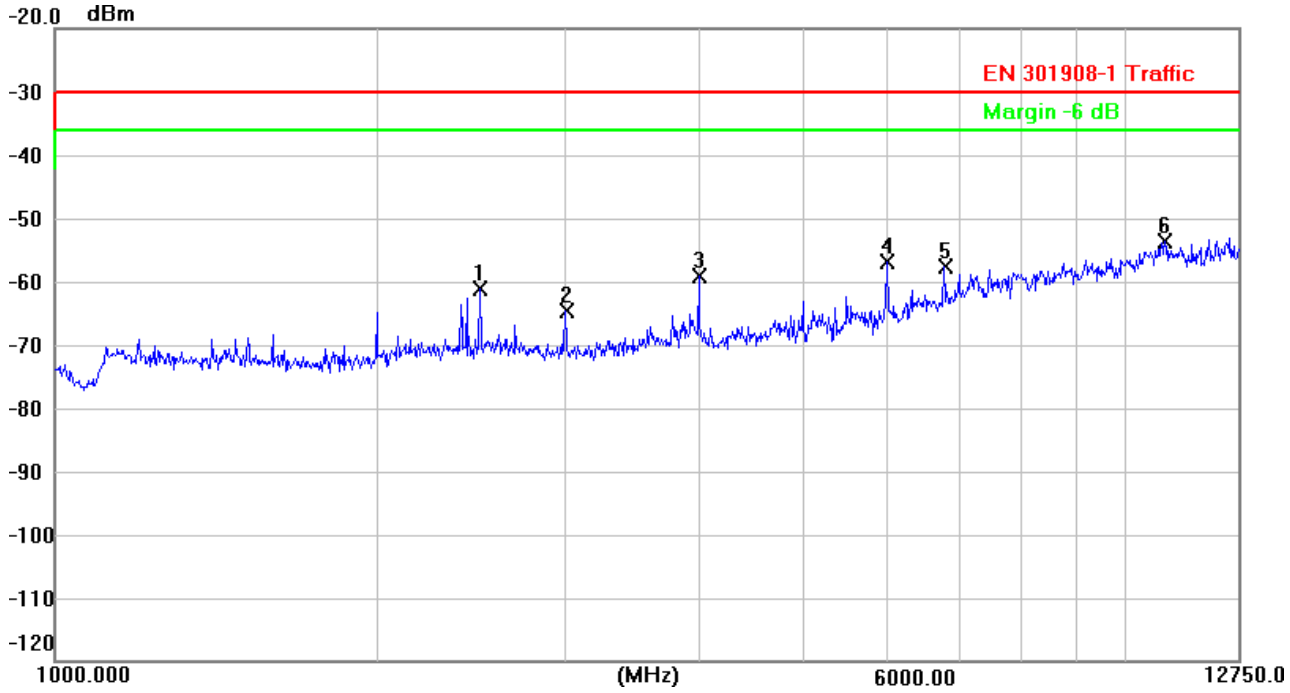
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2500.083	-57.37	-3.14	-60.51	-30.00	-30.51	peak	149	359	P	
2	4000.167	-60.16	0.54	-59.62	-30.00	-29.62	peak	149	359	P	
3	6001.583	-63.04	5.45	-57.59	-30.00	-27.59	peak	149	359	P	
4	6773.167	-62.38	7.97	-54.41	-30.00	-24.41	peak	149	359	P	
5	7000.333	-65.11	9.14	-55.97	-30.00	-25.97	peak	149	359	P	
6 *	10862.167	-68.61	14.62	-53.99	-30.00	-23.99	peak	149	359	P	

Radiated Emission Measurement

File :20231106 Data :#33 Date: 2023-11-07 Time: 9:09:21



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B8
Note:

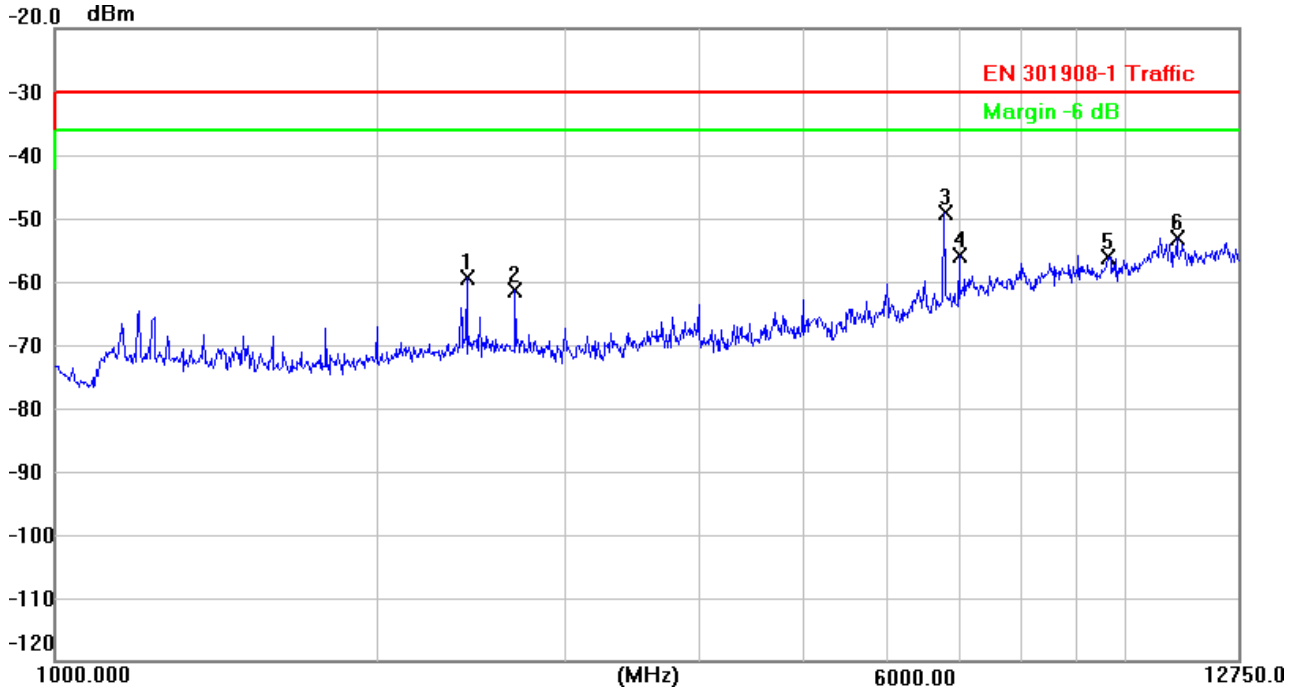
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2500.083	-57.94	-3.14	-61.08	-30.00	-31.08	peak	149	0	P	
2	3001.417	-62.54	-2.20	-64.74	-30.00	-34.74	peak	149	0	P	
3	4000.167	-59.61	0.54	-59.07	-30.00	-29.07	peak	149	0	P	
4	6001.583	-62.39	5.45	-56.94	-30.00	-26.94	peak	149	0	P	
5	6773.167	-65.62	7.97	-57.65	-30.00	-27.65	peak	149	0	P	
6 *	10901.333	-68.24	14.66	-53.58	-30.00	-23.58	peak	149	0	P	

Radiated Emission Measurement

File :20231106 Data :#34 Date: 2023-11-07 Time: 9:10:21



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B8
Note:

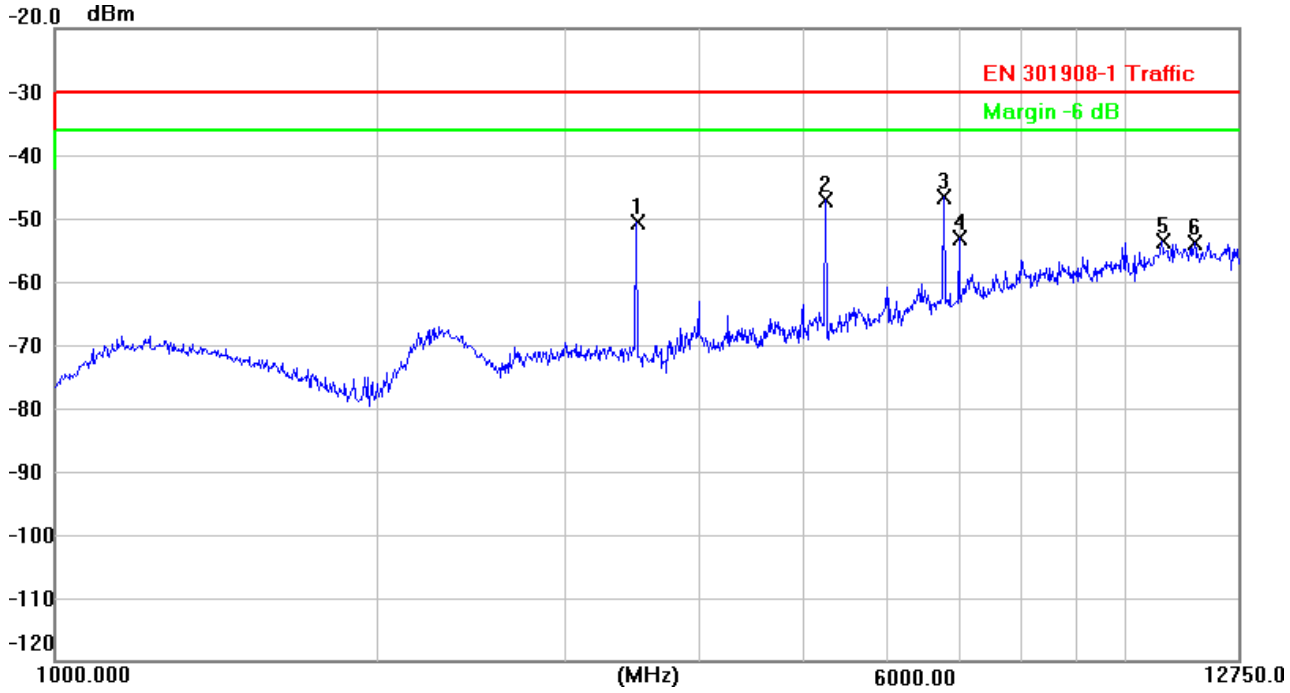
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2425.667	-56.13	-3.30	-59.43	-30.00	-29.43	peak	149	360	P	
2	2692.000	-58.62	-2.83	-61.45	-30.00	-31.45	peak	149	360	P	
3 *	6773.167	-57.01	7.97	-49.04	-30.00	-19.04	peak	149	360	P	
4	7000.333	-65.13	9.14	-55.99	-30.00	-25.99	peak	149	360	P	
5	9644.083	-68.71	12.70	-56.01	-30.00	-26.01	peak	149	360	P	
6	11183.333	-68.08	14.87	-53.21	-30.00	-23.21	peak	149	360	P	

Radiated Emission Measurement

File :20231106 Data :#35 Date: 2023-11-07 Time: 9:13:11



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B3
Note:

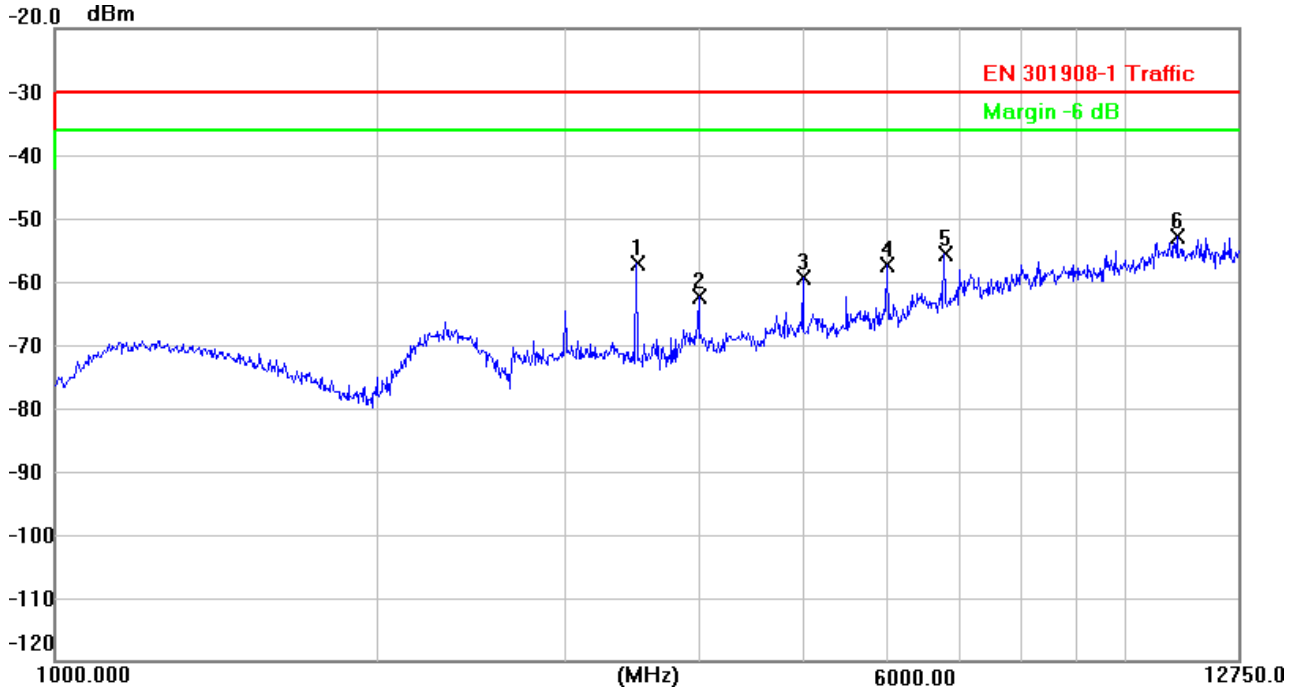
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	3494.917	-49.26	-1.29	-50.55	-30.00	-20.55	peak	149	0	P	
2	5253.500	-50.19	3.04	-47.15	-30.00	-17.15	peak	149	0	P	
3 *	6777.083	-54.69	7.99	-46.70	-30.00	-16.70	peak	149	0	P	
4	7000.333	-62.24	9.14	-53.10	-30.00	-23.10	peak	149	0	P	
5	10826.917	-68.10	14.59	-53.51	-30.00	-23.51	peak	149	0	P	
6	11622.000	-69.07	15.29	-53.78	-30.00	-23.78	peak	149	0	P	

Radiated Emission Measurement

File :20231106 Data :#36 Date: 2023-11-07 Time: 9:14:30



Site Radiation 3#
Limit: EN 301908-1 Traffic
EUT: Pending
Model: LED
Mode: B3
Note:

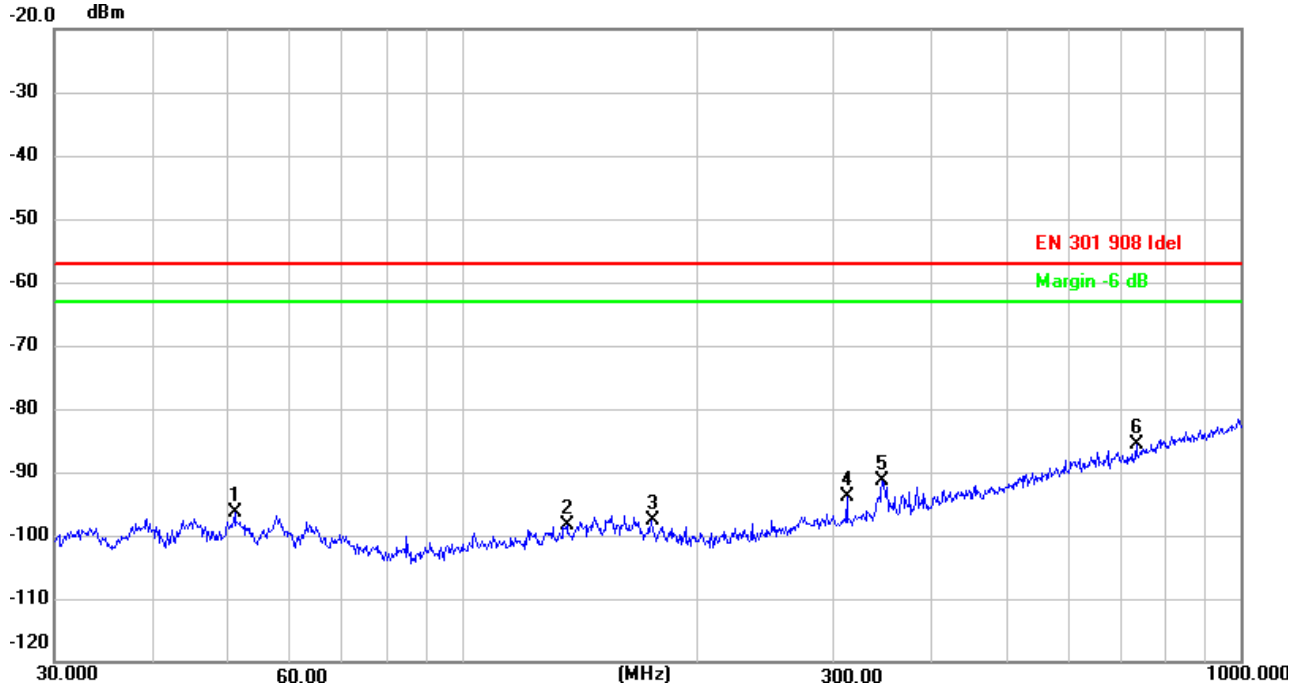
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	3494.917	-55.81	-1.29	-57.10	-30.00	-27.10	peak	149	359	P	
2	4000.167	-62.88	0.54	-62.34	-30.00	-32.34	peak	149	359	P	
3	4998.917	-61.84	2.36	-59.48	-30.00	-29.48	peak	149	359	P	
4	6001.583	-62.77	5.45	-57.32	-30.00	-27.32	peak	149	359	P	
5	6773.167	-63.61	7.97	-55.64	-30.00	-25.64	peak	149	359	P	
6 *	11195.083	-67.80	14.88	-52.92	-30.00	-22.92	peak	149	359	P	

Radiated Emission Measurement

File :20231103 Data :#9 Date: 2023/11/03 Time: 9:34:55



Site Chamber 2#
Limit: EN 301 908 Idel
EUT: LED
Model: Pending
Mode: NB IOT B3
Note:

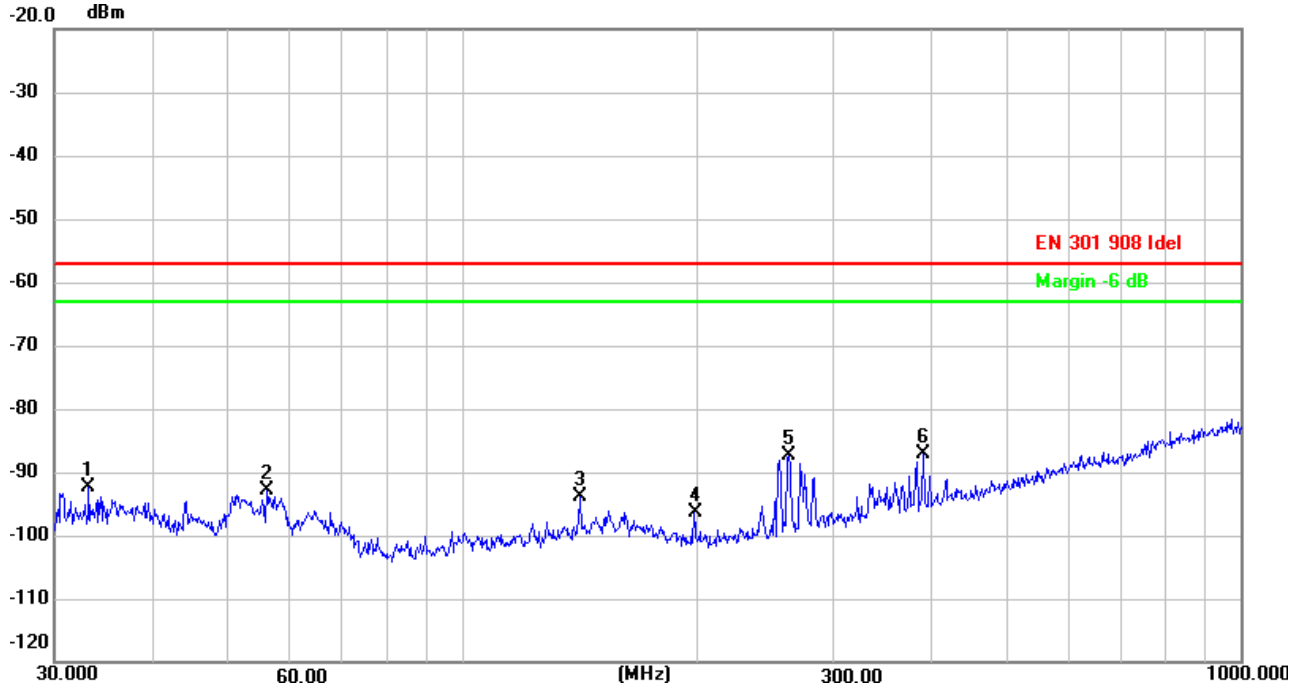
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 25(C)
Humidity: 56 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	51.1209	-78.42	-18.01	-96.43	-57.00	-39.43	peak	150	0	P	
2	136.4598	-79.08	-19.24	-98.32	-57.00	-41.32	peak	150	0	P	
3	175.0368	-78.29	-19.35	-97.64	-57.00	-40.64	peak	150	0	P	
4	312.1794	-76.61	-17.14	-93.75	-57.00	-36.75	peak	150	0	P	
5	346.8092	-75.18	-16.14	-91.32	-57.00	-34.32	peak	150	0	P	
6 *	734.4913	-77.66	-7.90	-85.56	-57.00	-28.56	peak	150	0	P	

Radiated Emission Measurement

File :20231103 Data :#10 Date: 2023/11/03 Time: 9:37:20



Site Chamber 2#
Limit: EN 301 908 Idel
EUT: LED
Model: Pending
Mode: NB IOT B3
Note:

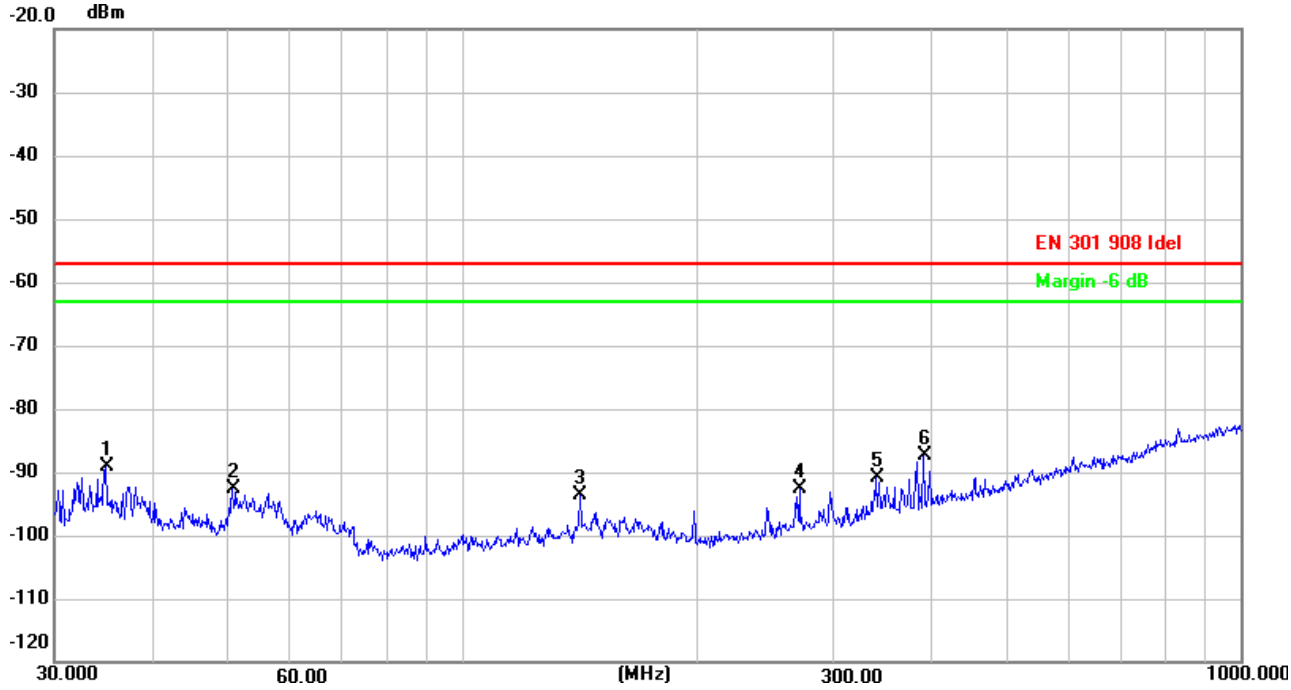
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 25(C)
Humidity: 56 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	33.2112	-73.60	-18.88	-92.48	-57.00	-35.48	peak	150	360	P	
2	56.1974	-74.35	-18.55	-92.90	-57.00	-35.90	peak	150	360	P	
3	141.8262	-75.33	-18.62	-93.95	-57.00	-36.95	peak	150	360	P	
4	198.5880	-75.57	-20.87	-96.44	-57.00	-39.44	peak	150	360	P	
5	262.8955	-68.56	-18.83	-87.39	-57.00	-30.39	peak	150	360	P	
6 *	390.7226	-72.15	-14.93	-87.08	-57.00	-30.08	peak	150	360	P	

Radiated Emission Measurement

File :20231103 Data :#11 Date: 2023/11/03 Time: 9:38:22



Site Chamber 2#
Limit: EN 301 908 Idel
EUT: LED
Model: Pending
Mode: NB IOT B8
Note:

Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 25(C)
Humidity: 56 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	34.8823	-70.42	-18.74	-89.16	-57.00	-32.16	peak	150	0	P	
2	50.7637	-74.72	-17.99	-92.71	-57.00	-35.71	peak	150	0	P	
3	141.8262	-74.90	-18.62	-93.52	-57.00	-36.52	peak	150	0	P	
4	271.3246	-74.15	-18.51	-92.66	-57.00	-35.66	peak	150	0	P	
5	340.7817	-74.58	-16.32	-90.90	-57.00	-33.90	peak	150	0	P	
6 *	392.0951	-72.48	-14.90	-87.38	-57.00	-30.38	peak	150	0	P	

Radiated Emission Measurement

File :20231103 Data :#12 Date: 2023/11/03 Time: 9:39:53



Site Chamber 2#
Limit: EN 301 908 Idel
EUT: LED
Model: Pending
Mode: NB IOT B8
Note:

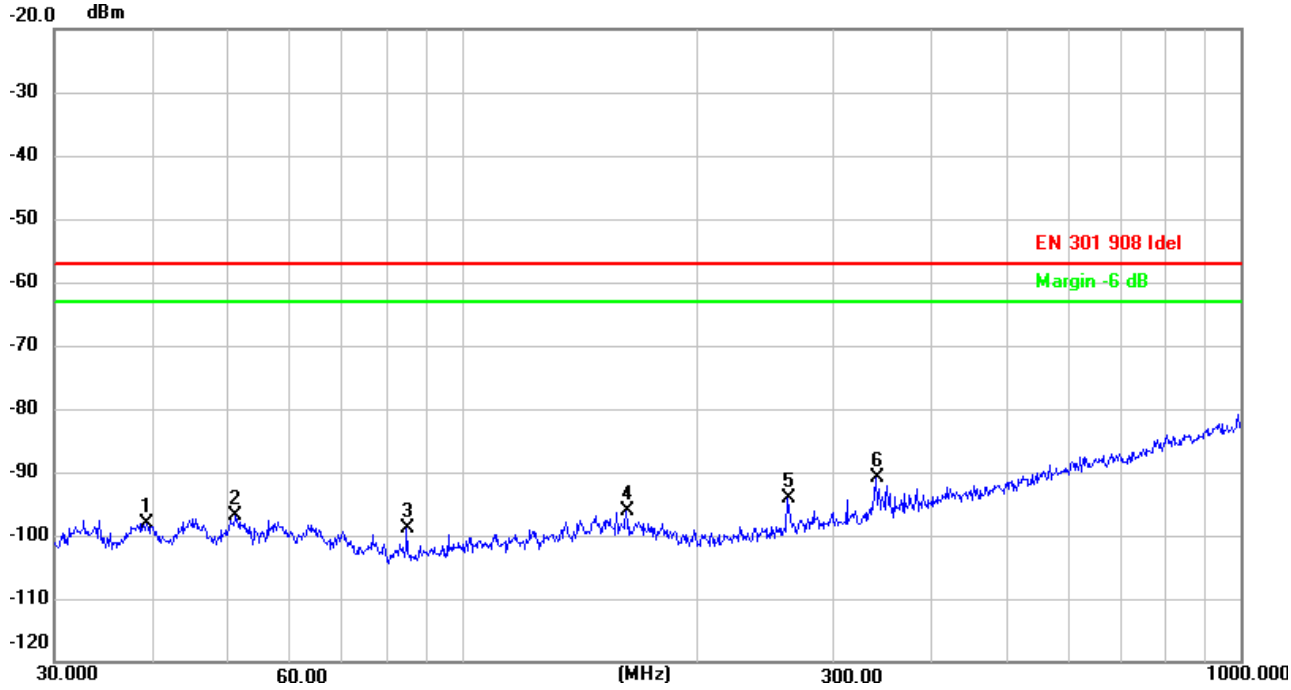
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 25(C)
Humidity: 56 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	44.9006	-78.37	-18.23	-96.60	-57.00	-39.60	peak	150	360	P	
2	154.8204	-78.91	-18.04	-96.95	-57.00	-39.95	peak	150	360	P	
3	263.8190	-75.93	-18.79	-94.72	-57.00	-37.72	peak	150	360	P	
4	312.1794	-75.87	-17.14	-93.01	-57.00	-36.01	peak	150	360	P	
5	344.3855	-72.80	-16.21	-89.01	-57.00	-32.01	peak	150	360	P	
6 *	793.3960	-78.12	-6.14	-84.26	-57.00	-27.26	peak	150	360	P	

Radiated Emission Measurement

File :20231103 Data :#13 Date: 2023/11/03 Time: 9:41:02



Site Chamber 2#
Limit: EN 301 908 Idel
EUT: LED
Model: Pending
Mode: NB IOT B20
Note:

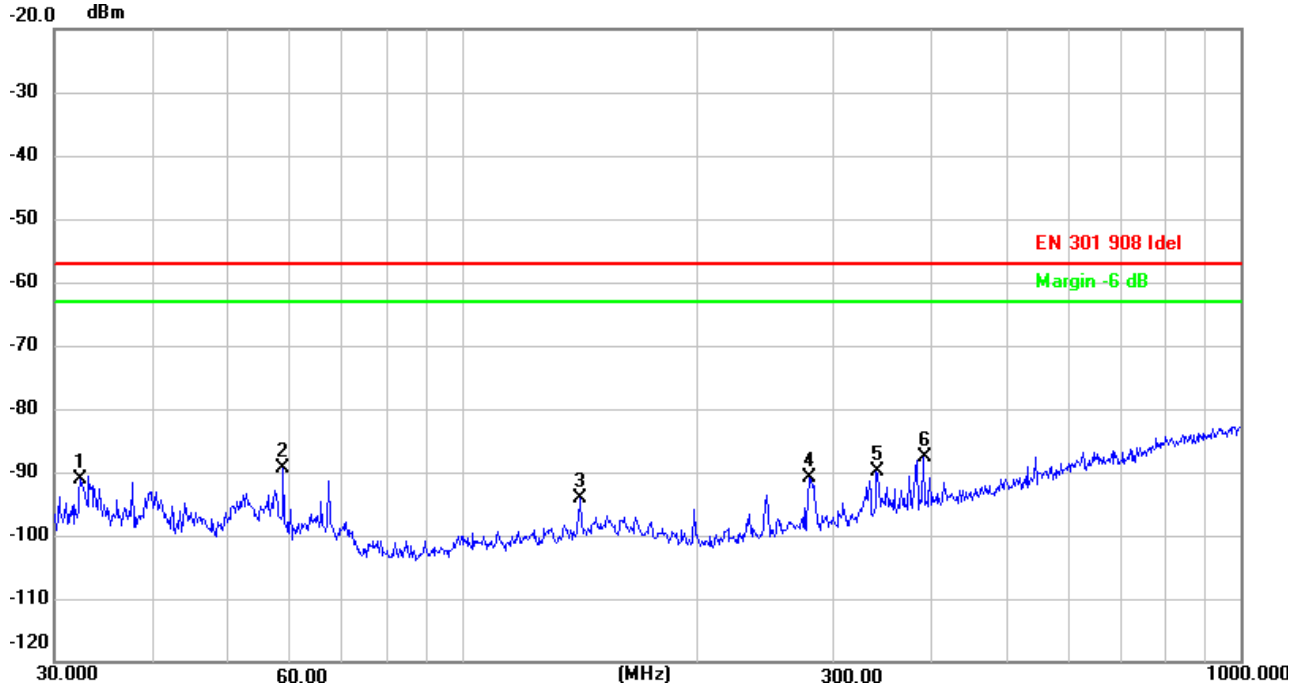
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 25(C)
Humidity: 56 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	39.4371	-79.94	-18.15	-98.09	-57.00	-41.09	peak	150	0	P	
2	51.3005	-78.95	-18.02	-96.97	-57.00	-39.97	peak	150	0	P	
3	84.9995	-76.41	-22.35	-98.76	-57.00	-41.76	peak	150	0	P	
4	162.6106	-77.57	-18.55	-96.12	-57.00	-39.12	peak	150	0	P	
5	261.9753	-75.18	-18.86	-94.04	-57.00	-37.04	peak	150	0	P	
6 *	340.7817	-74.62	-16.32	-90.94	-57.00	-33.94	peak	150	0	P	

Radiated Emission Measurement

File :20231103 Data :#14 Date: 2023/11/03 Time: 9:42:08



Site Chamber 2#
Limit: EN 301 908 Idel
EUT: LED
Model: Pending
Mode: NB IOT B20
Note:

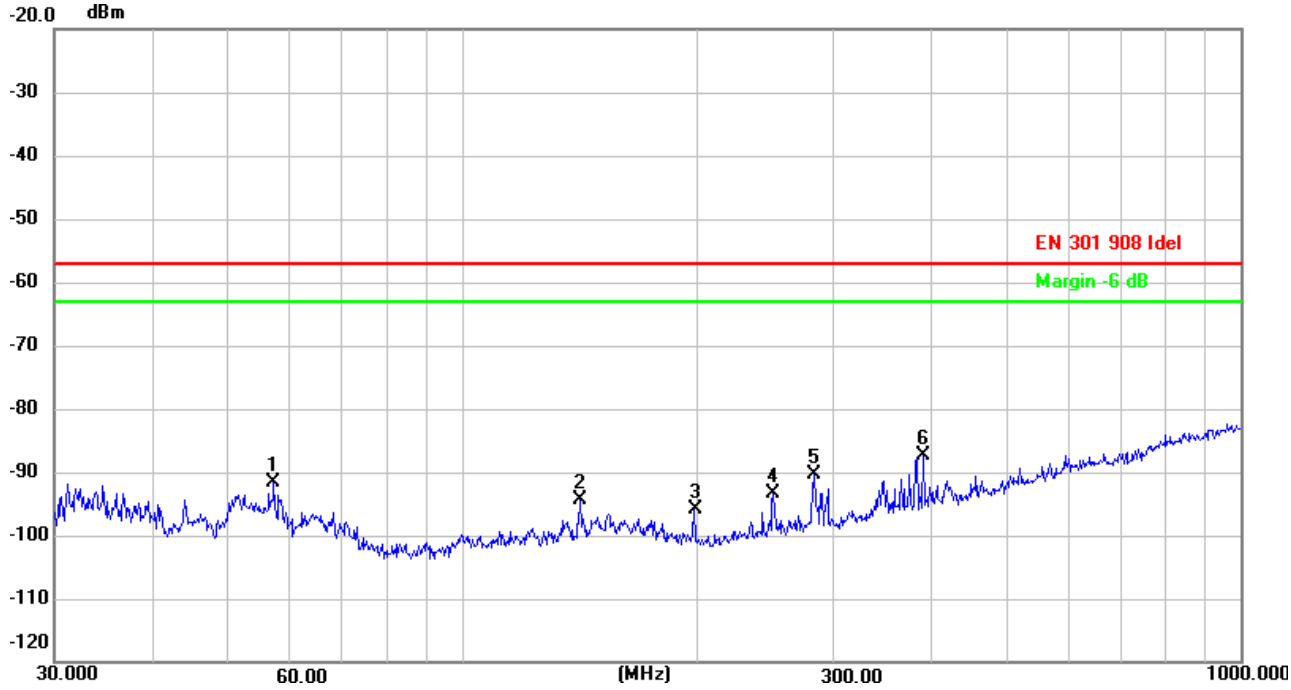
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 25(C)
Humidity: 56 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	32.4059	-72.13	-18.95	-91.08	-57.00	-34.08	peak	150	360	P	
2	59.0251	-70.51	-18.74	-89.25	-57.00	-32.25	peak	150	360	P	
3	141.8262	-75.58	-18.62	-94.20	-57.00	-37.20	peak	150	360	P	
4	280.0237	-72.56	-18.20	-90.76	-57.00	-33.76	peak	150	360	P	
5	340.7817	-73.54	-16.32	-89.86	-57.00	-32.86	peak	150	360	P	
6 *	392.0951	-72.72	-14.90	-87.62	-57.00	-30.62	peak	150	360	P	

Radiated Emission Measurement

File :20231103 Data :#15 Date: 2023/11/03 Time: 9:43:15



Site Chamber 2#
Limit: EN 301 908 Idel
EUT: LED
Model: Pending
Mode: NB IOT B28
Note:

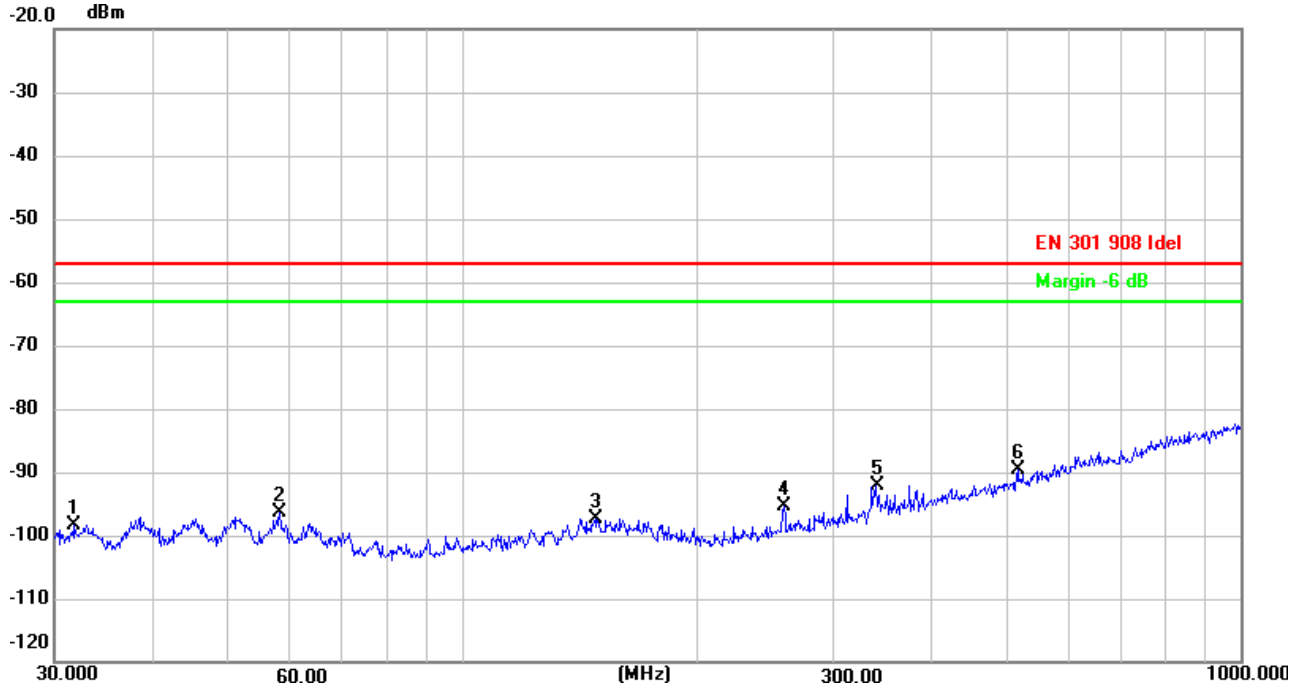
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 25(C)
Humidity: 56 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	57.3923	-72.92	-18.63	-91.55	-57.00	-34.55	peak	150	0	P	
2	141.8262	-75.78	-18.62	-94.40	-57.00	-37.40	peak	150	0	P	
3	198.5880	-75.04	-20.87	-95.91	-57.00	-38.91	peak	150	0	P	
4	251.1804	-74.25	-19.24	-93.49	-57.00	-36.49	peak	150	0	P	
5	283.9791	-72.43	-18.05	-90.48	-57.00	-33.48	peak	150	0	P	
6 *	390.7226	-72.51	-14.93	-87.44	-57.00	-30.44	peak	150	0	P	

Radiated Emission Measurement

File :20231103 Data :#16 Date: 2023/11/03 Time: 9:44:22



Site Chamber 2#
Limit: EN 301 908 Idel
EUT: LED
Model: Pending
Mode: NB IOT B28
Note:

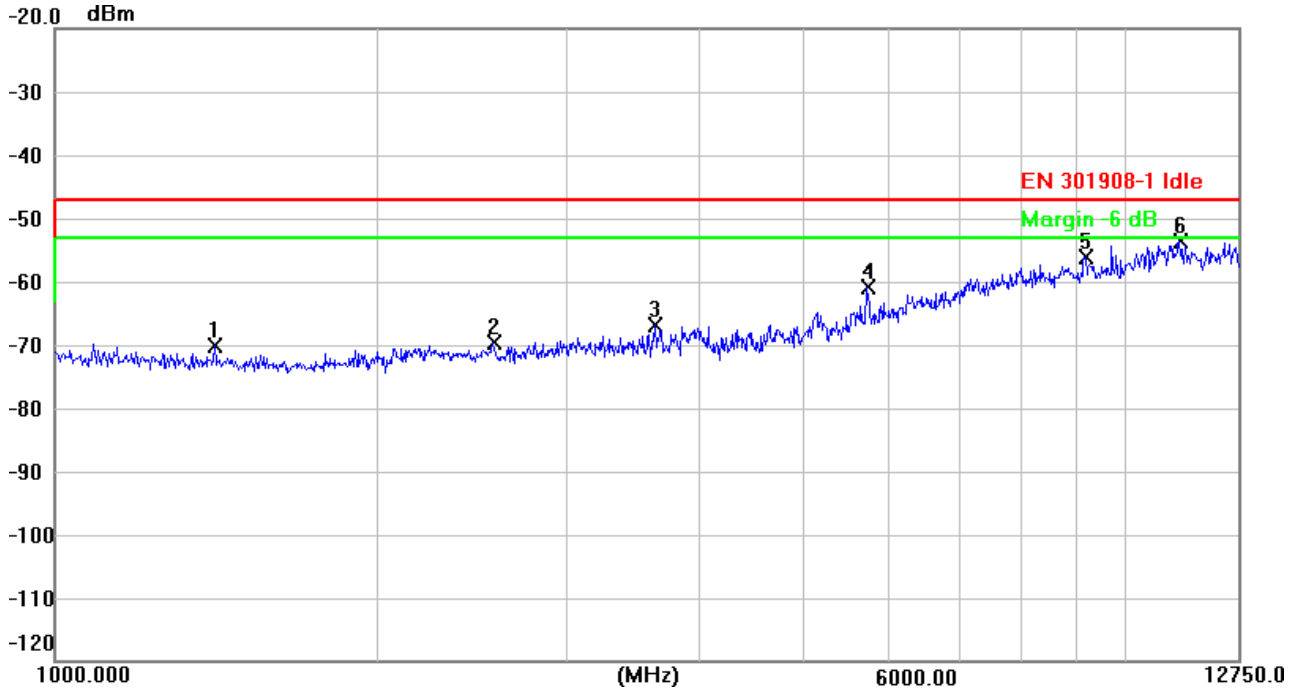
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 25(C)
Humidity: 56 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	31.8427	-79.30	-19.00	-98.30	-57.00	-41.30	peak	150	360	P	
2	58.4074	-77.78	-18.70	-96.48	-57.00	-39.48	peak	150	360	P	
3	148.4410	-79.46	-17.87	-97.33	-57.00	-40.33	peak	150	360	P	
4	259.2338	-76.37	-18.95	-95.32	-57.00	-38.32	peak	150	360	P	
5	340.7817	-75.77	-16.32	-92.09	-57.00	-35.09	peak	150	360	P	
6 *	517.2480	-77.90	-11.72	-89.62	-57.00	-32.62	peak	150	360	P	

Radiated Emission Measurement

File :20231106 Data :#13 Date: 2023-11-06 Time: 10:32:39



Site Radiation 3#
Limit: EN 301908-1 Idle
EUT: Pending
Model: LED
Mode: B3
Note:

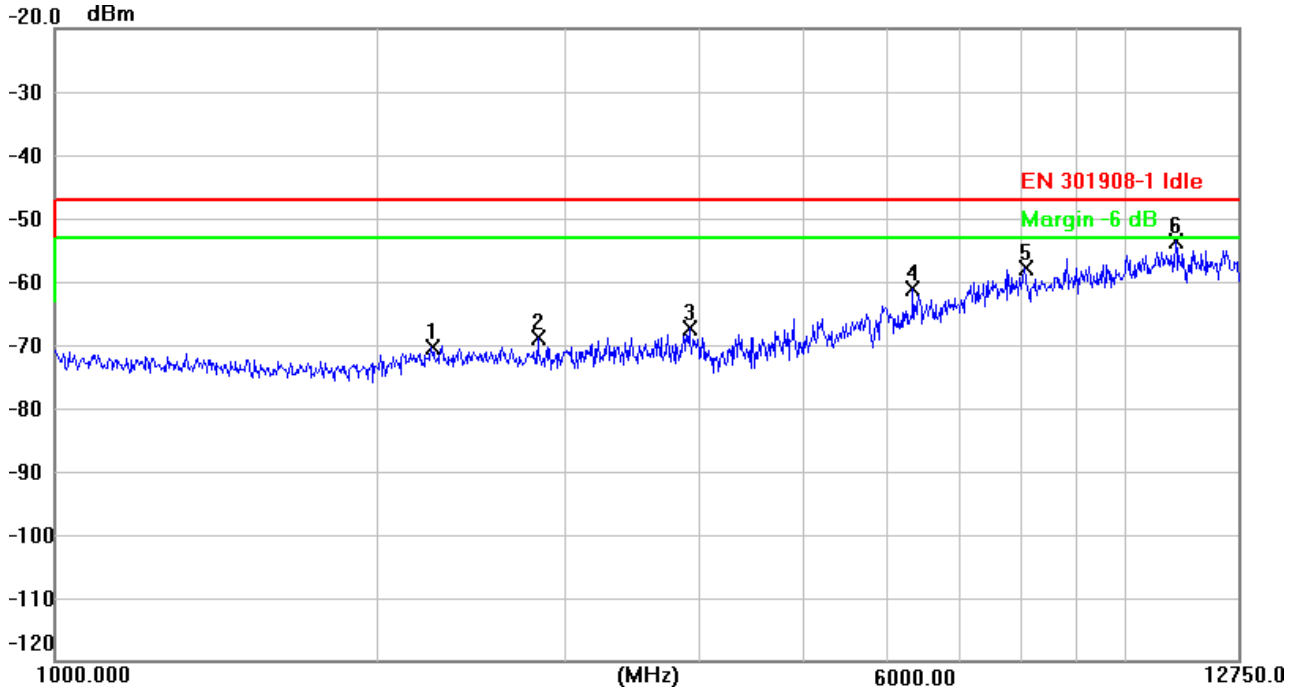
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	1411.250	-63.38	-6.83	-70.21	-47.00	-23.21	peak	150	1	P	
2	2566.667	-66.69	-3.00	-69.69	-47.00	-22.69	peak	150	1	P	
3	3639.833	-66.09	-0.75	-66.84	-47.00	-19.84	peak	150	1	P	
4	5739.167	-65.54	4.56	-60.98	-47.00	-13.98	peak	150	1	P	
5	9189.750	-68.51	12.27	-56.24	-47.00	-9.24	peak	150	1	P	
6 *	11238.167	-68.49	14.90	-53.59	-47.00	-6.59	peak	150	1	P	

Radiated Emission Measurement

File :20231106 Data :#14 Date: 2023-11-06 Time: 10:34:02



Site Radiation 3#
Limit: EN 301908-1 Idle
EUT: Pending
Model: LED
Mode: B3
Note:

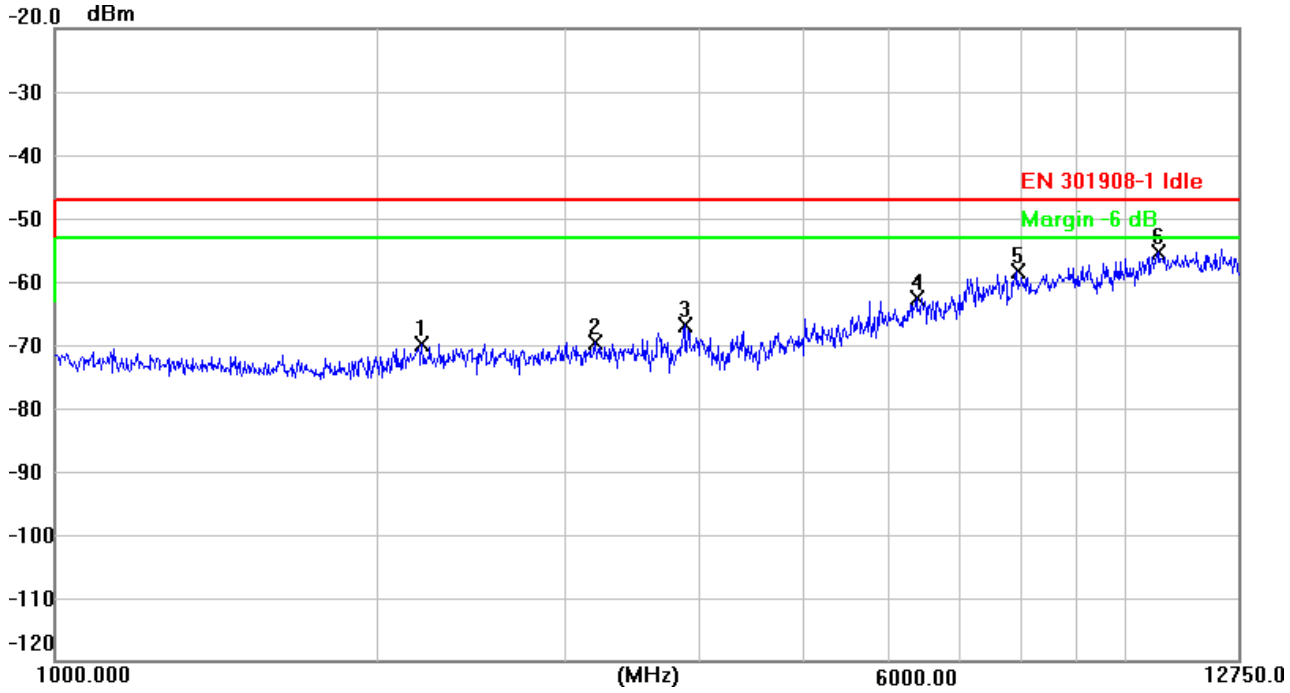
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2249.417	-66.76	-3.55	-70.31	-47.00	-23.31	peak	150	359	P	
2	2829.083	-66.33	-2.64	-68.97	-47.00	-21.97	peak	150	359	P	
3	3914.000	-67.59	0.20	-67.39	-47.00	-20.39	peak	150	359	P	
4	6346.250	-67.94	6.79	-61.15	-47.00	-14.15	peak	150	359	P	
5	8053.917	-68.72	10.73	-57.99	-47.00	-10.99	peak	150	359	P	
6 *	11152.000	-68.41	14.85	-53.56	-47.00	-6.56	peak	150	359	P	

Radiated Emission Measurement

File :20231106 Data :#15 Date: 2023-11-06 Time: 10:34:32



Site Radiation 3#
Limit: EN 301908-1 Idle
EUT: Pending
Model: LED
Mode: B8
Note:

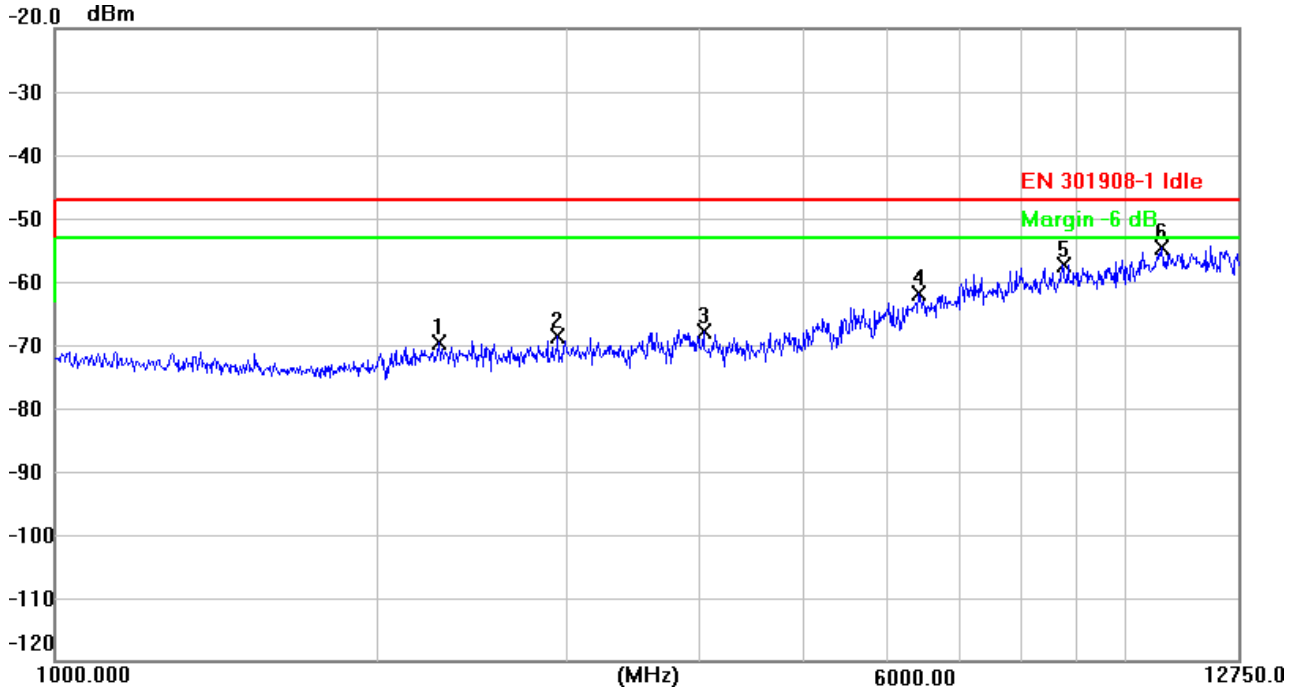
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2206.333	-66.30	-3.59	-69.89	-47.00	-22.89	peak	150	360	P	
2	3197.250	-67.56	-2.05	-69.61	-47.00	-22.61	peak	150	360	P	
3	3882.667	-66.91	0.07	-66.84	-47.00	-19.84	peak	150	360	P	
4	6385.417	-69.59	6.96	-62.63	-47.00	-15.63	peak	150	360	P	
5	7936.417	-69.19	10.74	-58.45	-47.00	-11.45	peak	150	360	P	
6 *	10772.083	-69.99	14.50	-55.49	-47.00	-8.49	peak	150	360	P	

Radiated Emission Measurement

File :20231106 Data :#16 Date: 2023-11-06 Time: 10:34:59



Site Radiation 3#
Limit: EN 301908-1 Idle
EUT: Pending
Model: LED
Mode: B8
Note:

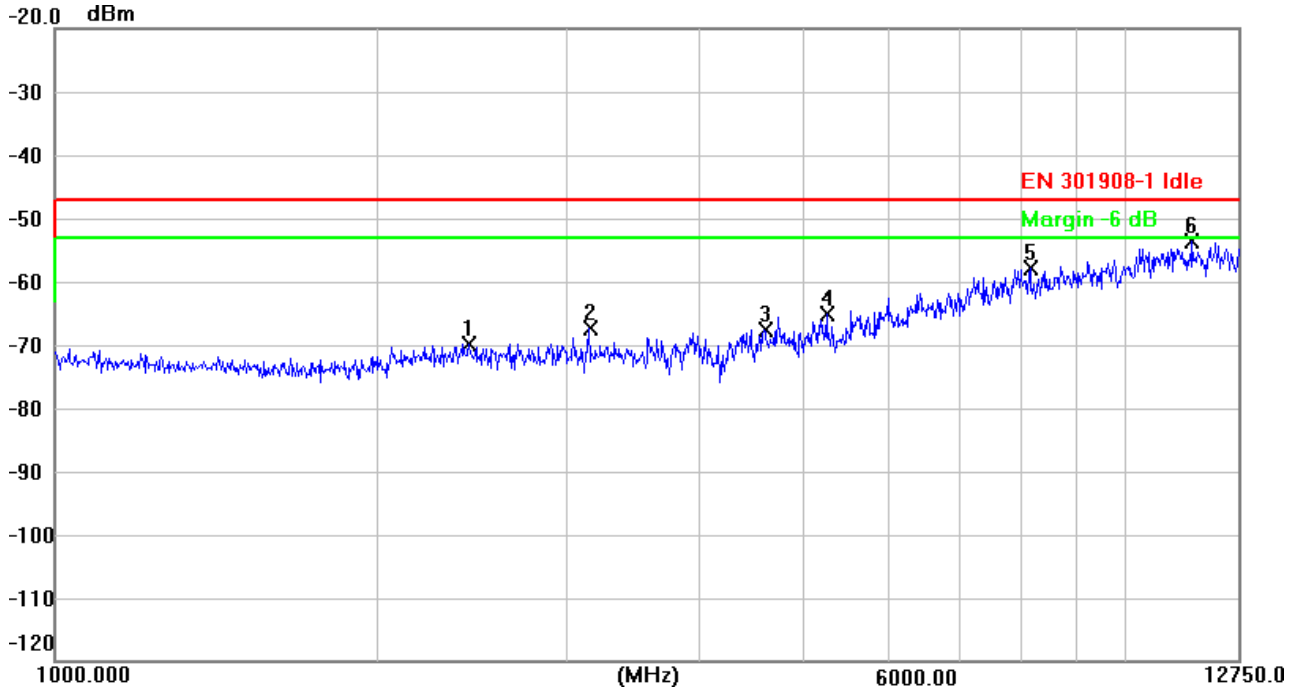
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2288.583	-66.04	-3.49	-69.53	-47.00	-22.53	peak	150	360	P	
2	2950.500	-66.19	-2.32	-68.51	-47.00	-21.51	peak	150	360	P	
3	4043.250	-68.48	0.58	-67.90	-47.00	-20.90	peak	150	360	P	
4	6424.583	-69.01	7.10	-61.91	-47.00	-14.91	peak	150	360	P	
5	8735.417	-68.70	11.21	-57.49	-47.00	-10.49	peak	150	360	P	
6 *	10807.333	-69.17	14.56	-54.61	-47.00	-7.61	peak	150	360	P	

Radiated Emission Measurement

File :20231106 Data :#17 Date: 2023-11-06 Time: 10:35:56



Site Radiation 3#
Limit: EN 301908-1 Idle
EUT: Pending
Model: LED
Mode: B20
Note:

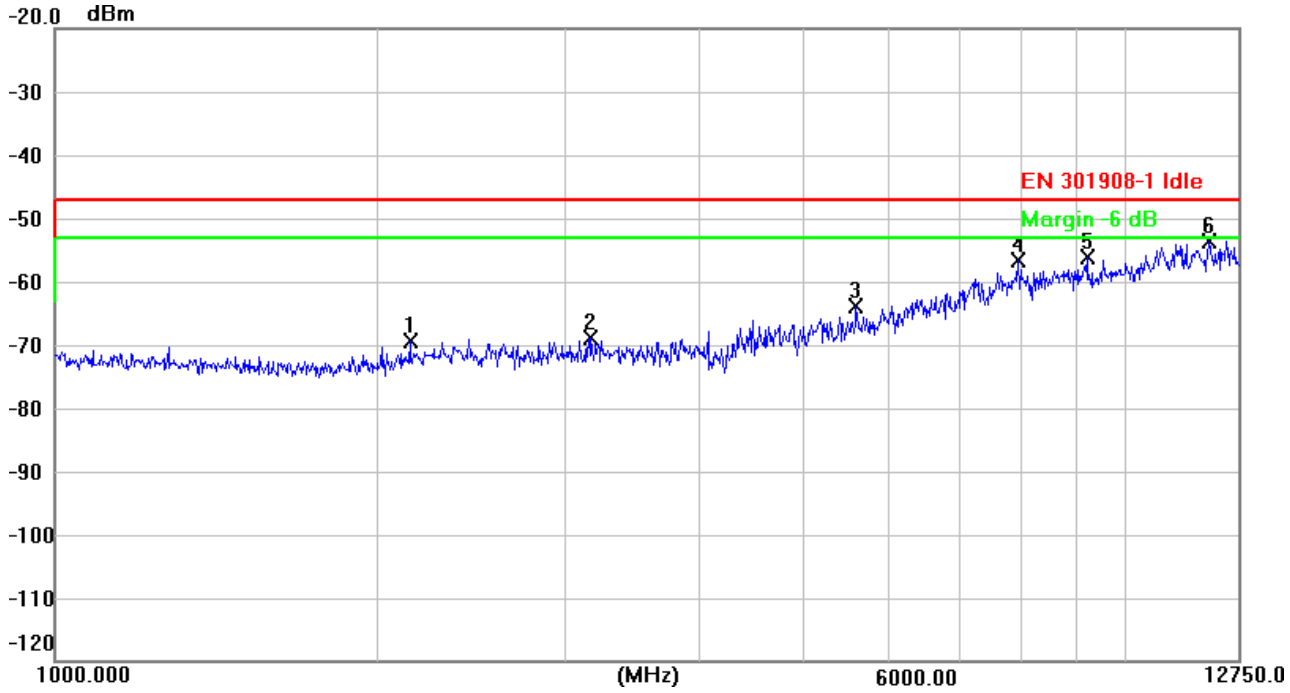
Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2441.333	-66.57	-3.27	-69.84	-47.00	-22.84	peak	150	360	P	
2	3154.167	-65.27	-2.08	-67.35	-47.00	-20.35	peak	150	360	P	
3	4599.417	-69.27	1.67	-67.60	-47.00	-20.60	peak	150	360	P	
4	5261.333	-68.16	3.06	-65.10	-47.00	-18.10	peak	150	360	P	
5	8163.583	-68.40	10.43	-57.97	-47.00	-10.97	peak	150	360	P	
6 *	11571.083	-68.92	15.23	-53.69	-47.00	-6.69	peak	150	360	P	

Radiated Emission Measurement

File :20231106 Data :#18 Date: 2023-11-06 Time: 10:36:31



Site Radiation 3#
Limit: EN 301908-1 Idle
EUT: Pending
Model: LED
Mode: B20
Note:

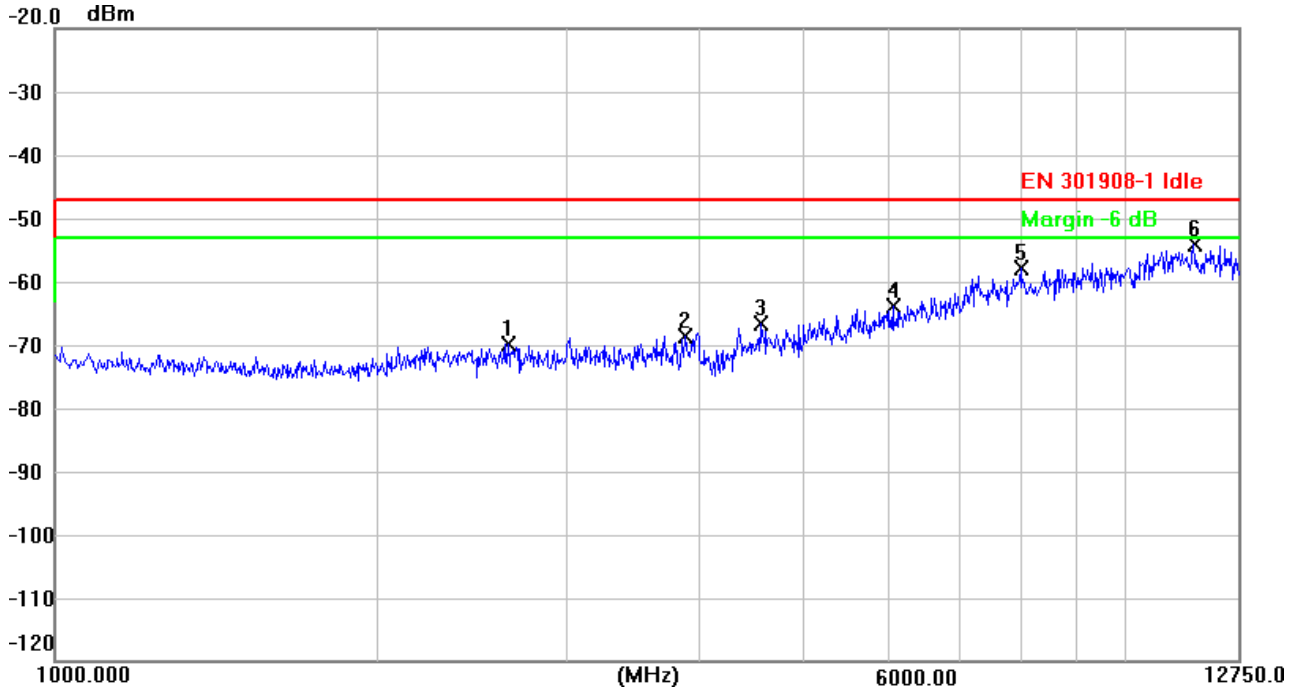
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2151.500	-65.29	-4.04	-69.33	-47.00	-22.33	peak	150	360	P	
2	3173.750	-66.76	-2.07	-68.83	-47.00	-21.83	peak	150	360	P	
3	5606.000	-67.90	4.05	-63.85	-47.00	-16.85	peak	150	360	P	
4	7952.083	-67.49	10.77	-56.72	-47.00	-9.72	peak	150	360	P	
5	9213.250	-68.48	12.32	-56.16	-47.00	-9.16	peak	150	360	P	
6 *	11982.333	-69.12	15.62	-53.50	-47.00	-6.50	peak	150	360	P	

Radiated Emission Measurement

File :20231106 Data :#19 Date: 2023-11-06 Time: 10:37:04



Site Radiation 3#
Limit: EN 301908-1 Idle
EUT: Pending
Model: LED
Mode: B28
Note:

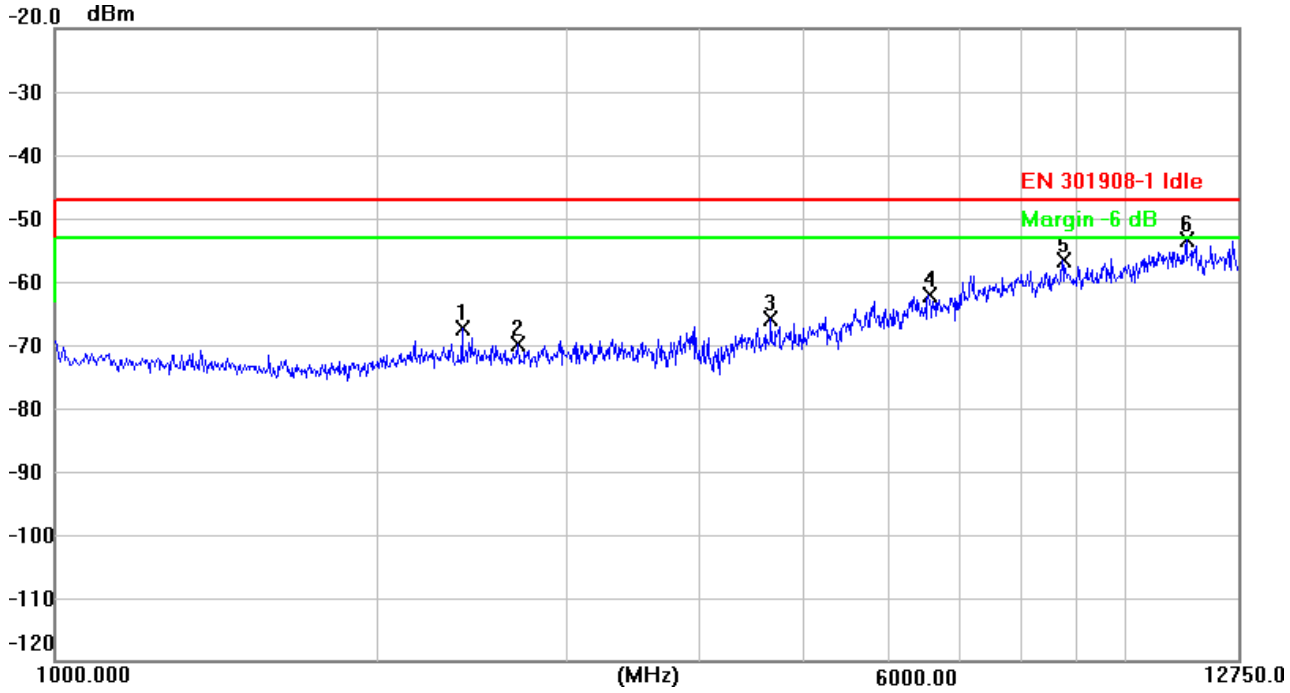
Polarization: ***Horizontal***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2656.750	-66.97	-2.87	-69.84	-47.00	-22.84	peak	150	360	P	
2	3882.667	-68.60	0.07	-68.53	-47.00	-21.53	peak	150	360	P	
3	4560.250	-68.18	1.56	-66.62	-47.00	-19.62	peak	150	360	P	
4	6072.083	-69.57	5.70	-63.87	-47.00	-16.87	peak	150	360	P	
5	7983.417	-68.82	10.84	-57.98	-47.00	-10.98	peak	150	360	P	
6 *	11590.667	-69.49	15.27	-54.22	-47.00	-7.22	peak	150	360	P	

Radiated Emission Measurement

File :20231106 Data :#20 Date: 2023-11-06 Time: 10:37:31



Site Radiation 3#
Limit: EN 301908-1 Idle
EUT: Pending
Model: LED
Mode: B28
Note:

Polarization: ***Vertical***
Power: DC9V
Distance: 3m

Temperature: 26(C)
Humidity: 54 %

No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2402.167	-64.04	-3.36	-67.40	-47.00	-20.40	peak	150	360	P	
2	2711.583	-67.06	-2.82	-69.88	-47.00	-22.88	peak	150	360	P	
3	4673.833	-67.72	1.82	-65.90	-47.00	-18.90	peak	150	360	P	
4	6549.917	-69.62	7.47	-62.15	-47.00	-15.15	peak	150	360	P	
5	8747.167	-67.88	11.23	-56.65	-47.00	-9.65	peak	150	360	P	
6 *	11422.250	-68.50	15.01	-53.49	-47.00	-6.49	peak	150	360	P	

Prüfbericht-Nr.: *Test report no.*:	**Part IV of CN2394C9 001**	**Auftrags-Nr.:** *Order no.*:	178186679	**Seite 1 von 26** *Page 1 of 26*
Kunden-Referenz-Nr.: *Client reference no.*:	N/A	**Auftragsdatum:** *Order date:*	2023-08-25	
Auftraggeber: *Client*:	Huizhou Duran Creative Technology Co., Ltd. Room 708, Innovation Building, No.1, Keji Road, Science and Technology Innovation Park, West Area of Daya Bay, Huizhou, Guangdong P.R.China			
Prüfgegenstand: *Test item*:	Portable Light			
Bezeichnung / Typ-Nr.: *Identification / Type no.*:	INSTAFLASH V16 G-001			
Auftrags-Inhalt: *Order content*:	RED report			
Prüfgrundlage: *Test specification*:	[Radio] EN 303 413 V1.2.1:2021			

Wareneingangsdatum: *Date of sample receipt*:	2023-08-25
Prüfmuster-Nr.: *Test sample no:*	178186679 001
Prüfzeitraum: *Testing period*:	2023-10-12 - 2023-11-27
Ort der Prüfung: *Place of testing*:	Refer to section 1.1
Prüflaboratorium: *Testing laboratory:*	TUV Rheinland / CCIC (Qingdao) Co., Ltd.
Prüfergebnis*: *Test result**:	Pass

Keine Fotodokumentation erforderlich

No photo documentation required

geprüft von: *tested by:*	x Geoff Guo Signed by: Geoff Guo	**genehmigt von:** *authorized by*:	x Hunter Yu Signed by: Hunter Yu
Datum: *Date:* 2023-12-07		**Ausstellungsdatum:** *Issue date:* 2023-12-07	
Stellung / Position:	Project manager	**Stellung** / Position:	Authorizer

Sonstiges / *Other*: This report only evaluates GNSS parts accroding to EN 303 413.

Zustand des Prüfgegenstandes bei Anlieferung: *Condition of the test item at delivery*: Prüfmuster vollständig und unbeschädigt *Test item complete and undamaged*

* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
** Legend:*	*P(ass) = passed a.m. test specification(s)*	*F(ail) = failed a.m. test specification(s)*	*N/A = not applicable*	*N/T = not tested*

Dieser Prüfbericht bezieht sich nur auf das o.g. Prüfmuster und darf ohne Genehmigung der Prüfstelle nicht auszugsweise vervielfältigt werden. Dieser Bericht berechtigt nicht zur Verwendung eines Prüfzeichens.
This test report only relates to the above mentioned test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any test mark.
V05

Anmerkungen
Remarks

1	Alle eingesetzten Prüfmittel waren zum angegebenen Prüfzeitraum gemäß eines festgelegten Kalibrierungsprogramms unseres Prüfhauses kalibriert. Sie entsprechen den in den Prüfprogrammen hinterlegten Anforderungen. Die Rückverfolgbarkeit der eingesetzten Prüfmittel ist durch die Einhaltung der Regelungen unseres Managementsystems gegeben. Detaillierte Informationen bezüglich Prüfkonditionen, Prüfequipment und Messunsicherheiten sind im Prüflabor vorhanden und können auf Wunsch bereitgestellt werden. *The equipment used during the specified testing period was calibrated according to our test laboratory calibration program. The equipment fulfils the requirements included in the relevant standards. The traceability of the test equipment used is ensured by compliance with the regulations of our management system.* *Detailed information regarding test conditions, equipment and measurement uncertainty is available in the test laboratory and could be provided on request.*
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3	Prüfklausel mit der Note * wurden an qualifizierte Unterauftragnehmer vergeben und sind unter der jeweiligen Prüfklausel des Berichts beschrieben. Abweichungen von Prüfspezifikation(en) oder Kundenanforderungen sind in der jeweiligen Prüfklausel im Bericht aufgeführt. *Test clauses with remark of * are subcontracted to qualified subcontractors and descripted under the respective test clause in the report. Deviations of testing specification(s) or customer requirements are listed in specific test clause in the report.*
4	Die Entscheidungsregel für Konformitätserklärungen basierend auf numerischen Messergebnisen in diesem Prüfbericht basiert auf der "Null-Grenzwert-Regel" und der "Einfachen Akzeptanz" gemäß ILAC G8:2019 und IEC Guide 115:2021, es sei denn, in der auf Seite 1 dieses Berichts genannten angewandten Norm ist etwas anderes festgelegt oder vom Kunden gewünscht. Dies bedeutet, dass die Messunsicherheit nicht berücksichtigt wird und daher auch nicht im Prüfbericht angegeben wird. Zu weiteren Informationen bezueglich des Risikos durch diese Entscheidungsregel siehe ILAC G8:2019. *The decision rule for statements of conformity, based on numerical measurement results, in this test report is based on the “Zero Guard Band Rule” and “Simple Acceptance” in accordance with ILAC G8:2019 and IEC Guide 115:2021, unless otherwise specified in the applied standard mentioned on Page 1 of this report or requested by the customer. This means that measurement uncertainty is not taken in account and hence also not declared in the test report. For additional information to the resulting risk based of this decision rule please refer to ILAC G8:2019.*

Test Summary

4.1.1 RECEIVER BLOCKING

Result:

Pass

4.1.2 RECEIVER SPURIOUS EMISSIONS

Result:

Pass

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1 Test Sites

1.1 Test Facilities

CTC Laboratories, Inc.
2/F., Building 1 and 1-2/F., Building 2, Jiaquan Building,
Guanlan High-Tech Park, Longhua District, Shenzhen, Guangdong, China

The used test equipment is in accordance with CISPR 16-1 series standards for measurement of radio interference.

1.2 List of Test and Measurement Instruments

Table 1: List of Test and Measurement Equipment

nTscend JS0806-2 Test system						
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Calibrated Date	Calibrated until
1	Spectrum Analyzer	Rohde & Schwarz	FSU26	100105	Dec. 26, 2022	Dec. 25, 2023
2	Spectrum Analyzer	Rohde & Schwarz	FUV40-N	101331	Mar. 16, 2022	Mar. 15, 2023
3	MXG Vector Signal Generator	Agilent	N5182A	MY47420864	Dec. 26, 2022	Dec. 25, 2023
4	Signal Generator	Agilent	E8257D	MY46521908	Dec. 26, 2022	Dec. 25, 2023
5	Power Sensor	Agilent	U2021XA	MY5365004	Dec. 26, 2022	Dec. 25, 2023
6	Power Sensor	Agilent	U2021XA	MY5365006	Dec. 26, 2022	Dec. 25, 2023
7	Simultaneous Sampling DAQ	Agilent	U2531A	TW54493510	Dec. 26, 2022	Dec. 25, 2023
8	Wideband Radio Communication Tester	Rohde & Schwarz	CMW500	116410	Dec. 26, 2022	Dec. 25, 2023
9	Climate Chamber	ESPEC	MT3065	/	Dec. 26, 2022	Dec. 25, 2023
10	300328 V2.2.2 test system	TONSCEND	v2.6	/	/	/

Transmitter spurious emissions & Receiver spurious emissions						
Item	Test Equipment	Manufacturer	Model No.	Serial No.	Calibrated Date	Calibrated until
1	EMI Test Receiver	Rohde & Schwarz	ESCI	100658	Dec. 26, 2022	Dec. 25, 2023
2	High pass filter	micro-tranics	HPM50111	142	Dec. 26, 2022	Dec. 25, 2023
3	Log-Bicon Antenna	Schwarzbeck	CBL6141A	4180	Dec. 26, 2022	Dec. 25, 2023
4	Ultra-Broadband Antenna	ShwarzBeck	BBHA9170	25841	Dec. 26, 2022	Dec. 25, 2023
5	Loop Antenna	LAPLAC	RF300	9138	Dec. 26, 2022	Dec. 25, 2023
6	Spectrum Analyzer	Rohde & Schwarz	FSU26	100105	Dec. 26, 2022	Dec. 25, 2023
7	Horn Antenna	Schwarzbeck	BBHA 9120D	647	Dec. 26, 2022	Dec. 25, 2023
8	Pre-Amplifier	HP	8447D	1937A03050	Dec. 26, 2022	Dec. 25, 2023
9	Pre-Amplifier	EMCI	EMC051835	980075	Dec. 26, 2022	Dec. 25, 2023
10	Antenna Mast	UC	UC3000	N/A	N/A	N/A
11	Turn Table	UC	UC3000	N/A	N/A	N/A
12	Cable Below 1GHz	Schwarzbeck	AK9515E	33155	Dec. 26, 2022	Dec. 25, 2023
13	Cable Above 1GHz	Hubersuhner	SUCOFLEX 102	DA1580	Dec. 26, 2022	Dec. 25, 2023

1.3 Uncertainty of Measurement

Table 2: Emission Measurement Uncertainty

Test Items	Measurement Uncertainty	Notes
Maximum transmit power	±1.5dB	(1)
Power Spectral Density	±1.5dB	(1)
Duty Cycle, Tx-sequence, Tx-gap	±5%	(1)
Accumulated Transmit Time, Frequency Occupation and Hopping Sequence	±5%	(1)
Hopping Frequency Separation	±5%	(1)
Medium Utilisation (MU) factor	±5%	(1)
Adaptively	±5%	(1)
Occupied Channel Bandwidth	±5%	(1)
Transmitter unwanted emissions in the out-of-band domain	±2.8dB	(1)
Transmitter unwanted emissions in the spurious domain	±2.8dB	(1)
Receiver spurious emissions	±2.8dB	(1)
Receiver Blocking	±2.8dB	(1)
Radiated Emissions 30~1000MHz	±4.51 dB	(1)
Radiated Emissions 1~18GHz	±5.84 dB	(1)

Note: (1) This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=1.96.

2 General Product Information

2.1 Product Function and Intended Use

The EUT (equipment under test) is a Portable Light which supports NB-lot and GNSS functions.
For details refer to the User Manual, Circuit Diagram.

The aim of this report is to evalute the RF characteristics of the GNSS of the EUT.

2.2 Ratings and System Details

General	
Product	Portable Light
Product type	G-001
Input voltage	DC 9V from battery
Radio Technology used	GPS, GLONASS, BDS
Software Version	LC76FANR01A01
GPS	
Frequency Range	1559 ~ 1610MHz
GLONASS	
Frequency Range	1559 ~ 1610MHz
BDS	
Frequency Range	1559 ~ 1610MHz

2.3 Independent Operation Modes

A. On，receive mode
B. Off

2.4 Noise Generating and Noise Suppressing Parts

Refer to the circuit diagram and photo documentation for further information.

2.5 Submitted Documents

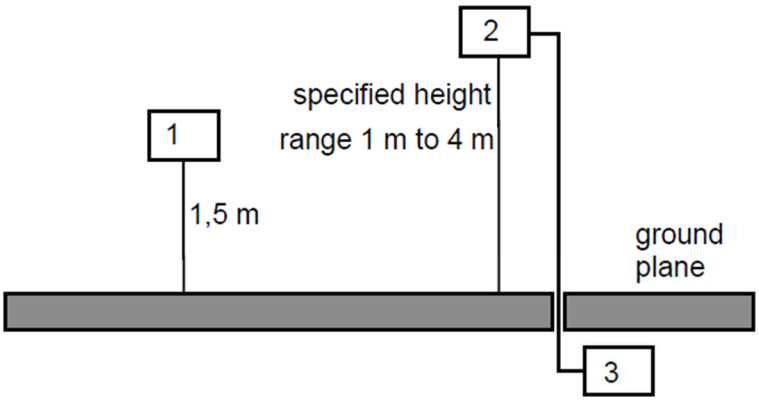
- Block Diagram
- PCB Layout
- Rating Label
- Parts List
- Schematics
- User Manual

3 Test Set-up and Operation Modes

3.1 Principle of Configuration Selection

Radiated measurements:

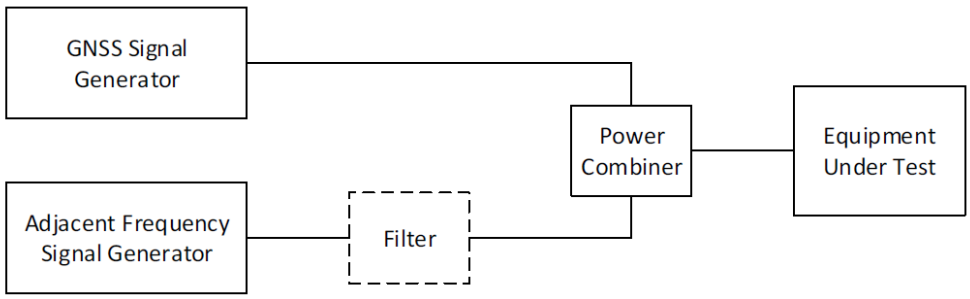
The radiated emissions from the UUT are performed in a modified semi-anechoic chamber. The UUT is placed on a non-conductive turntable. The UUT is powered using the UUT's battery.



1) UUT
2) Measurement antenna
3) Measurement equipment

Conducted measurements:

For EUT with an external, detachable, antenna, the EUT shall be connected to the test bed by means of the antenna port. A conceptual block diagram for conducted measurements is shown in figure below. The test bed shall be calibrated so that the adjacent frequency signal power levels of the test signals specified in EN 303 413 V1.2.1 are presented to the input of the EUT.



3.2 General Test Conditions / Configurations

Test Environment:

Environment Parameter	Selected Values During Tests	
	Temperature	Relative Humidity
Tnom & Vnom	Ambient: 15 °C to 35 °C	Ambient: 20% to 75%

3.3 Test Operation and Test Software

Refer to the relevant clause for further information.

3.4 Special Accessories and Auxiliary Equipment

None.

3.5 Countermeasures to achieve Compliance

None.

4 Radio frequency measurement results

4.1 Receiver

4.1.1 Receiver Blocking

Result:	**Pass**

Date of testing	:	2023-11-17
Test procedure	:	Clause 5.4.3 of EN 303 413 V1.2.1
Test environment	:	Normal
Operation mode	:	A
Ambient temperature	:	25°C
Relative humidity	:	51%
Atmospheric pressure	:	101kPa

Essential Conformance Requirements:
A GUE utilizing the RNSS band 1 559 MHz to 1 610 MHz shall be presumed to conform to this technical requirements specification if the C/N_0, as reported by the GUE for each declared GNSS constellation and GNSS signal, does not degrade by more than the value given in equation $\Delta C/N_0 \leq 1$dB in the presence of the adjacent frequency signals in table below.

Frequency band (MHz)	Test point centre frequency (MHz)	Blocking signal power level (dBm)	Comments
1 518 to 1 525	1 524	-65	MSS (space-to-Earth) band
1 525 to 1 549	1 548	-95	MSS (space-to-Earth) band
1 549 to 1 559	1 554	-105	MSS (space-to-Earth) band
1 559 to 1 610	GUE RNSS band under test		
1 610 to 1 626	1 615	-105	MSS (Earth-to-space) band
1 626 to 1 640	1 627	-85	MSS (Earth-to-space) band

A GUE also utilizing the RNSS band(s) 1164 MHz to 1300 MHz shall be presumed to conform to this technical requirements specification if the C/N_0, as reported by the GUE for each declared GNSS constellation and GNSS signal, also does not degrade by more than the value given in equation $\Delta C/N_0 \leq 1$dB in the presence of the adjacent frequency signals in table below.

Frequency band (MHz)	Test point centre frequency (MHz)	Blocking signal power level (dBm)	Comments
960 to 1 164	1 154	-75	AM(R)S, ARNS band
1 164 to 1 215	GUE RNSS band under test		
1 215 to 1 260	GUE RNSS band under test		
1 260 to 1 300	GUE RNSS band under test		
1 300 to 1 350	1 310	-85	Radiolocation, ARNS, RNSS (Earth-to-space) band

Table 3: Test Result of Receiver Blocking for GPS mode

Frequency band (MHz)	Test point centre frequency (MHz)	Adjacent frequency signal power level (dBm)	No interfering signal	With interfering signal	Decrease of C/N0	Limit	Result
1 518 to 1 525	1524	-65	45	45	0	≤1dB	Pass
1 525 to 1 549	1548	-95	45	45	0	≤1dB	Pass
1 549 to 1 559	1554	-105	45	45	0	≤1dB	Pass
1 610 to 1 626	1615	-105	45	45	0	≤1dB	Pass
1 626 to 1 640	1627	-85	45	45	0	≤1dB	Pass

Table 4: Test Result of Receiver Blocking for GLONASS mode

Frequency band (MHz)	Test point centre frequency (MHz)	Adjacent frequency signal power level (dBm)	No interfering signal	With interfering signal	Decrease of C/N0	Limit	Result
1 518 to 1 525	1524	-65	44	44	0	≤1dB	Pass
1 525 to 1 549	1548	-95	44	44	0	≤1dB	Pass
1 549 to 1 559	1554	-105	44	44	0	≤1dB	Pass
1 610 to 1 626	1615	-105	44	44	0	≤1dB	Pass
1 626 to 1 640	1627	-85	44	44	0	≤1dB	Pass

Table 5: Test Result of Receiver Blocking for BDS mode

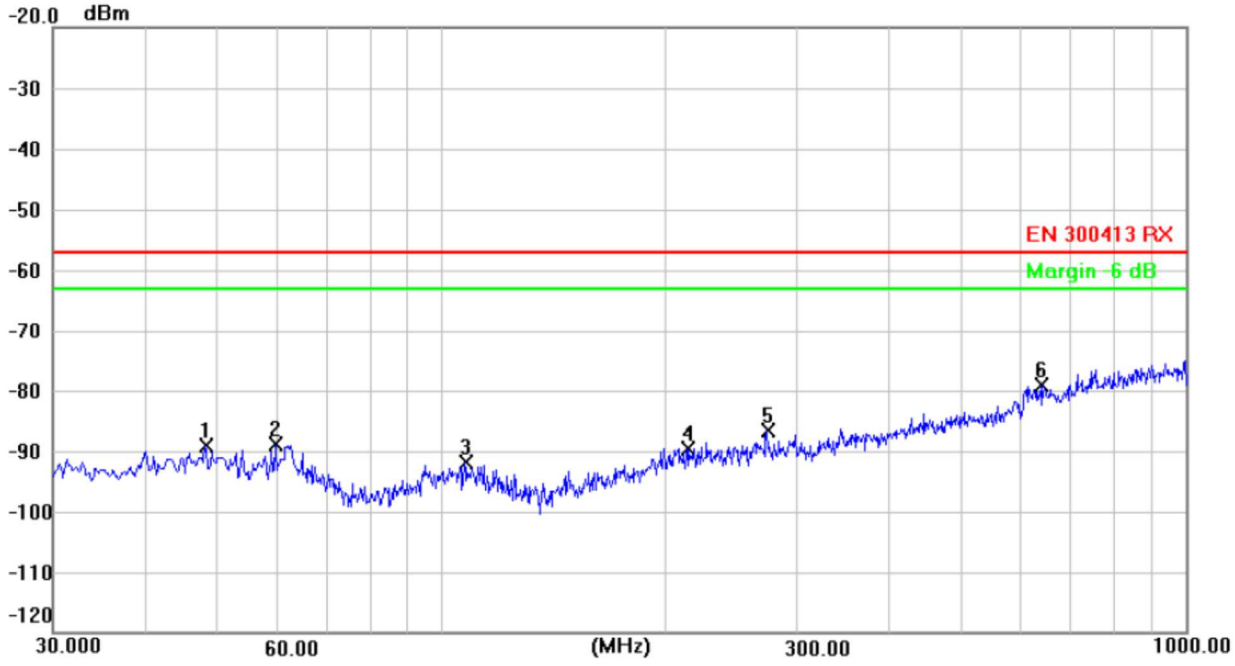
Frequency band (MHz)	Test point centre frequency (MHz)	Adjacent frequency signal power level (dBm)	No interfering signal	With interfering signal	Decrease of C/N0	Limit	Result
1 518 to 1 525	1524	-65	45	45	0	≤1dB	Pass
1 525 to 1 549	1548	-95	45	45	0	≤1dB	Pass
1 549 to 1 559	1554	-105	45	45	0	≤1dB	Pass
1 610 to 1 626	1615	-105	45	45	0	≤1dB	Pass
1 626 to 1 640	1627	-85	45	45	0	≤1dB	Pass

4.1.2 Receiver Spurious Emissions

Result:	**Pass**

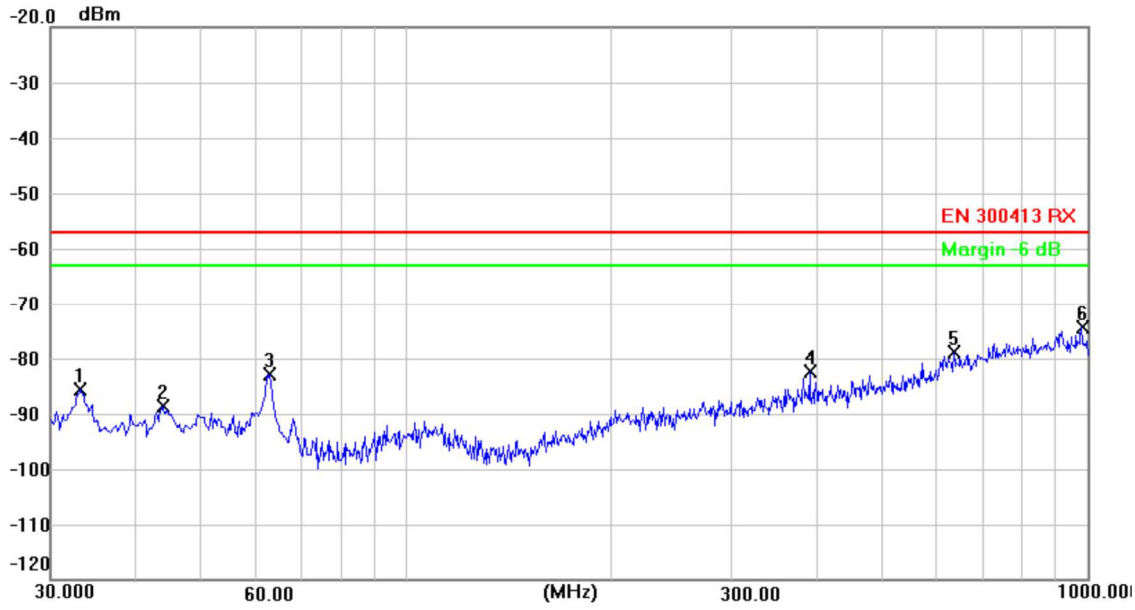
Date of testing	:	2023-11-06 ~ 2023-11-12
Test procedure	:	Clause 5.5.2 of EN 303 413 V1.2.1
Test environment	:	Normal
Operation mode	:	A
Ambient temperature	:	25°C
Relative humidity	:	50%
Atmospheric pressure	:	101KPa

Figure 1: Spectral diagrams and measurement results for 30-1000MHz, Horizontal polarization, BDS



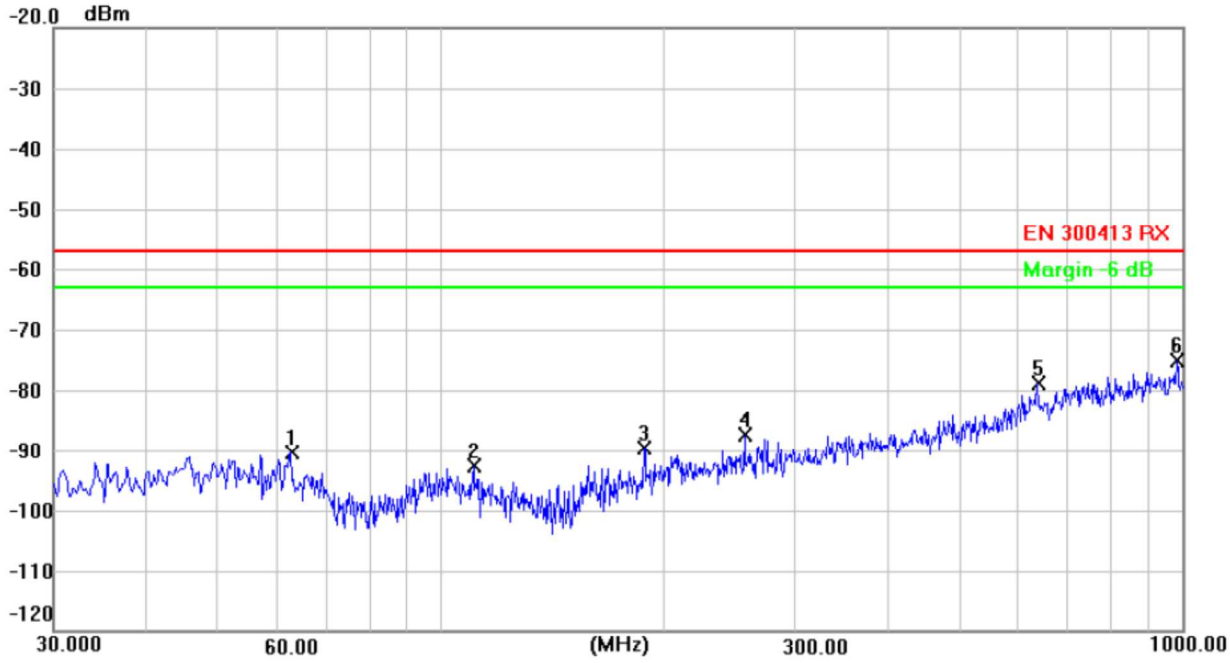
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	48.1067	-75.42	-13.82	-89.24	-57.00	-32.24	peak	150	358	P	
2	59.7467	-73.57	-15.32	-88.89	-57.00	-31.89	peak	150	358	P	
3	107.2767	-76.10	-15.87	-91.97	-57.00	-34.97	peak	150	358	P	
4	214.3000	-73.98	-15.55	-89.53	-57.00	-32.53	peak	150	358	P	
5	274.1167	-72.67	-14.01	-86.68	-57.00	-29.68	peak	150	358	P	
6 *	640.1300	-72.47	-6.59	-79.06	-57.00	-22.06	peak	150	358	P	

Figure 2: Spectral diagrams and measurement results for 30-1000MHz, Vertical polarization, BDS



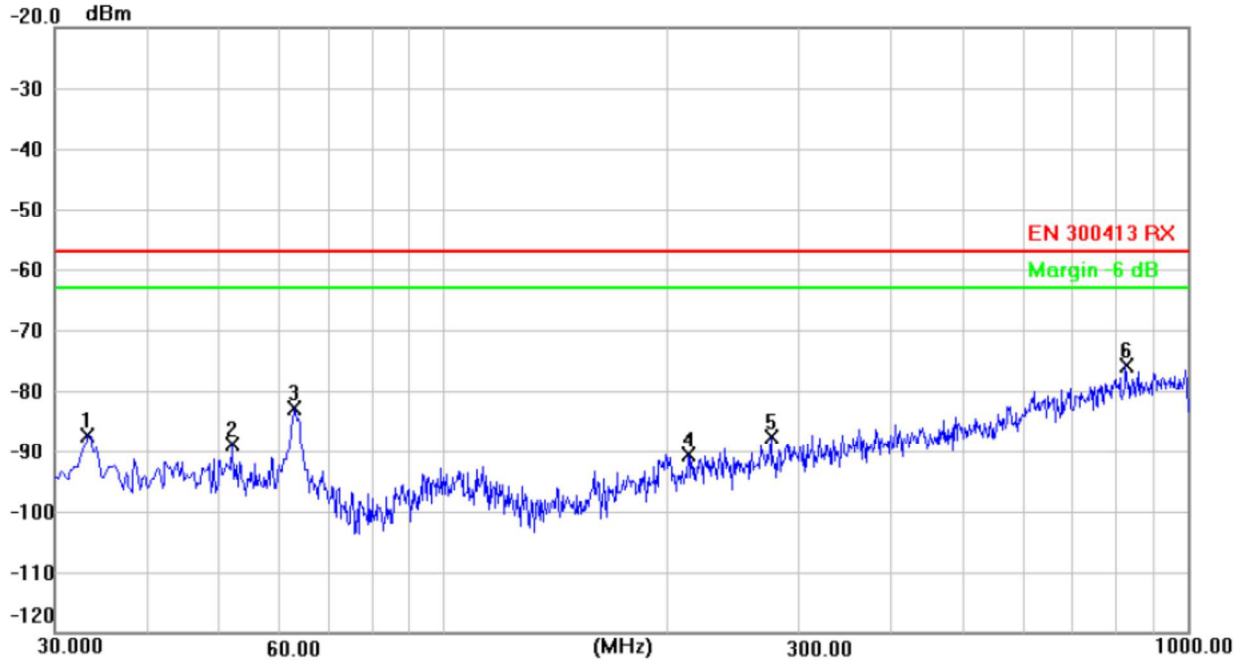
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	33.2333	-69.98	-15.67	-85.65	-57.00	-28.65	peak	150	4	P	
2	43.9033	-74.81	-13.81	-88.62	-57.00	-31.62	peak	150	4	P	
3	62.9800	-67.13	-15.80	-82.93	-57.00	-25.93	peak	150	4	P	
4	391.4867	-71.10	-11.24	-82.34	-57.00	-25.34	peak	150	4	P	
5	638.1900	-72.24	-6.62	-78.86	-57.00	-21.86	peak	150	4	P	
6 *	979.3067	-72.24	-2.22	-74.46	-57.00	-17.46	peak	150	4	P	

Figure 3: Spectral diagrams and measurement results for 30-1000MHz, Horizontal polarization, GPS



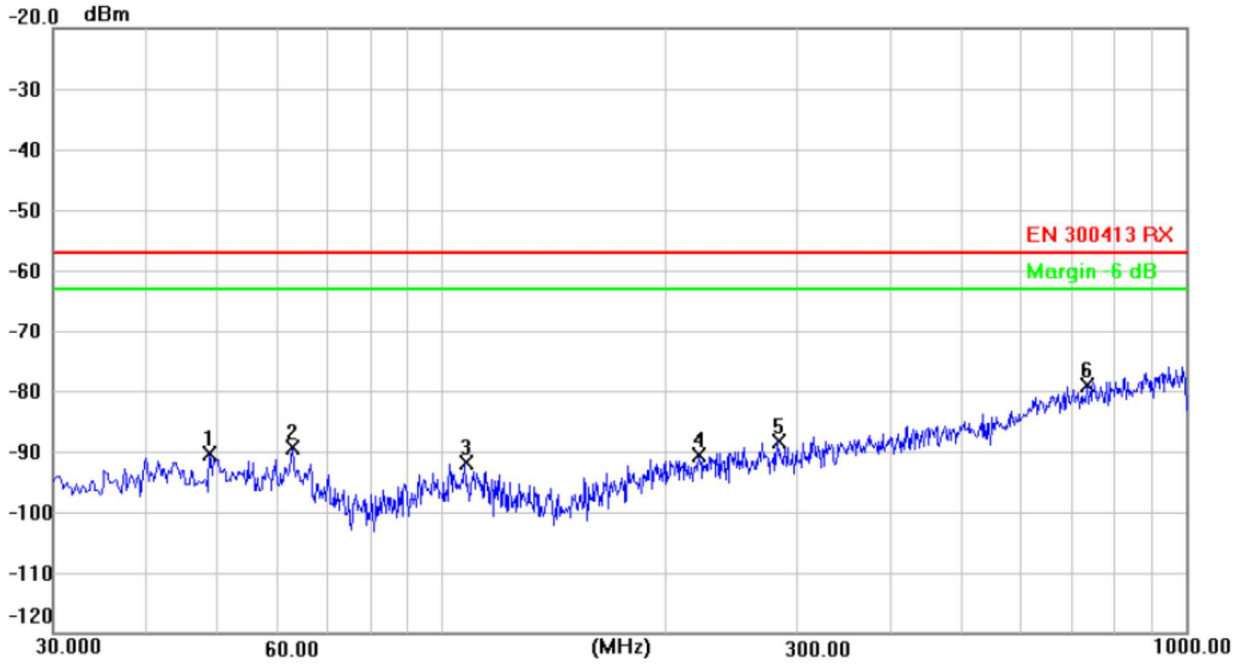
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	62.6567	-74.63	-15.75	-90.38	-57.00	-33.38	peak	150	360	P	
2	110.8333	-76.51	-16.13	-92.64	-57.00	-35.64	peak	150	360	P	
3	188.1100	-72.46	-17.11	-89.57	-57.00	-32.57	peak	150	360	P	
4	257.3032	-73.03	-14.36	-87.39	-57.00	-30.39	peak	150	360	P	
5	637.8667	-72.34	-6.62	-78.96	-57.00	-21.96	peak	150	360	P	
6 *	983.8333	-72.99	-2.16	-75.15	-57.00	-18.15	peak	150	360	P	

Figure 4: Spectral diagrams and measurement results for 30-1000MHz, Vertical polarization, GPS



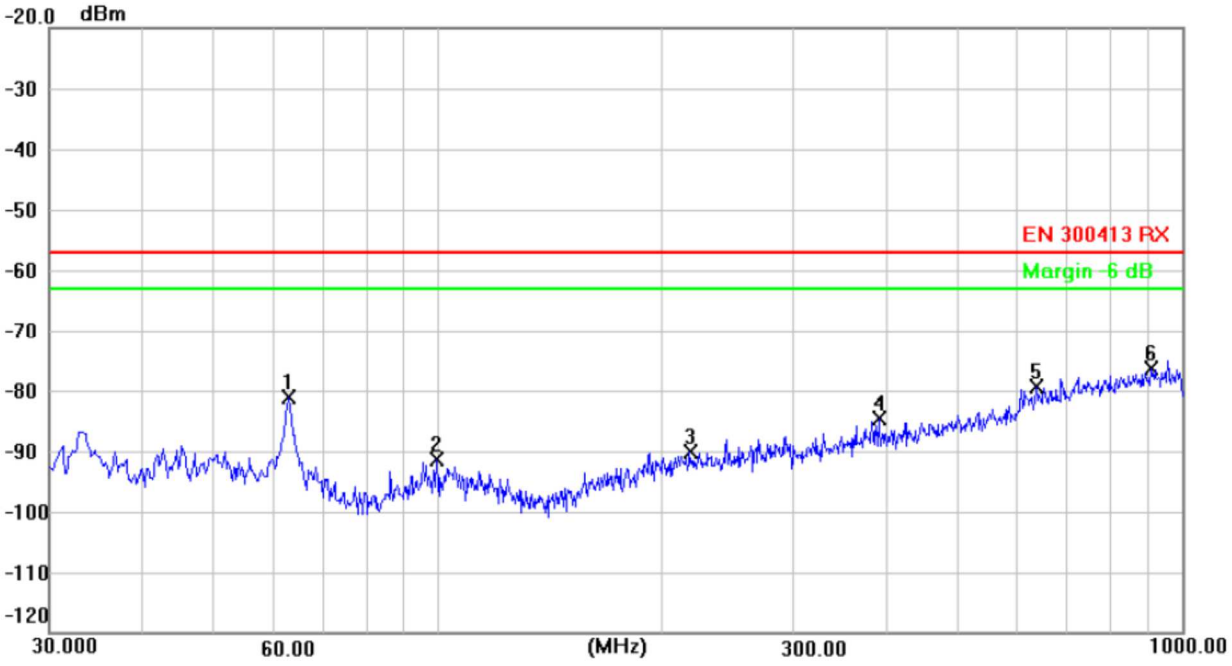
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	33.2333	-71.67	-15.67	-87.34	-57.00	-30.34	peak	150	360	P	
2	51.9867	-74.69	-14.06	-88.75	-57.00	-31.75	peak	150	360	P	
3	62.9800	-67.09	-15.80	-82.89	-57.00	-25.89	peak	150	360	P	
4	213.3300	-74.99	-15.57	-90.56	-57.00	-33.56	peak	150	360	P	
5	275.4100	-73.65	-13.98	-87.63	-57.00	-30.63	peak	150	360	P	
6 *	826.0467	-71.84	-3.98	-75.82	-57.00	-18.82	peak	150	360	P	

Figure 5: Spectral diagrams and measurement results for 30-1000MHz, Horizontal polarization, GLONASS



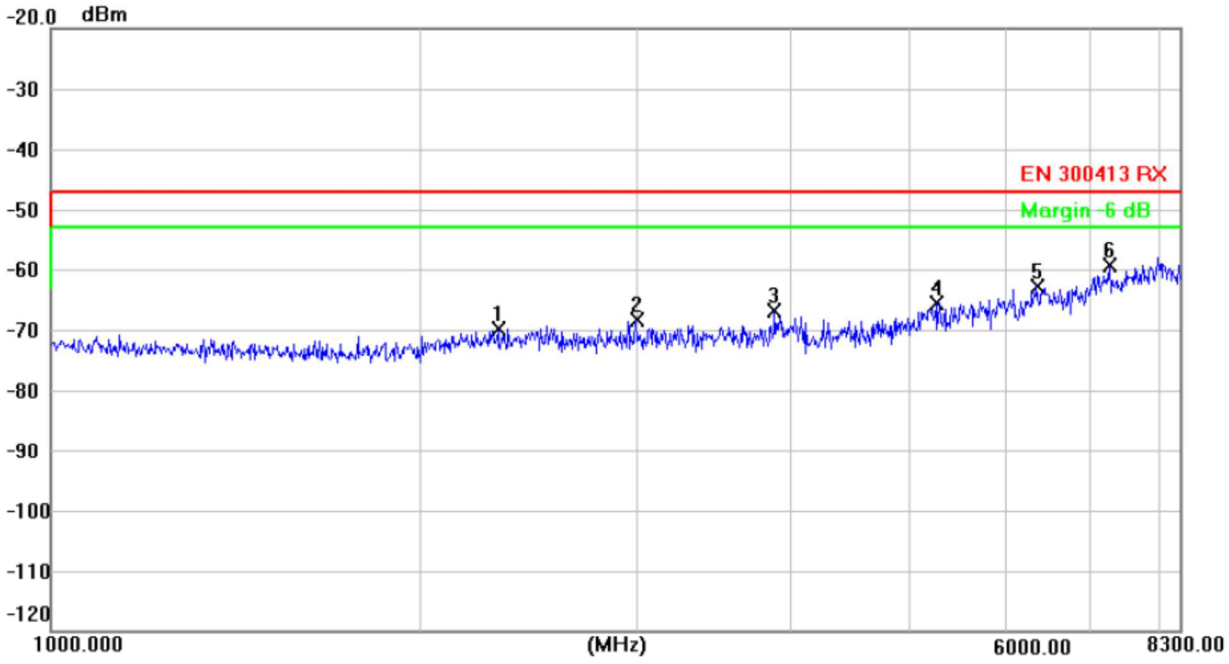
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	48.7533	-76.54	-13.85	-90.39	-57.00	-33.39	peak	150	360	P	
2	62.9800	-73.49	-15.80	-89.29	-57.00	-32.29	peak	150	360	P	
3	107.2767	-76.09	-15.87	-91.96	-57.00	-34.96	peak	150	360	P	
4	222.0600	-75.28	-15.33	-90.61	-57.00	-33.61	peak	150	360	P	
5	282.8467	-74.64	-13.83	-88.47	-57.00	-31.47	peak	150	360	P	
6 *	739.3933	-74.14	-5.08	-79.22	-57.00	-22.22	peak	150	360	P	

Figure 6: Spectral diagrams and measurement results for 30-1000MHz, Vertical polarization, GLONASS



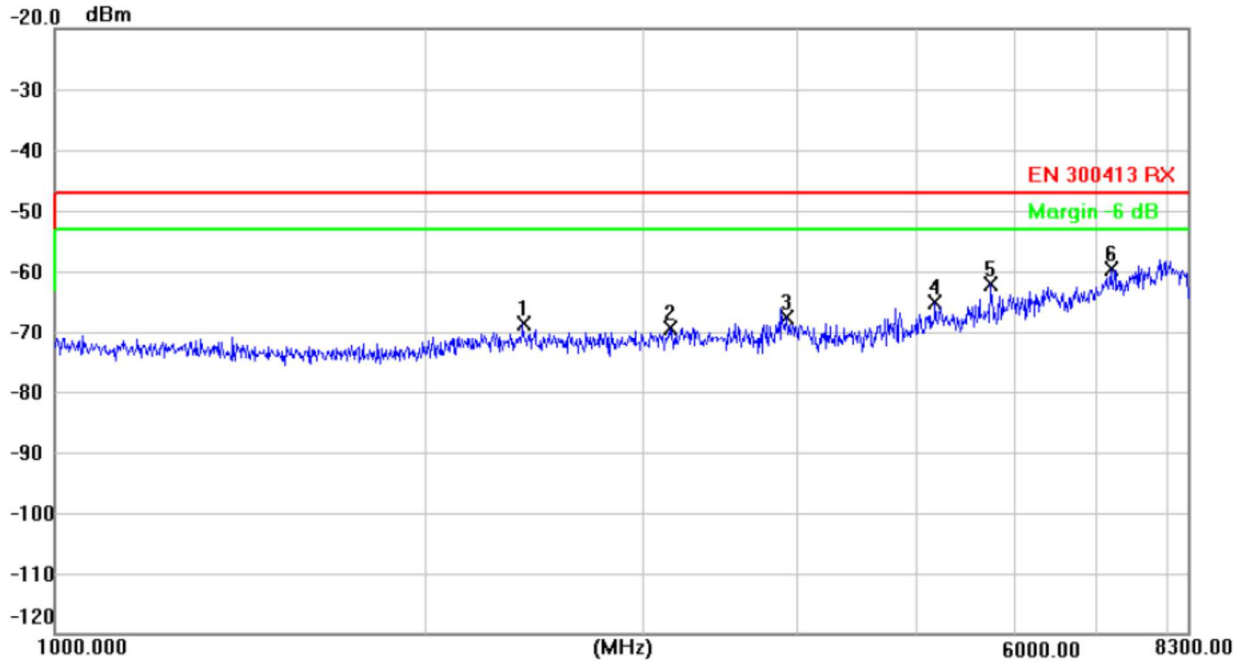
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	62.9800	-65.37	-15.80	-81.17	-57.00	-24.17	peak	150	360	P	
2	99.8399	-75.21	-16.09	-91.30	-57.00	-34.30	peak	150	360	P	
3	217.8567	-74.60	-15.46	-90.06	-57.00	-33.06	peak	150	360	P	
4	391.4867	-73.26	-11.24	-84.50	-57.00	-27.50	peak	150	360	P	
5	638.1900	-72.74	-6.62	-79.36	-57.00	-22.36	peak	150	360	P	
6 *	909.1433	-73.50	-2.90	-76.40	-57.00	-19.40	peak	150	360	P	

Figure 7: Spectral diagrams and measurement results for 1-8.3GHz, Horizontal polarization, BDS



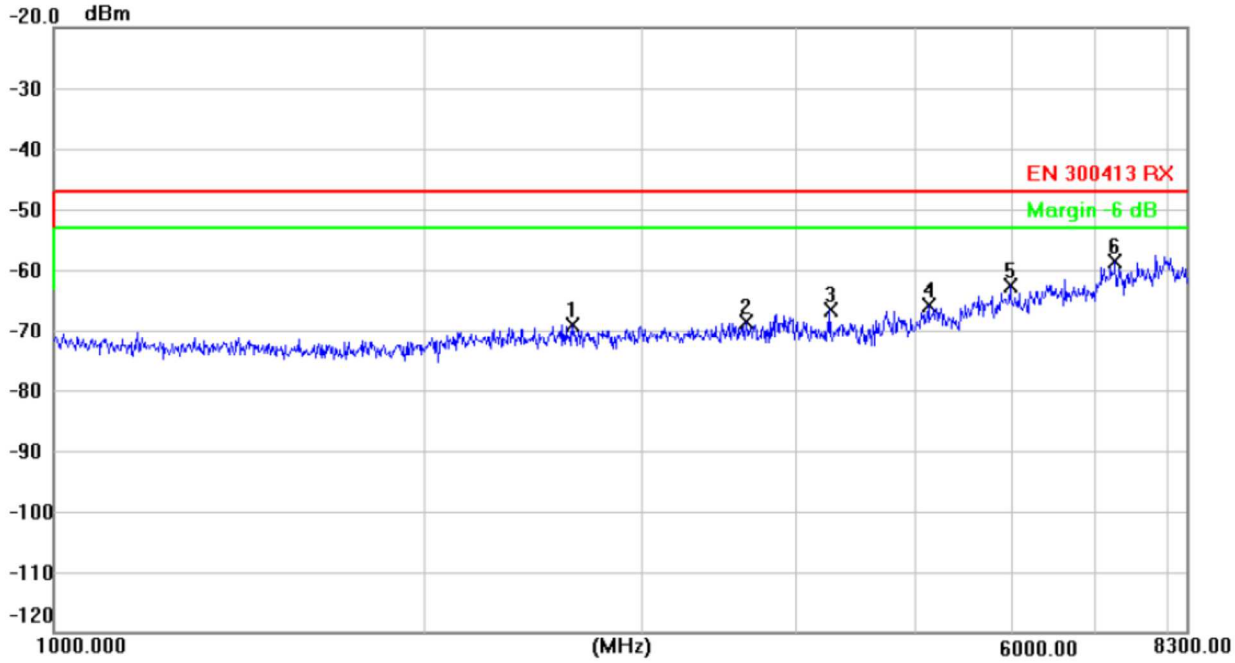
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2316.433	-66.42	-3.47	-69.89	-47.00	-22.89	peak	150	360	P	
2	2997.767	-66.28	-2.20	-68.48	-47.00	-21.48	peak	150	360	P	
3	3883.500	-66.87	0.07	-66.80	-47.00	-19.80	peak	150	360	P	
4	5258.333	-68.69	3.05	-65.64	-47.00	-18.64	peak	150	360	P	
5	6348.467	-69.77	6.80	-62.97	-47.00	-15.97	peak	150	360	P	
6 *	7287.733	-69.52	10.20	-59.32	-47.00	-12.32	peak	150	360	P	

Figure 8: Spectral diagrams and measurement results for 1-8.3GHz, Vertical polarization, BDS



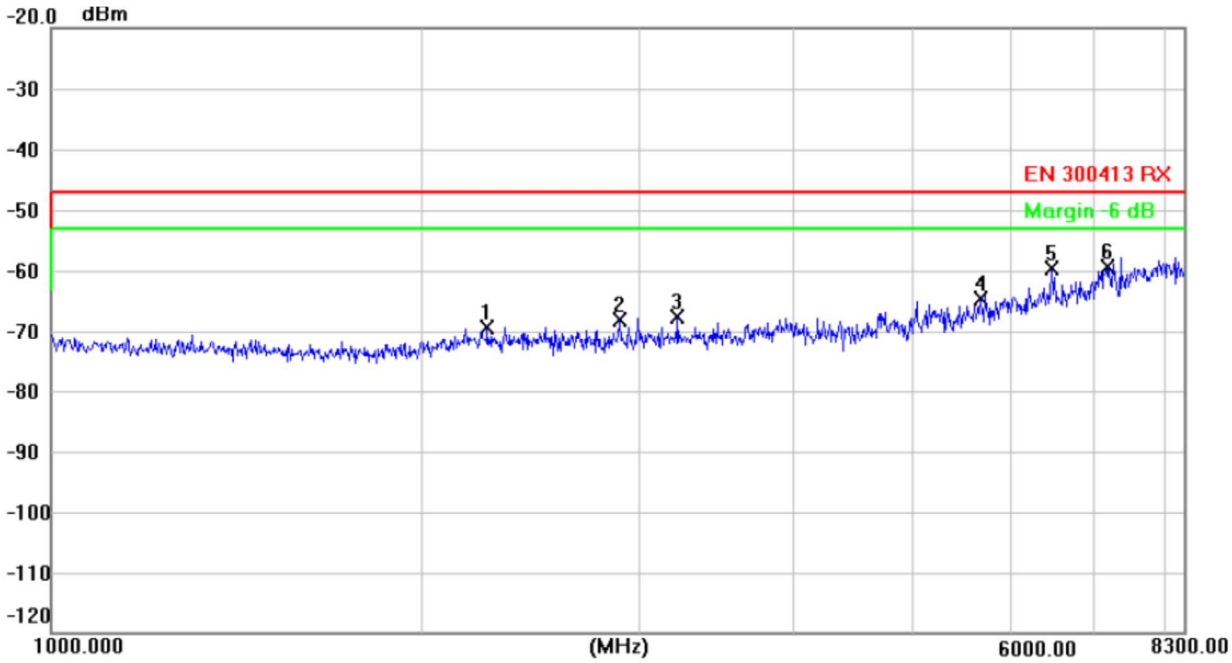
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2399.167	-65.22	-3.36	-68.58	-47.00	-21.58	peak	150	360	P	
2	3160.800	-67.28	-2.08	-69.36	-47.00	-22.36	peak	150	360	P	
3	3929.733	-67.85	0.25	-67.60	-47.00	-20.60	peak	150	360	P	
4	5185.333	-68.11	2.88	-65.23	-47.00	-18.23	peak	150	360	P	
5	5745.000	-66.63	4.58	-62.05	-47.00	-15.05	peak	150	360	P	
6 *	7197.700	-69.80	10.17	-59.63	-47.00	-12.63	peak	150	360	P	

Figure 9: Spectral diagrams and measurement results for 1-8.3GHz, Horizontal polarization, GPS



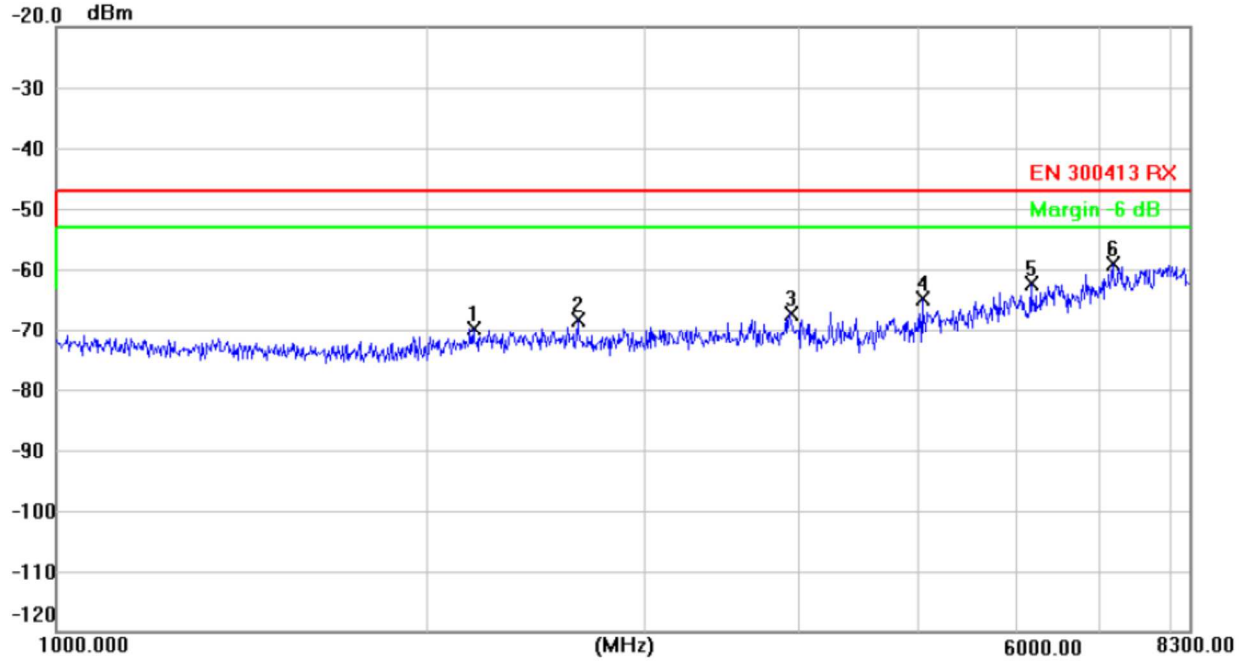
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2637.633	-66.33	-2.89	-69.22	-47.00	-22.22	peak	150	360	P	
2	3649.900	-67.89	-0.72	-68.61	-47.00	-21.61	peak	150	360	P	
3	4267.967	-67.60	0.90	-66.70	-47.00	-19.70	peak	150	360	P	
4	5124.500	-68.56	2.70	-65.86	-47.00	-18.86	peak	150	360	P	
5	5976.167	-67.99	5.37	-62.62	-47.00	-15.62	peak	150	360	P	
6 *	7270.700	-68.92	10.19	-58.73	-47.00	-11.73	peak	150	360	P	

Figure 10: Spectral diagrams and measurement results for 1-8.3GHz, Vertical polarization, GPS



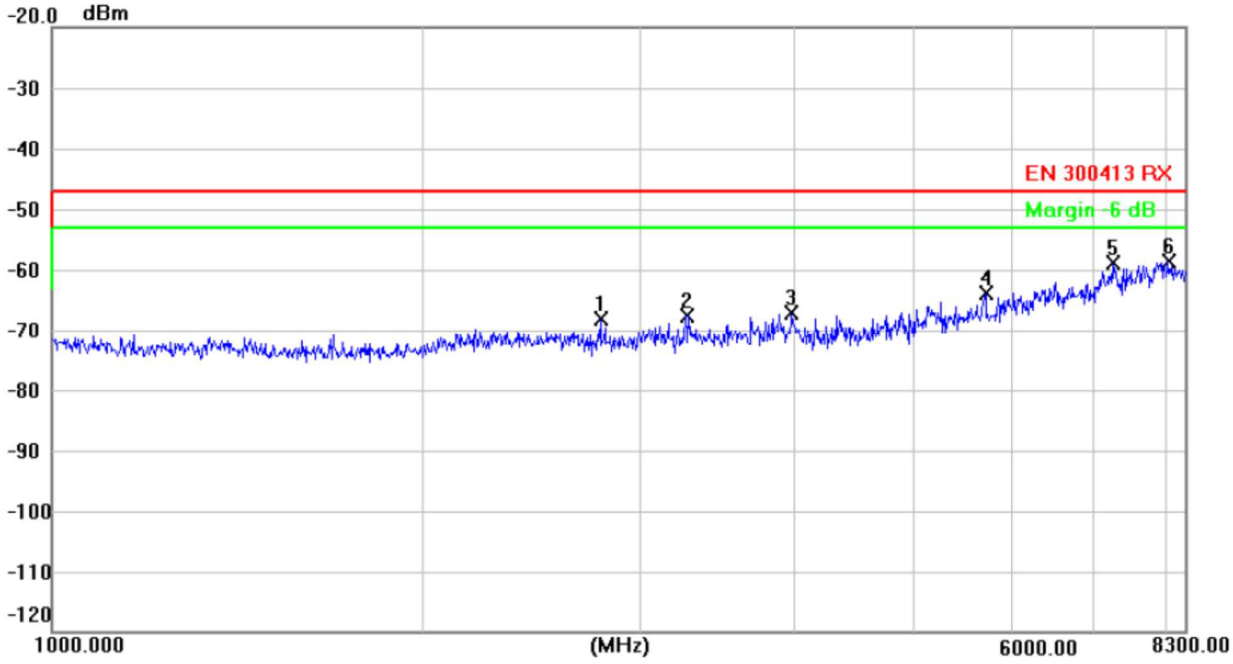
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2262.900	-65.84	-3.52	-69.36	-47.00	-22.36	peak	150	360	P	
2	2898.000	-65.62	-2.46	-68.08	-47.00	-21.08	peak	150	360	P	
3	3226.500	-65.70	-2.01	-67.71	-47.00	-20.71	peak	150	360	P	
4	5672.000	-69.04	4.30	-64.74	-47.00	-17.74	peak	150	360	P	
5	6482.300	-66.84	7.26	-59.58	-47.00	-12.58	peak	150	360	P	
6 *	7200.133	-69.53	10.18	-59.35	-47.00	-12.35	peak	150	360	P	

Figure 11: Spectral diagrams and measurement results for 1-8.3GHz, Horizontal polarization, GLONASS



No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2180.167	-66.13	-3.79	-69.92	-47.00	-22.92	peak	150	360	P	
2	2652.233	-65.46	-2.87	-68.33	-47.00	-21.33	peak	150	360	P	
3	3939.467	-67.65	0.30	-67.35	-47.00	-20.35	peak	150	360	P	
4	5041.767	-67.32	2.47	-64.85	-47.00	-17.85	peak	150	360	P	
5	6175.700	-68.43	6.05	-62.38	-47.00	-15.38	peak	150	360	P	
6 *	7197.700	-69.39	10.17	-59.22	-47.00	-12.22	peak	150	360	P	

Figure 12: Spectral diagrams and measurement results for 1-8.3GHz, Vertical polarization, GLONASS



No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	2788.500	-65.43	-2.73	-68.16	-47.00	-21.16	peak	150	360	P	
2	3267.867	-65.70	-1.94	-67.64	-47.00	-20.64	peak	150	360	P	
3	3985.700	-67.67	0.48	-67.19	-47.00	-20.19	peak	150	360	P	
4	5720.667	-68.37	4.49	-63.88	-47.00	-16.88	peak	150	360	P	
5	7253.667	-69.06	10.19	-58.87	-47.00	-11.87	peak	150	360	P	
6 *	8042.067	-69.27	10.75	-58.52	-47.00	-11.52	peak	150	360	P	

5 Photographs of the Test Set-Up

Photograph 1: Set-up for radiated spurious emission below 1GHz



Photograph 2: Set-up for radiated spurious emission above 1GHz



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